NON-RECIPROCAL EFFECTS IN GATED SUPERCONDUCTOR/SEMICONDUCTOR PLANAR JOSEPHSON JUNCTIONS

by

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To my teachers,

my family,

and everyone who helped me reach this point.

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INTRODUCTION

In 1962, Brian D. Josephson predicted that a dissipationless current, called a supercurrent, can flow between two superconductors (S) separated by a thin insulating layer (I) [1]. The resulting device, now called a Josephson junction (JJ), supports a direct current at zero applied voltage, the Josephson effect. This prediction was experimentally verified in the following year [2], triggering decades of research activities, where JJs were applied to large-scale superconducting circuits.

Josephson junctions serve as a building block for superconducting devices such as magnetic sensors, voltage standards, high-speed digital logic, etc. [3, 4]. In particular, when placed in a loop, JJs can be used to build superconducting quantum interference devices (SQUIDs). They are highly sensitive to magnetic field changes, allowing the detection of magnetic fields as small as 10^{-14}T [5]. Furthermore, due to their non-dissipative nonlinear behavior as a circuit element, JJs are crucial for the realization and manipulation of superconducting qubits [6]. The JJ nonlinearity introduces anharmonicity in the energy levels, allowing the isolation of the two lowest energy states (qubit) from higher energy states [7]. On the other hand, the lack of dissipation helps increase the qubit coherence [8]. For this reason, superconducting qubits are among the leading contenders in the quest to create quantum computers capable of solving highly complex problems believed to be intractable with classical (digital) computers [9]. Superconducting qubits built with JJs are customarily used in IBM quantum processors [10, 11]. Most recently, the 2025 Nobel Prize in Physics recognized macroscopic quantum tunneling and energy quantization in superconducting circuits built from JJs, underscoring their central role in today's quantum technologies [12].

Conventional S/I/S JJs have been extensively studied over the past several decades. However, because the weak link is formed by an insulating tunnel barrier, these junctions usually have low transparency and limited post-fabrication tunability. More recently, proximitized planar Josephson junctions have emerged as a highly tunable alternative [13]. In these devices, superconducting electrodes are placed in contact with a semiconductor quantum well, allowing superconducting correlations to extend into the semiconductor through the proximity effect [14]. As a result, a supercurrent can flow through a gate-tunable semiconductor weak link. As detailed in Chapters 1 and 2, this proximity-induced superconductivity provides the basis for realizing planar JJs in semiconductor two-dimensional electron gases. For simplicity, we refer to these proximitized semiconductor-superconductor Josephson junctions as planar JJs throughout this thesis.

The main goal of this thesis is to theoretically study nonreciprocal Josephson transport in semiconductor–superconductor planar JJs. In particular, we focus on the superconducting diode effect (SDE) [15–21], where the magnitudes of the forward and reverse critical currents are different. In the mechanism investigated in this thesis, inversion symmetry is broken by Rashba and/or Dresselhaus spin–orbit coupling, while time-reversal symmetry is broken by the Zeeman interaction. Their interplay can produce a nonreciprocal current–phase relation and a finite SDE. Junction transparency, crystallographic orientation, electrostatic gating, and disorder can further modify the magnitude, magnetic-field dependence, and polarity of the diode response.

This thesis is organized as follows. Chapter 1 provides the general background on superconductivity, semiconductor–superconductor systems, and the Josephson effect.

Chapter 2 introduces planar JJs and discusses how the superconducting proximity effect enables supercurrent transport through a semiconductor weak link. Chapter 3 presents the Bogoliubov–de Gennes formalism and the analytical and numerical methods used to calculate the transport properties of these junctions. Chapter 4 examines the SDE in planar JJs, with emphasis on its symmetry requirements, phenomenological description, and microscopic origin. Chapter 5 investigates the magnetocurrent–phase relation and its phase-resolved signatures of the SDE. Chapter 6 analyzes the magnetic and crystalline anisotropies arising from Rashba and Dresselhaus spin–orbit couplings. Chapter 7 examines how electrostatic gating, junction transparency, and disorder modify the magnitude and polarity of the diode response. Finally, Chapter 8 summarizes the main conclusions of this thesis.

1 GENERAL BACKGROUND

1.1 Superconductivity

Superconductivity is one of the most interesting macroscopic quantum phenomena in condensed matter physics. It was discovered by the Dutch physicist Heike Kamerlingh Onnes in 1911, while studying the low-temperature resistance of mercury using liquid helium. He observed that, below a critical temperature T_c, the electrical resistance of mercury abruptly vanished, as shown in Fig. 1 [22]. Throughout the 1920s and 1930s, the magnetic characteristics of superconductors received considerable interest. A breakthrough was made in 1933 when Meissner and Ochsenfeld demonstrated that the flux inside a superconductor was expelled under magnetic fields below a critical field value, H_c [23]. This perfect diamagnetic behavior is called the Meissner effect.

Later on, in 1957, Bardeen, Cooper, and Schrieffer developed the first microscopic theory to explain superconductivity [24]. In a normal metal, electrons scatter independently. However, near the Fermi surface, below the critical temperature, an attractive interaction can arise due to phonon exchange. This leads to the formation of Cooper pairs (CPs) composed of electrons with opposite momenta and spin $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ [25]. The CPs behave as bosons and can eventually condense into a Bose-Einstein condensate below T_c. The microscopic origin of the attractive interaction between electrons in a superconductor was first proposed by Fröhlich, who recognized the role of the electron–phonon coupling in mediating an effective attraction between charge carriers [26]. A qualitative picture can be drawn as follows: as an electron moves through the ionic lattice, it exerts a Coulomb force on the surrounding ions, locally distorting the lattice and producing a region of

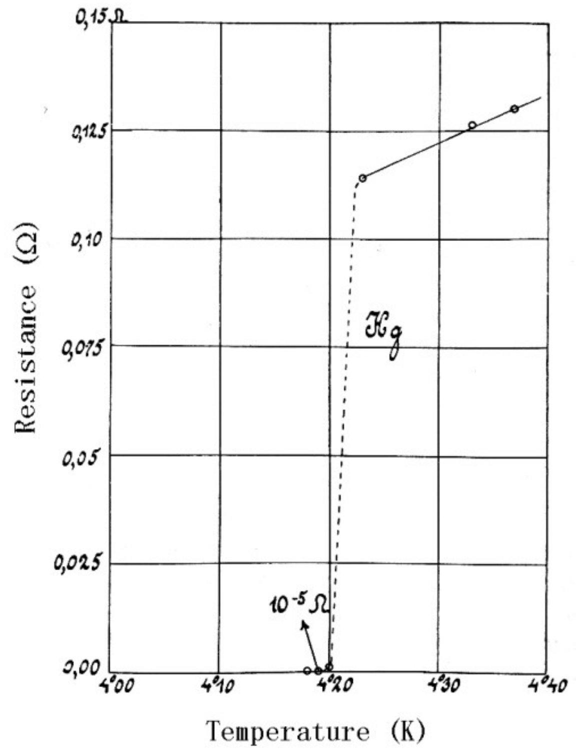

Figure 1: Resistance of mercury as a function of temperature. As the system cools, we can observe that the resistance decreases. At about $4.2K$, the resistance abruptly drops to zero, turning the system into a superconductor. This graph is adapted from the novel work of Kamerlingh Onnes [22].

increased positive charge density. This charge cloud effectively screens the electron's negative charge and acts as an attractive potential for other electrons passing through the same region. When another electron, typically with opposite momentum and spin, passes through this region before the lattice distortion relaxes, it experiences this attractive potential and becomes correlated with the first electron, forming a Cooper pair. These pairs are not localized but extend coherently over many lattice sites, establishing the macroscopic phase coherence that defines the superconducting state. The formation of CPs lowers the energy of the system, opening an energy gap Δ in the (single-particle) electronic density of states at the Fermi level [27]. This gap represents the energy required to break a Cooper pair and is responsible for the absence of low-energy excitations and electrical resistance. From a quantum-mechanical perspective, the lattice vibrations responsible for this interaction correspond to the emission and absorption of phonons, which serve as mediators of the electron–electron attraction.

The effective Hamiltonian incorporating the phonon-mediated attraction between electrons of opposite momentum and spin can be written as

$$H = \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}}\, c^\dagger_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma} - V \sum_{\mathbf{k},\mathbf{k}'} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}, \tag{1.1}$$

where $\epsilon_{\mathbf{k}}$ is the single-particle energy measured relative to the Fermi level, and $V > 0$ represents the effective phonon-mediated attraction between time-reversed electronic states within an energy window of width $\hbar\omega_D$ (the Debye energy), and $c^\dagger_{k\sigma}(c_{k\sigma})$ is the particle creation (annihilation) operator [28].

The interaction term in Eq. (1.1) contains four fermionic operators and is therefore difficult to diagonalize exactly. In the mean-field approximation, fluctuations around the average pairing amplitude are neglected, allowing the two-body interaction to be replaced by an effective pairing field [29]. The superconducting pairing potential is then defined as

$$\Delta = V \sum_{\mathbf{k}'} \langle c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} \rangle. \tag{1.2}$$

The interaction Hamiltonian then reduces to

$$H_{\mathrm{MF}} = \sum_{\mathbf{k},\sigma} \epsilon_{\mathbf{k}}\, c^\dagger_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma} - \sum_{\mathbf{k}} \left(\Delta\, c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} + \Delta^*\, c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \right) + \frac{|\Delta|^2}{V}. \tag{1.3}$$

The mean-field Hamiltonian in Eq. (1.3) is quadratic in the fermionic operators and can therefore be diagonalized by introducing Bogoliubov quasiparticle operators. For each pair of time-reversed states $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$, the Bogoliubov transformation is defined as

$$\gamma_{\mathbf{k}\uparrow} = u_{\mathbf{k}} c_{\mathbf{k}\uparrow} - v_{\mathbf{k}} c^{\dagger}_{-\mathbf{k}\downarrow}, \tag{1.4}$$

$$\gamma^{\dagger}_{-\mathbf{k}\downarrow} = v^{*}_{\mathbf{k}} c_{\mathbf{k}\uparrow} + u^{*}_{\mathbf{k}} c^{\dagger}_{-\mathbf{k}\downarrow}, \tag{1.5}$$

where

$$|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1. \tag{1.6}$$

The coefficients $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ describe the mixing of particle and hole states caused by superconducting pairing. Consequently, a Bogoliubov-de Gennes (BdG) quasiparticle is a coherent superposition of particle-like and hole-like components.

After this transformation, the mean-field Hamiltonian takes the diagonal form

$$H_{\mathrm{MF}} = \sum_{\mathbf{k}} E_{\mathbf{k}} \left(\gamma^{\dagger}_{\mathbf{k}\uparrow} \gamma_{\mathbf{k}\uparrow} + \gamma^{\dagger}_{-\mathbf{k}\downarrow} \gamma_{-\mathbf{k}\downarrow} \right) + \mathrm{const.}, \tag{1.7}$$

where

$$E_{\mathbf{k}} = \sqrt{\epsilon^2_{\mathbf{k}} + |\Delta|^2}. \tag{1.8}$$

Here, $E_{\mathbf{k}}$ is the energy associated with a Bogoliubov quasiparticle of momentum $\mathbf{k}$. The constant term is independent of the quasiparticle occupation and does not affect the excitation spectrum.

In the BdG representation, the quasiparticle spectrum contains two particle–hole-related branches,

$$E_{\pm}(\mathbf{k}) = \pm E_{\mathbf{k}} = \pm\sqrt{\epsilon_{\mathbf{k}}^2 + |\Delta|^2}. \tag{1.9}$$

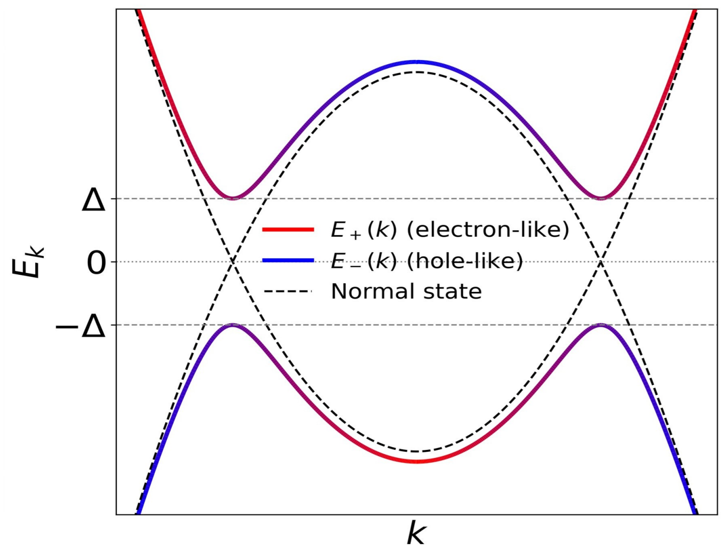

Figure 2: Quasiparticle spectrum in the Bogoliubov–de Gennes formalism. The dashed black curves represent the normal-state particle and hole dispersions, $\epsilon_{\mathbf{k}}$ and $-\epsilon_{\mathbf{k}}$. Superconducting pairing hybridizes these states and produces the positive- and negative-energy quasiparticle branches $E_+(\mathbf{k})$ and $E_-(\mathbf{k})$. At the Fermi momenta, the two branches are separated by an energy $2|\Delta|$ [28].

Figure 2 illustrates the opening of the superconducting quasiparticle gap. In the normal-state BdG representation, the particle and hole dispersions intersect at zero energy at the Fermi momenta. Superconducting pairing removes these crossings. At the Fermi momenta, where $\epsilon_{\mathbf{k}} = 0$, the quasiparticle energies are $E_{\pm} = \pm|\Delta|$. Therefore, the minimum energy required to create a quasiparticle excitation is $|\Delta|$.

The quasiparticle spectrum discussed above corresponds to a spatially uniform superconductor, for which the pairing potential can be treated as a constant. In nonuniform superconducting systems, such as JJs, both the magnitude and phase of the pairing potential may vary in space. The pairing potential is therefore written more generally as

$$\Delta(\mathbf{r}) = |\Delta(\mathbf{r})|e^{i\phi(\mathbf{r})}, \tag{1.10}$$

where $|\Delta(\mathbf{r})|$ characterizes the local pairing strength and $\phi(\mathbf{r})$ is the superconducting phase. The absolute phase of an isolated superconductor is not directly observable. However, when two superconductors are connected through a weak link, the phase difference between them becomes physically relevant and determines the Josephson supercurrent, as discussed in Sec. 1.3.

1.2 Semiconductors and Superconductors

In this section, we review the basic properties of semiconductor-superconductor hybrid systems in the context of the Al/InAs heterostructures studied in this thesis.

Two-Dimensional Electron Gas

A two-dimensional electron gas (2DEG) is formed when electrons are confined in one spatial direction but remain free to move in the other two directions. In semiconductor heterostructures, this confinement is produced by band bending and quantum-well formation. The resulting electronic states are quantized in the growth direction, while the carriers remain mobile in the plane of the heterostructure. In this thesis, we focus on 2DEGs formed in III–V semiconductor quantum wells, such as InAs-based heterostructures. These systems are useful for planar JJs because they can provide high carrier mobility and electrostatic gate tunability. Depending on the material and device design, they can also exhibit sizable spin-orbit coupling (SOC) and large effective g factors. These properties are important for controlling the normal-state transport and spin structure of the weak link.

A planar Josephson junction requires superconducting regions coupled to a semiconductor. In such a hybrid system, superconducting correlations are induced in the semiconductor through the superconducting proximity effect, which will be discussed in the following subsection. Therefore, the role of the 2DEG here is to provide a gate-tunable conducting channel through which proximity-induced superconductivity and Josephson transport can occur.

In an InAs-based heterostructure, semiconductor layers with different band gaps are grown on top of one another, as shown schematically in Fig. 3(a). The resulting conduction-band profile creates a quantum well in the InAs layer. This quantum well confines carriers in the growth direction while allowing them to move in the plane of the device. The confined conducting channel formed in this way is the 2DEG, as illustrated in Fig. 3(b).

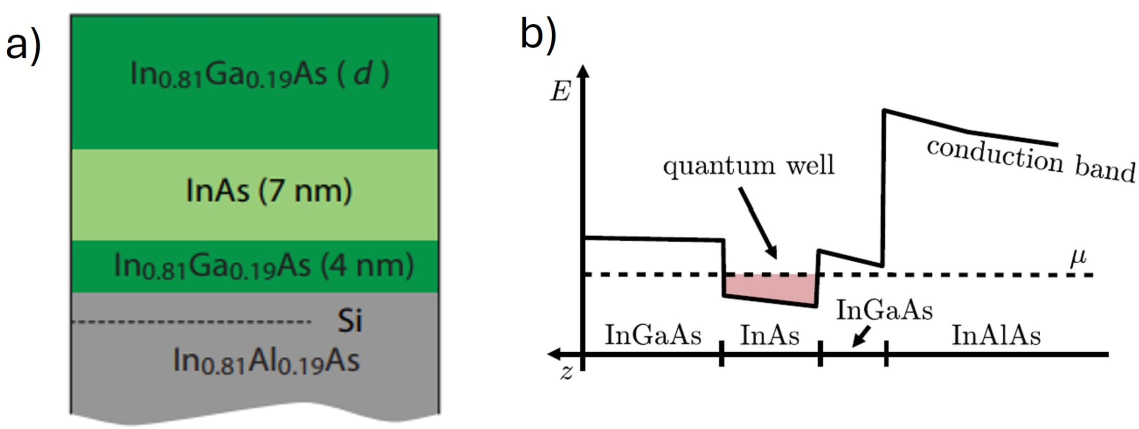

Figure 3: **(a)** Schematics of a semiconductor heterostructure, where a 2DEG forms in the InAs layer **(b)** Diagram of the conduction band profile of the considered heterostructure. The quantum well in the InAs layer confines carriers in the growth direction and hosts the 2DEG. Figures are adapted from Ref. [30] and [31].

Spin-orbit coupling describes the coupling between the spin and orbital motion of charge carriers. In semiconductor heterostructures, SOC plays an important role because the motion of electrons is influenced both by the crystal structure and by electric fields associated with the confining potential. This results in effective SOC fields that can be several orders of magnitude stronger than the atomic SOC. Two important forms of SOC in III–V semiconductor quantum wells are Dresselhaus and Rashba SOCs. Dresselhaus SOC originates from bulk inversion asymmetry (BIA) of the crystal structure [32, 33], while Rashba SOC originates from structure inversion asymmetry (SIA) of the heterostructure [33, 34]. Rashba and Dresselhaus SOCs and their role in planar JJs are discussed in Chapter 3.

Superconductor-Semiconductor Interface

Let us consider the interface between a superconductor and a normal (not superconducting) material, like a semiconductor. When a semiconductor is brought into contact with a superconductor, the combined system can host a rich variety of hybrid electronic states. The emergence of these hybrid excitations is rooted in the process of Andreev reflection, the proximity effect, and the formation of Andreev bound states (ABSs) in confined geometries. These mechanisms form the microscopic basis of supercurrent transport in superconductor–semiconductor JJs and are crucial for understanding transport phenomena in planar junctions, as explained below.

Andreev Reflection

Consider a normal–superconductor (NS) interface, as shown in Fig. 4(a). An electron with up spin is incident on the NS interface from the normal region with an energy $|E| < \Delta$. Because quasiparticle states are forbidden inside the superconducting gap, the electron cannot directly enter the superconductor as a single particle. Instead, it can

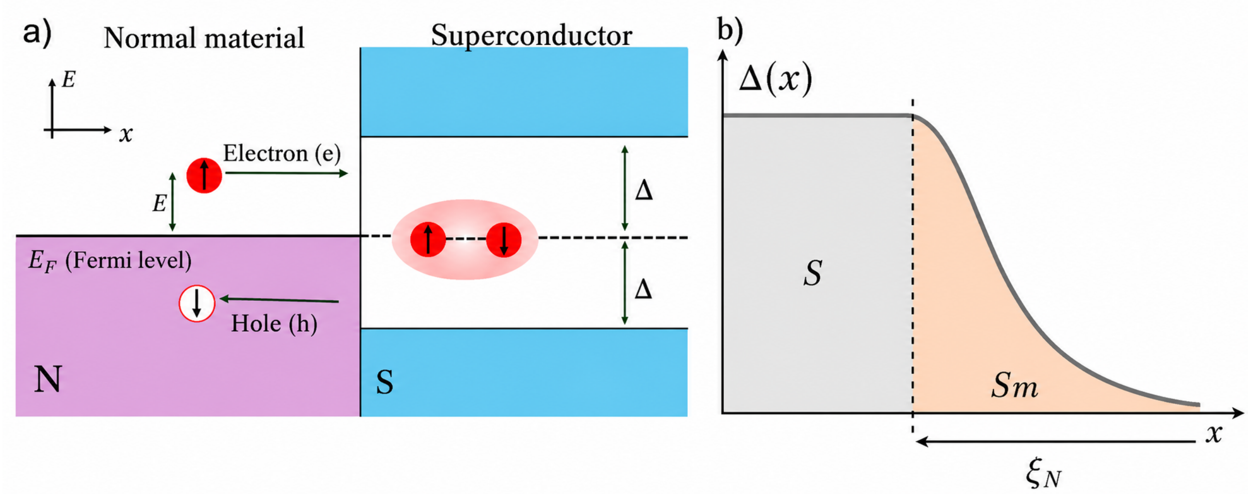

Figure 4: **(a)** Andreev reflection at a normal-superconductor (NS) interface. An electron with spin up and energy $|E| < \Delta$ incident from the normal region is retroreflected as a hole with opposite spin. This process transfers a Cooper pair into the superconducting condensate. **(b)** Schematic illustration of the superconducting proximity effect at a superconductor-semiconductor (S/Sm) interface. Superconducting correlations extend from the superconductor into the semiconductor and decay over a characteristic length scale ξ_N [31, 35].

contribute to a Cooper pair in the superconductor by drawing a second electron (of opposite momentum and spin) from the Fermi sea. This process leaves behind a hole in the semiconductor with spin and momentum opposite to those of the incident electron, causing the hole to retrace the electron's original trajectory. This electron–hole retroreflection is known as Andreev reflection. Thus, a charge 2e is transferred into the superconductor in each Andreev reflection process. This is the essential mechanism through which phase coherence leaks into the semiconductor, enabling superconducting transport across non-superconducting (normal) regions. The efficiency of Andreev reflection depends sensitively on the transparency of the interface and on the spin polarization of the non-superconducting region.

Superconducting Proximity Effect

When a normal metal or a semiconductor is brought into contact with a superconductor, superconducting correlations can extend into the non-superconducting side. This is known as the *superconducting proximity effect*. At the microscopic level, it originates from the process of Andreev reflection described earlier.

In a simple picture, the superconducting order parameter $\Delta(x)$ does not change abruptly at the interface but rather decays smoothly from its bulk value inside the superconductor into the normal region, as depicted in Fig. 4(b). The decay length depends on the transport regime: in a clean or ballistic system, it is of order $\xi_N = \hbar v_F / 2\pi k_B T$, while in the diffusive regime it becomes $\xi_N = \sqrt{\hbar D / 2\pi k_B T}$, where D is the diffusion coefficient [13, 36]. Within this region, Cooper-pair correlations survive even though the intrinsic pairing interaction is absent. As a result, the semiconductor acquires an effective superconducting gap, usually called the *proximity-induced gap*[13, 35].

Andreev Bound States

An SNS JJ is typically composed of a normal layer sandwiched between two superconducting layers. Multiple Andreev reflections can occur at the two N/S interfaces, leading to interference between counterpropagating Andreev-reflected electrons and holes [see Fig. 5(a)]. This interference leads to the formation of standing waves and the emergence of discrete subgap resonant states of entangled electron-hole pairs, confined between the S regions [see Fig. 5(a)]. Such states are known as Andreev bound states (ABS) and are analogous to the resonant standing-wave modes of a Fabry–Pérot interferometer [37].

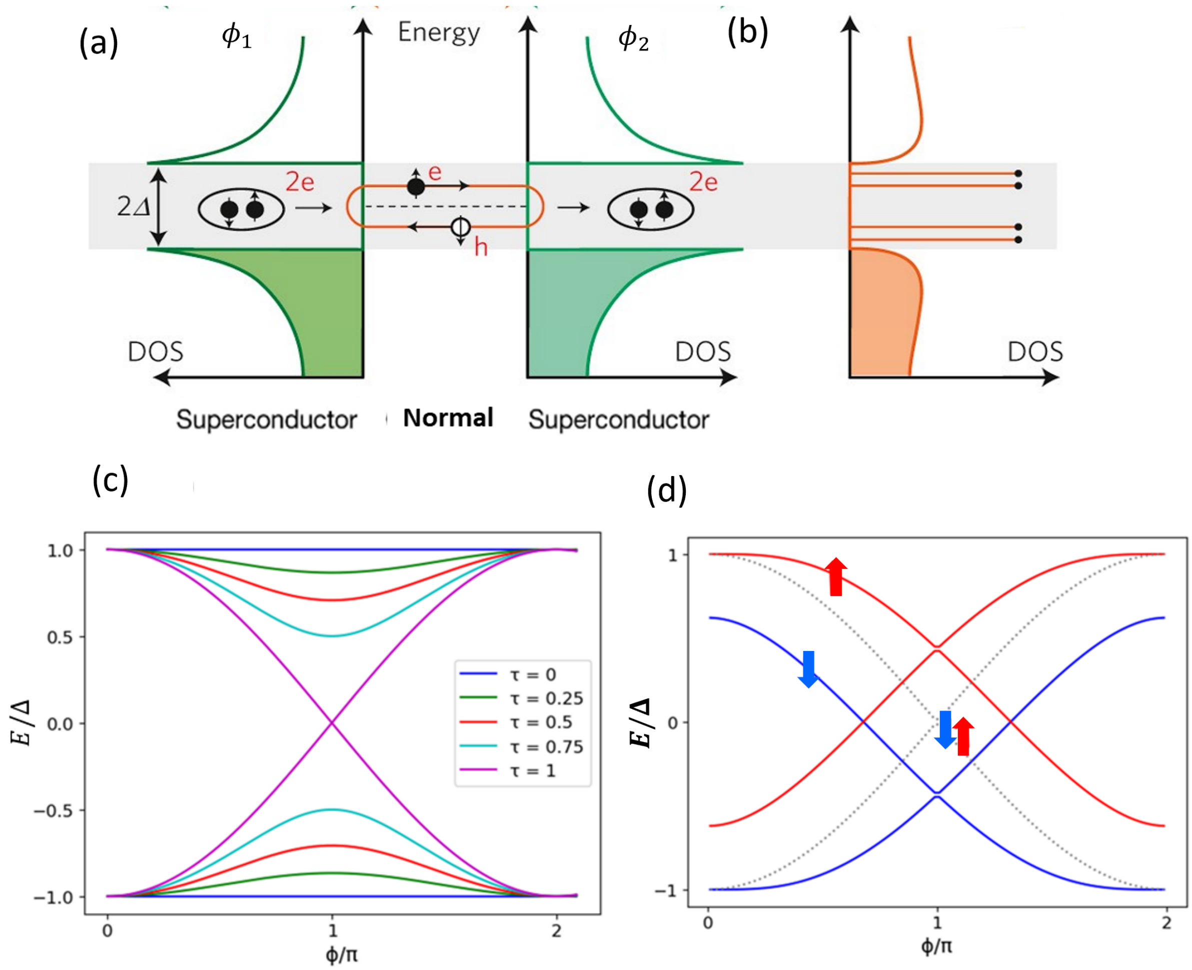

Figure 5: **(a)** Density of states (DOS) in a JJ and formation of ABS, Phase coherent electron-like (e) and hole-like (h) quasi-particles with energy inside the superconducting gap (Δ) undergo Andreev reflection and interfere with each other to form resonant states of entangled e-h pairs, strongly localized in the N region of the JJ, and called ABS [37]. **(b)** ABS energies for a single-channel JJ at a fixed phase difference, $\phi = \phi_2 - \phi_1$. Due to the particle-hole symmetry of the superconductor, ABS form in pairs. The presence of a magnetic field breaks the spin degeneracy, leading to the formation of four sub-gap ABS (orange straight lines inside the gray-dashed superconducting gap). **(c)** ABS energy spectrum as a function of the phase difference, ϕ, for different values of the junction transparency (τ). **(d)** ABS spectrum as a function of ϕ, the grey-dotted lines represent the two-fold spin-degenerate spectrum that would exist in the absence of a magnetic field. In the presence of a magnetic field, the spin-degeneracy is broken and each gray-dotted line splits into two non-degenerate spin states (red and blue lines). The zero-energy crossings between spin-up and spin-down states are accompanied by changes in the fermion parity of the junction and under some conditions may indicate the emergence of topological phase transition [38, 39]

Andreev bound states strongly depend on the superconducting phase difference, $\phi = \phi_2 - \phi_1$, across the junction and are largely responsible for carrying Josephson currents in SNS JJs. Note that, due to the particle-hole symmetry of the superconductor, ABS always occur in pairs. For a single-channel JJ in the absence of a magnetic field, two spin-degenerate ABS appear within the superconducting gap. A magnetic field breaks the spin degeneracy, resulting in four ABS with distinct energies [see Fig. 5]. Figure 5(d) illustrates the ABS spectrum of a perfectly transparent JJ as a function of the superconducting phase difference. The gray dotted lines represent the doubly degenerate ABS spectrum at zero magnetic field. In a finite magnetic field, the spin-up (red lines) and spin-down (blue lines) energy curves shift in opposite directions, resulting in spin-nondegenerate ABS.

The phase-dependent energy of spin-degenerate ABS in a single-channel JJ at zero magnetic field is given by [40–42],

$$E_{\pm}(\phi) = \pm\Delta\sqrt{1 - \tau \sin^2(\phi/2)}\ , \tag{1.11}$$

where Δ is the proximity-induced superconducting gap, ϕ is the phase difference between the S regions, and τ is the junction transparency. Equation (1.11) shows that the ABS spectrum depends strongly on the junction transparency. In the low-transparency regime, $\tau \ll 1$, the ABS energies remain close to the superconducting gap edges, $\pm\Delta$, and exhibit only a weak dependence on the phase difference. As the transparency increases, the phase dependence becomes stronger. In the fully transparent limit, $\tau = 1$, the two ABS branches cross at zero energy when $\phi = \pi$, as shown in Fig. 5(c). Under some conditions, gap crossings and subsequent reopenings may be accompanied by phase transitions into the topological superconducting state [38, 39].

1.3 Josephson Effect

The Josephson effect is one of the most striking macroscopic manifestations of quantum mechanics in condensed matter systems. To illustrate the basic idea, let us consider a conventional SIS junction, as shown in Fig. 6. The thin red region represents the insulating barrier, typically an oxide layer, that separates the two superconducting electrodes. Although the barrier suppresses the flow of individual electrons, the macroscopic phase coherence of the superconducting condensate allows Cooper pairs to tunnel across it without dissipation. This coherent tunneling gives rise to a supercurrent even in the absence of an applied voltage, which is the essence of the Josephson effect. Let the superconducting

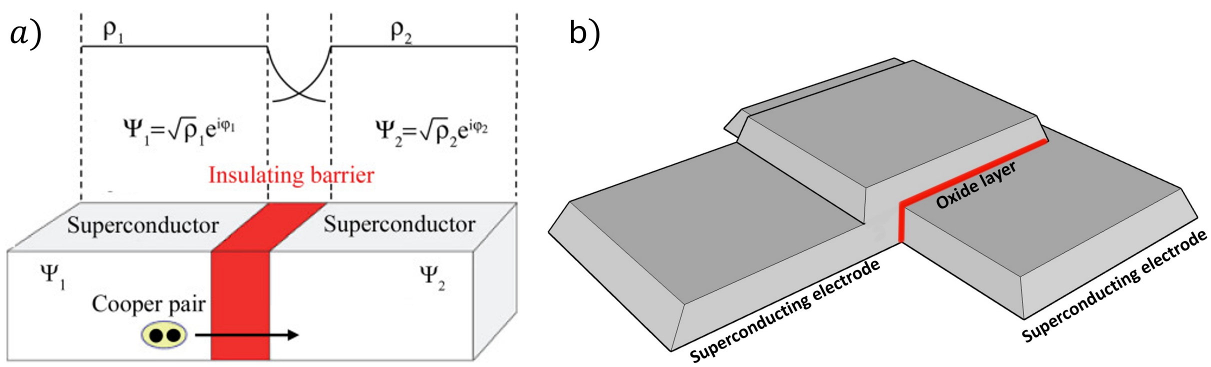

Figure 6: Scheme of an S/I/S junction: **(a)**Two superconductors are separated by a thin insulating barrier (colored red), forming a tunnel junction. The macroscopic wavefunctions from both superconductors overlap in the insulating layer creating a nonzero probability for the Cooper pairs to tunnel across the junction [43]. **(b)** Schematic of a typical S/I/S JJ. The system is composed of two superconducting layers (gray) separated by an insulating oxide layer (red)

order parameters on both sides of the junction be,

$$\Psi_1 = |\Psi_1| e^{i\phi_1}, \quad \Psi_2 = |\Psi_2| e^{i\phi_2}$$

where ϕ_1 and ϕ_2 are the macroscopic phases in the left and right superconductors, respectively. The phase difference

$$\phi = \phi_2 - \phi_1$$

drives the supercurrent,

$$I = I_c \sin\phi \ , \tag{1.12}$$

where I_c is the maximum supercurrent that can be sustained in the junction, called the critical current. If a constant voltage V is applied across the junction, the superconducting phase acquires a time dependence determined by the relation,

$$\frac{d\phi}{dt} = \frac{2eV}{\hbar} \ . \tag{1.13}$$

This shows that a finite voltage causes the phase to evolve linearly in time, leading to an oscillating supercurrent.

To understand the energy exchange associated with this process, we consider the instantaneous electrical power delivered to the junction, given by

$$P(t) = I(t)\, V(t). \tag{1.14}$$

If $E(\phi)$ denotes energy stored in the junction (equivalently, the free energy at $T \to 0$), then energy conservation implies,

$$\frac{dE}{dt} = P = I\,V = I\,\frac{\hbar}{2e}\,\frac{d\phi}{dt} \quad \Longrightarrow \quad \frac{dE}{d\phi} = \frac{\hbar}{2e}\,I(\phi). \tag{1.15}$$

Therefore, the supercurrent can be determined from the derivative of the free energy using,

$$I(\phi) = \frac{2e}{\hbar}\frac{dF(\phi)}{d\phi} \ . \tag{1.16}$$

This is called the current phase relation (CPR) for the JJ, and the factor $2e$ comes from the charge of the Cooper pair.

In SNS JJs, the Josephson current is primarily determined by the phase dependence of the ABS, which constitutes the microscopic mechanism responsible for supercurrent transport in planar JJs.

2 PLANAR JOSEPHSON JUNCTION

The fabrication of S/I/S JJs typically involves the use of conventional superconducting materials and an oxide layer as the insulating barrier, as depicted in Fig. 6(b). Because of the presence of the oxide potential barrier, S/I/S JJs have low transparency. In contrast to S/I/S JJs, proximitized planar JJs can exhibit very high transparency, as demonstrated in recent experiments [44]. This opens new possibilities for exploring the high-transparency regime, in which novel phenomena like zero-energy level crossings, topological superconductivity supporting the formation of Majorana bound states (MBS) [44–46], and SDE sign changes [20, 47] are expected to emerge. Furthermore, a top gate with potential, V_g, located above the N region [see Fig. 7(b)] allows for tuning the junction all the way from the low to the high-transparency limits, and for varying the strength of the SOC [33, 48]. Thus, the top gate can serve as an additional control knob for realizing new functionalities in JJ-based devices such as superconducting diodes and superconducting transistors [49–51] (also see the discussion in Ch. 4). Moreover, under certain conditions, the top gate can be used to self-tune the superconducting phase difference without the need for a magnetic flux or SQUID-type superconducting loop. This could be relevant for improving superconducting qubits' performance by reducing the circuit complexity and decoherence resulting from cross-talk between qubits. Table 1 compares SIS and SNS JJs.

Planar JJs built using semiconductor-superconductor heterostructures have been experimentally realized by using HgTe, InAs, and InSb quantum wells, where hard proximity-induced superconducting gaps, with amplitudes close to the native gap of the proximitizing superconductor, have been observed. These structures have strong SOC and large effective

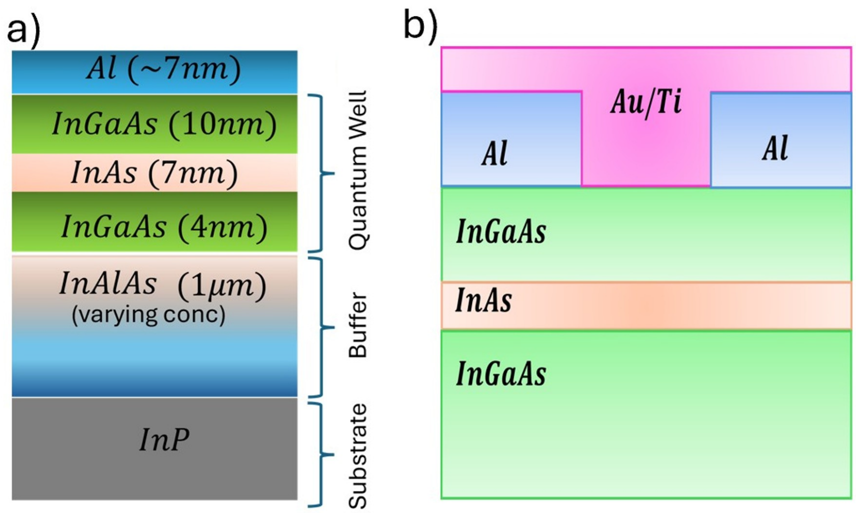

Figure 7: **(a)** Schematic of a heterostructure typically used for building proximitized JJs. **(b)** Schematic of a planar JJ fabricated on an Al-InAs based heterostructure. Light blue layers represent the superconducting Al electrodes. Due to the proximity effect, the two-dimensional electron gas confined to the InAs (light-orange layer) acquires superconducting properties in the regions underneath the Al electrodes. The JJ forms in the InAs layer. The top gate (pink) controls the transparency of the junction by tuning the chemical potential in the normal (not proximitized) region of the InAs layer.

g-factors, allowing for spin manipulation by electrical and/or magnetic fields. The symmetry and properties of the SOC fields depend on the specific quantum well materials and structure. Two types of SOCs are typically present in the above-mentioned structures: i) Rashba SOC, which originates from the structural inversion asymmetry, and ii) Dresselhaus SOC, which originates from the lack of bulk inversion symmetry [33, 39, 52]. The SOC in InAs-based heterostructures is predominantly of Rashba type, leading to sizable anisotropic effects with respect to the magnetic field direction [20, 53]. On the other hand, HgTe and InSb-based heterostructures can exhibit comparable strengths of Rashba and Dresselhaus SOC and are, therefore, good candidates to investigate both magnetic and crystalline anisotropic effects [54].

The planar Josephson junction is fabricated by first depositing an InAlAs buffer layer on top of an InP substrate, followed by the InGaAs–InAs–InGaAs layers. The InAs layer forms a quantum well where the carriers behave as a 2DEG. Subsequently, an aluminum (Al) layer is deposited on top of the heterostructure and selectively etched to form a device geometry similar to that shown in Fig. 7(b). Through the proximity effect, the Al superconducting contacts induce superconductivity in the InAs 2DEG, thus creating a planar JJ [20, 53]. Since both the S and N regions are built on the same high-mobility semiconductor [InAs in the case of Fig. 7(b)], the resulting planar JJ may exhibit high transparency [48, 55]. As discussed earlier, in planar JJs, the ABS governs the transport properties. The ABS spectrum depends sensitively on the junction transparency [see Fig. 5(c)], which can be tuned experimentally by adjusting the top-gate voltage. In addition, the energy of the ABS also depends on the orientation and magnitude of the in-plane magnetic field. Together, these control parameters allow us to manipulate the magnetic, electric, and even thermal response of the planar JJ. This tunability makes planar JJs a versatile platform for exploring non-reciprocal superconducting transport. We discuss the transport properties of the planar JJ in the following chapter.

Feature	**SIS Junction (Tunnel-type)**	**Planar Josephson Junction**
Barrier	Thin insulating layer ($\sim$1–2 nm) separating two superconductors [56]	Normal metal or semiconductor region ($\sim$10–100 nm) between superconducting leads [37, 40]
Coupling mechanism	Quantum tunneling of Cooper pairs through the insulating barrier [56]	Andreev reflection and proximity-induced superconductivity in the normal or semiconducting region [40, 41]
Transparency (τ)	Very low ($\tau \ll 1$), dominated by tunneling probability [56]	Tunable and often high ($\tau \lesssim 1$), depending on interface quality and gate voltage [42]
Control parameters	Fixed during fabrication, limited post-growth tunability [56]	Gate voltage, carrier density, and magnetic field can be used to control transparency [41, 42]
Phase-difference control	The phase difference is tuned by embedding the JJ in a superconducting loop and applying a magnetic flux.	The phase difference can also be tuned without a superconducting loop or magnetic flux by using a gate voltage and/or an in-plane magnetic field.
Current–phase relation	Nearly sinusoidal (weak link limit) [40, 56]	Strongly non-sinusoidal in short junction or high-transparency limit [40, 41]
Applications	Voltage standards, SQUIDs, and conventional Josephson qubits [56]	Gatemons, topological superconductivity, and superconducting diode effect studies [54, 57]

Table 1: Comparison between conventional tunnel-type SIS junctions and planar JJs [37, 40–42, 54, 57, 58].

3 BOGOLIUBOV-DE GENNES FORMALISM

In this chapter, we describe the theoretical framework used to model and analyze the transport properties of planar JJs. The general geometry of the planar JJ and the coordinate system are illustrated in Figs. 8(a) and (b), respectively. We consider a two-dimensional electron gas proximitized by two superconducting leads of width W_S, separated by a normal region of width W_N. The $\hat{x}$ axis is taken along the direction of the current, while $\hat{y}$ denotes the transverse direction within the 2DEG plane.

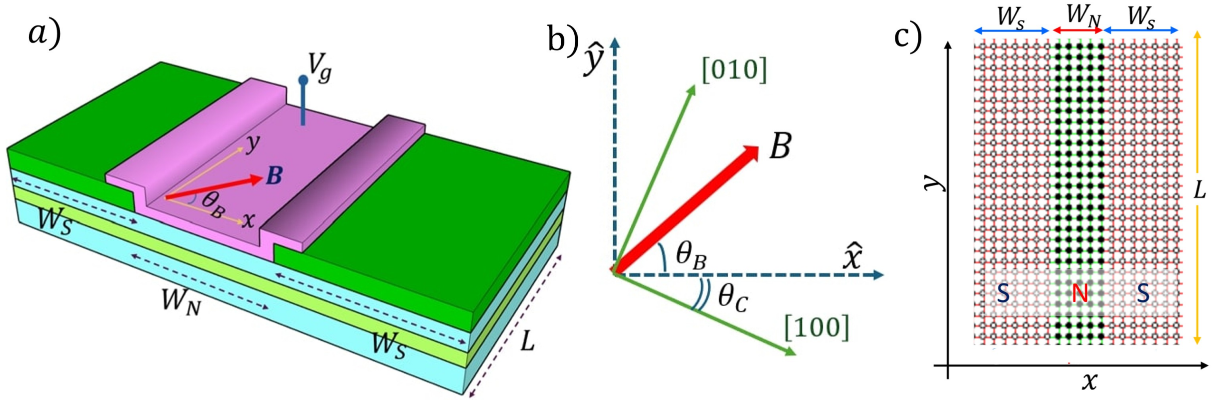

Figure 8: **(a)** Schematic of a gated planar JJ built on a non-centrosymmetric semiconductor 2DEG (light green, middle layer). The uppermost layer, colored green, are the superconducting leads, which induce superconductivity in the 2DEG through the proximity effect. The $\hat{x}$ direction is taken as the current direction and the $\hat{y}$ axis is oriented along the junction. The transparency of the junction can be modulated by applying a gate voltage (V_g) over the normal region. **(b)** θ_B and θ_C represent the angles describing the directions of the magnetic field ($\mathbf{B}$) and the [100] crystallographic direction with respect to the $\hat{x}$ axis, respectively. **(c)** Square-lattice discretization of the planar JJ used in the numerical tight-binding calculations. The dots represent lattice sites.

Two angles, θ_B and θ_C, characterize the orientations relevant to the anisotropic response of the junction [see Fig. 8(b)]. The magnetic-field angle θ_B is measured between the in-plane magnetic field $\mathbf{B}$ and the current direction $\hat{x}$. The crystallographic angle θ_C is measured between the device axis $\hat{x}$ and the semiconductor crystallographic direction

[100]. These angles control the relative orientation between the Zeeman field and the effective SOC fields, thereby modifying the Andreev spectrum and the resulting current–phase relation.

Although the semiconductor heterostructure is three-dimensional, the quantum well hosting the proximitized 2DEG in experimental heterostructures, such as the one shown in Fig. 7, is typically only a few nanometers thick. The electrons are therefore strongly confined along the growth direction. Consequently, the low-energy electronic motion can be described using an effective two-dimensional model in the x–y plane. Figure 8(c) shows the square-lattice discretization used in the numerical model.

To study the physical mechanism and SOC-induced anisotropic effects on nonreciprocal effects in planar JJs, we consider two complementary methods. First, we develop an analytical model based on a short-junction approximation within the Bogoliubov–de Gennes formalism, which allows us to extract simplified analytical expressions for the Andreev spectrum and the critical current in the high-transparency and low-field limits. Second, we implement a fully numerical tight-binding model using the Kwant package [59], enabling simulations of realistic junction geometries that include arbitrary spin–orbit coupling strengths, finite magnetic fields, and disorder. The comparison between these two approaches and the already established experimental results provides both physical insight and validation of the superconducting diode effect studied in later chapters.

Bogoliubov-de Gennes Formalism

The mathematical formulation of the Bogoliubov-de Gennes (BdG) Hamiltonian is central to understanding the interplay between superconductivity, SOC, external electromagnetic fields, and non-reciprocal effects in planar JJs. The BdG formalism has been shown to be adequate for treating non-homogeneous superconducting systems, where the superconducting pairing potential $\Delta(\mathbf{r})$, among other parameters, is position-dependent. Therefore, the BdG formalism, which can also incorporate spin-orbit and Zeeman interactions, is typically considered for microscopic theoretical models of JJs [60].

To include electron and hole degrees of freedom within the BdG formalism, we use the Nambu spinor:

$$\Psi(\mathbf{r}) = \begin{pmatrix} \psi_\uparrow(\mathbf{r}) \\ \psi_\downarrow(\mathbf{r}) \\ -\psi_\downarrow^*(\mathbf{r}) \\ \psi_\uparrow^*(\mathbf{r}) \end{pmatrix}, \tag{3.1}$$

and the BdG Hamiltonian is then expressed in block form as:

$$\mathcal{H}_{\mathrm{BdG}} = \begin{pmatrix} H_e(\mathbf{r}) & \Delta(\mathbf{r}) \\ \Delta^*(\mathbf{r}) & H_h(\mathbf{r}) \end{pmatrix}, \tag{3.2}$$

where $H_e(\mathbf{r})$ is the single-particle Hamiltonian for electrons, including the kinetic energy, SOC, Zeeman field, and electrostatic potential terms. The hole sector is defined through time-reversal symmetry as:

$$H_h(\mathbf{r}) = -\mathcal{T} H_e(\mathbf{r}) \mathcal{T}^{-1}, \tag{3.3}$$

with $\mathcal{T} = i\sigma_y\mathcal{K}$ being the time-reversal operator acting on spin-$\frac{1}{2}$ states. This structure ensures particle-hole symmetry.

The BdG equation is therefore,

$$H_{BdG}\Psi(x,y) = E\Psi(x,y) \ , \tag{3.4}$$

and the BdG Hamiltonian can be rewritten using Pauli matrices as follows,

$$H_{BdG} = \sigma_z \otimes (H_0 + H_{\rm so}) + H_{\rm z} + \Delta(x,\phi)\tau_+ + \Delta^*(x,\phi)\tau_- \ . \tag{3.5}$$

The first term, H_0 is the spin-independent part of the single-particle Hamiltonian,

$$H_0 = \left[\frac{p_x^2 + p_y^2}{2m^*} - \mu(\mathbf{r})\right]\sigma_0 \ , \tag{3.6}$$

with σ_0 denoting the 2×2 identity matrix, (p_x, p_y) are the in-plane momentum components, m^* is the effective electron mass, $\mu(\mathbf{r})$ is the chemical potential.

The second term is the SOC Hamiltonian; the specific form of the SOC depends on the quantum well growth direction, junction crystallographic orientation and structural symmetry. Spin-orbit coupling originates from special relativistic corrections to the electron's motion in an electrostatic potential. In the electron's instantaneous rest frame, the electric field $\mathbf{E} = -\nabla V/e$ transforms into an effective magnetic field, which couples to the electron spin through a Zeeman-like interaction. Equivalently, SOC emerges from the expansion of the Dirac Hamiltonian, yielding (to order $1/c^2$) the SOC term

$$H_{\rm so} = \frac{\hbar}{4m_0^2c^2}\,\boldsymbol{p}\cdot[\boldsymbol{\sigma}\times\nabla V(\mathbf{r})]\,, \tag{3.7}$$

where m_0 is the free-electron mass and $V(\mathbf{r})$ is the electrostatic potential [33].

In semiconductor heterostructures, the effective SOC is determined by the inversion asymmetries of both the underlying crystal and the heterostructure. For zinc-blende quantum wells grown along the $[001]$ crystallographic direction, the two relevant contributions are Rashba and Dresselhaus SOCs, which arise due to the lack of inversion symmetry [33, 38, 54].

In the heterostructure shown in Fig. 3(a), the layers above and below the InAs quantum well are not equivalent, resulting in an asymmetric confining potential $V(z)$ along the growth direction. For a quantum well grown along the $\mathbf{z} \parallel [001]$ direction, this asymmetry breaks inversion symmetry with respect to the xy plane at the 2DEG layer. Such structural inversion asymmetry (SIA) may arise from gating, asymmetric barriers, or band bending and generates Rashba SOC in the semiconductor heterostructures [33].

The effective Rashba SOC Hamiltonian is

$$H_R = \frac{\alpha}{\hbar}\left(p_y \sigma_x - p_x \sigma_y\right), \tag{3.8}$$

where α is the Rashba SOC strength, tunable via gate voltages or carrier density. The Rashba SOC is invariant under rotations in the plane of the 2DEG.

In contrast, Dresselhaus SOC arises from the bulk inversion asymmetry (BIA) of the zinc-blende crystal. In the bulk, the Dresselhaus SOC is cubic in momentum and, unlike the Rashba SOC, is not invariant under in-plane rotations. In the crystallographic reference frame defined by $\hat{\mathbf{X}} \parallel [100]$, $\hat{\mathbf{Y}} \parallel [010]$, and $\hat{\mathbf{Z}} \parallel [001]$, the bulk Dresselhaus SOC

Hamiltonian is

$$H_{\rm D}^{\rm bulk} = \gamma\big[k_x\big(k_y^2 - k_z^2\big)\,\sigma_x + k_y\big(k_z^2 - k_x^2\big)\,\sigma_y + k_z\big(k_x^2 - k_y^2\big)\,\sigma_z\big]\,, \tag{3.9}$$

where γ is a material parameter.

For a quantum well grown along the $[001]$ crystallographic direction, confinement along the growth direction produces an effective two-dimensional Dresselhaus SOC Hamiltonian. After averaging over the confined motion, the leading term, which is linear in the in-plane momentum, becomes

$$H_{\rm D} = \frac{\beta}{\hbar}\,\big(p_x\sigma_x - p_y\sigma_y\big), \tag{3.10}$$

where β is the effective Dresselhaus SOC strength. Since the presence of the Dresselhaus SOC breaks the invariance under in-plane rotations, it constitutes a source of crystalline anisotropic effects. To study these effects we conveniently set the x-axis along the supercurrent direction at an angle θ_C with respect to the $[100]$ crystallographic direction of the quantum well semiconductor [see Fig. 8(b)]. In this θ_C-rotated frame, the linearized Dresselhaus SOC becomes,

$$H_{\rm D} = \frac{\beta}{\hbar}\big[\big(p_x\sigma_x - p_y\sigma_y\big)\cos 2\theta_c - \big(p_x\sigma_y + p_y\sigma_x\big)\sin 2\theta_c\big]\,, \tag{3.11}$$

which is the form used throughout this work. When both SIA and BIA are present, the effective SOC Hamiltonian reads

$$H_{\text{so}} = H_{\text{R}} + H_{\text{D}} \,. \tag{3.12}$$

The third term in Eq. (3.5) is the Zeeman interaction due to the in-plane magnetic field, $\mathbf{B} = (|\mathbf{B}|\cos\theta_B, |\mathbf{B}|\sin\theta_B, 0)$,

$$H_{\text{z}} = -\frac{g^*\mu_B}{2}\mathbf{B}\cdot(\sigma_0 \otimes \boldsymbol{\sigma}) \,, \tag{3.13}$$

with μ_B representing the Bohr magneton and g^* as the effective g-factor.

The last two terms in Eq. (3.5) describe the superconducting pairing. In this work we model the proximity effect by considering a pair potential with the spatial dependence

$$\Delta(x,\phi) = \Delta\, e^{-i\,\text{sgn}(x)\phi/2}\,\Theta(|x| - W_N/2), \tag{3.14}$$

where Δ is the proximity-induced superconducting gap extracted from experimental measurements, ϕ is the superconducting phase difference, and $\Theta(x)$ is the Heaviside step function. This form assumes a finite and spatially uniform induced superconducting gap in the proximitized regions of the quantum well beneath the top Al leads, while the gap is assumed to vanish in the normal (non-proximitized) region [see Fig. 8(a)].

An accurate theoretical determination of the proximity-induced superconducting gap (Δ) requires a self-consistent treatment of the proximity effect, in which the superconducting gap and the BdG equations are solved simultaneously. Such calculations are computationally demanding and are beyond the scope of the present study. Instead, we adopt phenomenological values of the proximity-induced gap estimated from previous experiments [53, 55, 58].

The matrices τ_i appearing in the BdG Hamiltonian are the Nambu matrices acting in particle–hole space. They are defined as

$$\tau_j = \sigma_j \otimes \sigma_0 \;, \qquad \tau_\pm = \frac{1}{2}\left(\tau_x \pm i\tau_y\right) , \tag{3.15}$$

where σ_j denote the Pauli matrices with $j = x, y, z$.

3.1 Transport Properties

As discussed in Sec. 1.3, the supercurrent flowing through the junction is determined by the phase dependence of the free energy, which, in turn, is governed by the BdG quasiparticle spectrum. The free energy is given by,

$$F = -\sum_{E_n(\phi)>0} \left[E_n(\phi) + 2k_BT \ln\left(1 + e^{-E_n(\phi)/k_BT}\right)\right] + F_0(\mathbf{B}, T) \;, \tag{3.16}$$

where $F_0(\mathbf{B}, T)$ is a phase-independent contribution, $E_n(\phi)$ denotes the energy spectrum of the BdG Hamiltonian of the junction, and T is the temperature. In short ballistic junctions, the dominant phase dependence of the free energy arises from the discrete Andreev bound states.

One can then use Eq. (1.16) to evaluate the supercurrent after computing the free energy from the BdG energy spectrum. Combining Eqs. (1.16)and (3.16) yields,

$$I(\phi, T) = -\frac{e}{\hbar} \sum_{E_n>0} \tanh\!\left(\frac{E_n(\phi)}{2k_BT}\right) \frac{\partial E_n(\phi)}{\partial \phi}. \tag{3.17}$$

In the zero-temperature limit, the hyperbolic tangent approaches unity and the supercurrent expression reduces to,

$$I(\phi,0) = -\frac{e}{\hbar}\sum_{E_n>0}\frac{\partial E_n(\phi)}{\partial\phi}. \tag{3.18}$$

In the absence of SOC and magnetic field, the Andreev spectrum of a short, single-channel JJ with transparency τ can be approximated by Eq. (1.11), leading to the well-known current–phase relation (CPR),

$$I(\phi,T) = \frac{e\Delta}{2\hbar}\frac{\tau\sin\phi}{\sqrt{1-\tau\sin^2(\phi/2)}}\tanh\!\left(\frac{\Delta\sqrt{1-\tau\sin^2(\phi/2)}}{2k_BT}\right). \tag{3.19}$$

In multichannel junctions, the total supercurrent is obtained by summing the contributions of all transport channels, each characterized by a transparency τ_n, i.e.,

$$I(\phi,T) = \sum_n \frac{e\Delta}{2\hbar}\frac{\tau_n\sin\phi}{\sqrt{1-\tau_n\sin^2(\phi/2)}}\tanh\!\left(\frac{\Delta\sqrt{1-\tau_n\sin^2(\phi/2)}}{2k_BT}\right). \tag{3.20}$$

Hence, in the zero-temperature limit, the CPR becomes

$$I(\phi,0) = \sum_n \frac{e\Delta}{2\hbar}\frac{\tau_n\sin\phi}{\sqrt{1-\tau_n\sin^2(\phi/2)}}. \tag{3.21}$$

The forward and reverse critical currents are defined as

$$I_c^+ = \max_\phi I(\phi), \qquad I_c^- = \left|\min_\phi I(\phi)\right|. \tag{3.22}$$

In systems with both spatial inversion and time-reversal symmetries, $I(\phi)$ is antisymmetric,

$I(-\phi) = -I(\phi)$, leading to $I_c^+ = I_c^-$. However, the simultaneous presence of SOC and the Zeeman interaction breaks these symmetries, resulting in different amplitudes for the forward and reverse supercurrents, i.e., $I_c^+ \neq I_c^-$. This asymmetry between the forward and reverse critical currents is the SDE investigated in the following chapter.

3.1.1 Analytical Model

The complexity of the BdG equations for the system under consideration generally precludes an analytical treatment. Nevertheless, analytical solutions can be obtained in certain limiting cases. Here, we consider the Dirac-delta model, in which the junction width is assumed to be much smaller than the superconducting coherence length, ξ. The problem is further simplified in the limit $W_S \to \infty$, which is a good approximation for junctions with $W_S \gg \xi$. Within these approximations ($W_N \ll \xi \ll W_S$), the BdG Hamiltonian reduces to,

$$\begin{aligned} H_{BdG}(\mathbf{B},\phi) &= \sigma_z \otimes \Big[\left(\frac{\mathbf{p}^2}{2m^*} - \mu_S + V_0 W_N \delta(x) \right) \sigma_0 + \frac{\alpha}{\hbar}(p_y\sigma_x - p_x\sigma_y) \Big] \\ &- E_z\delta(x)W_N(\sigma_0 \otimes \sigma_y) + \Delta(x,\phi)\tau_+ + \Delta^*(x,\phi)\tau_- \ , \end{aligned} \tag{3.23}$$

where $E_z = g^*\mu_B|B|/2$, is the Zeeman energy and $V_0 = \mu_S - \mu_N$.

For simplicity, we assume that the magnetic field is aligned with the junction direction. This choice is particularly convenient because spin is conserved at $k_y = 0$. As a result, the BdG equations decouple into independent sectors for each spin component, greatly simplifying the analysis.

Taking into account the translational invariance along the y direction and applying appropriate boundary conditions, we find analytical solutions for the eigenfunctions and eigenenergies. The obtained expressions are lengthy and we omit them here. However, a simplified expression for the energy spectrum at $k_y = 0$ can be obtained within the Andreev approximation ($\Delta \ll \mu_s$) [39],

$$E_{\lambda,\sigma}(k_y = 0, \phi) = \frac{\mathrm{sgn}[f_\sigma^\lambda(Z_0, Zy, \phi)]\Delta}{\sqrt{1 + [f_\sigma^\lambda(Z_0, Zy, \phi)]^2}} \tag{3.24}$$

where,

$$f_\sigma^\lambda(Z_0, Z_y, \phi) = \frac{2\lambda Z_y + \sigma\sqrt{2[1 + Z_0^2 + Z_y^2 + (Z_y^2 - Z_0^2)\cos\phi - \cos 2\phi]}}{2 + Z_0^2 - Z_y^2 + 2\cos\phi}, \tag{3.25}$$

$k_F = \sqrt{2m^*\mu_S/\hbar^2}$ is the Fermi wavevector in the S region, and $\lambda, \sigma = \pm 1$.

In the equations above, we have introduced the following dimensionless parameters:

$$\begin{aligned} Z_y &= \frac{2m^* E_Z W_N}{\hbar^2 k_F}, \\ Z_0 &= \frac{2m^* V_0 W_N}{\hbar^2 k_F}. \end{aligned} \tag{3.26}$$

In the presence of translational invariance along the junction, the y-component of the momentum, $p_y = \hbar k_y$ is a good quantum number and the free energy in the zero-temperature limit can be written as,

$$\begin{aligned} F &= \sum_{\lambda, k_y, \sigma = \pm} E_{\lambda\sigma}^<(k_y) \approx \frac{L}{2\pi} \sum_{\lambda,\sigma=\pm} \int_{-k_{F+}}^{k_{F+}} E_{\lambda\sigma}^<(k_y) dk_y \\ &= \frac{L}{\pi} \sum_\lambda \left\{ \int_0^{k_{F-}} \left[E_{\lambda+}^<(k_y) + E_{\lambda-}^<(k_y) \right] dk_y \right. \\ &+ \left. \int_{k_{F-}}^{k_{F+}} E_{\lambda+}^<(k_y) dk_y \right\}, \end{aligned} \tag{3.27}$$

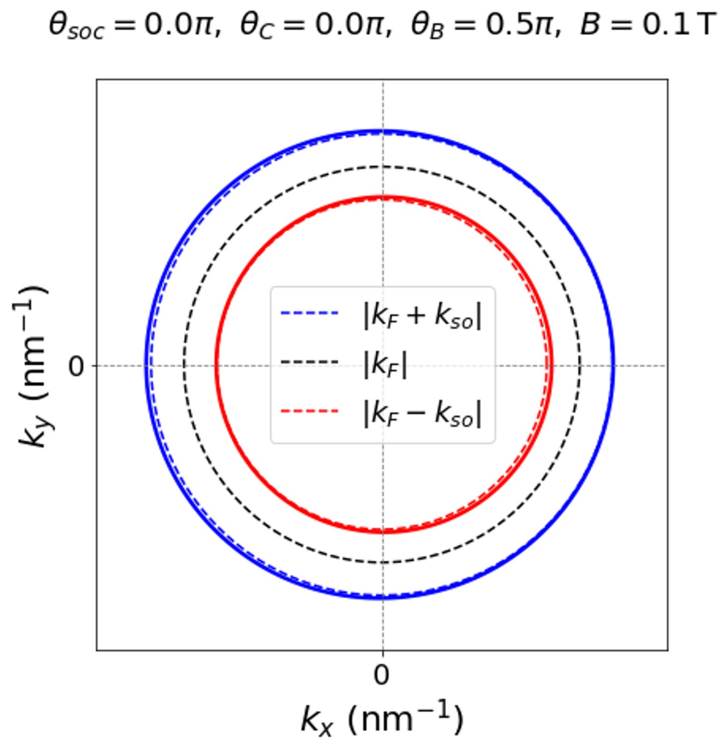

Figure 9: Comparison between the exact and approximate spin-split Fermi contours of a 2DEG with Rashba SOC and an in-plane magnetic field applied along the y direction. The solid red and blue curves represent the exact Fermi contours obtained from the single-particle Hamiltonian. The dashed red and blue circles represent the weak-SOC, low-field approximation in which the contours are assigned the radii $k_{F-} = k_F - k_{\rm so}$ and $k_{F+} = k_F + k_{\rm so}$, respectively. The black dashed circle represents the spin-degenerate Fermi contour of radius k_F in the absence of SOC and magnetic field.

where L is the length of the junction and the upper integration limit has been approximated by the largest (k_{F+}) of the two Fermi wavevectors of the single-particle Rashba bands,

$$k_{F\pm} = k_F \pm k_{so}\ . \tag{3.28}$$

where $k_{so} = m^*\alpha/\hbar^2$. Figure 9 compares the exact spin-split Fermi contours with the circular approximation used in Eq. (3.28). In the weak-SOC and low-field regime, the field-induced deformation of the contours is small, and the exact contours are closely approximated by circles with radii $k_{F\pm} = k_F \pm k_{\rm so}$. This approximation allows the intersections of the two spin-split contours with the k_y axis to be used as the integration limits in Eq. (3.27). To evaluate the integrals in Eq. (3.27) analytically, the energy branches are expanded in a Taylor series around $k_y = 0$, yielding

$$\begin{aligned} F &\approx \frac{L}{\pi} \sum_{\lambda} \sum_{n=0}^{\infty} \left\{ \frac{1}{(n+1)!} \left[\left. \frac{\partial^n E^{<}_{\lambda+}}{\partial k_y^n} \right|_{k_y=0} (k_{F+})^{n+1} \right.\right. \\ &+ \left.\left. \left. \frac{\partial^n E^{<}_{\lambda-}}{\partial k_y^n} \right|_{k_y=0} (k_{F-})^{n+1} \right] \right\} . \end{aligned} \tag{3.29}$$

We consider the case of weak SOC ($k_{so} \ll k_F$). Since the $k_{F\pm}$ already account for the Rashba SOC strength, the free energy up to first order in SOC can be approximated as [61],

$$\begin{aligned} F &\approx \frac{L}{\pi} \sum_{\lambda} \left[E^{<}_{\lambda+}(0)\; k_{F+} + E^{<}_{\lambda-}(0)\; k_{F-} \right] \\ &\approx \frac{L}{\pi} \sum_{\lambda} \left[E^{<}_{\lambda+}(0)\; (k_F + k_{so}) + E^{<}_{\lambda-}(0)\; (k_F - k_{so}) \right] . \end{aligned} \tag{3.30}$$

3.1.2 Numerical Model

The BdG eigenvalue problem in Eq. (3.4) corresponds to a set of four coupled, second-order differential equations. In general, solving this problem for realistic junction geometries requires a numerical approach. We therefore discretize the continuum model on a square lattice with lattice spacing a [see Fig. 8(c)], and approximate first and second derivatives using finite differences,

$$\frac{\partial\psi(x,y)}{\partial x} \approx \frac{\psi(x+a,y)-\psi(x-a,y)}{2a}, \tag{3.31}$$

$$\frac{\partial^2\psi(x,y)}{\partial x^2} \approx \frac{\psi(x+a,y)+\psi(x-a,y)-2\psi(x,y)}{a^2}, \tag{3.32}$$

and similarly for the derivatives with respect to y. This allows for transforming the system of coupled differential equations into a system of algebraic equations.

Introducing the discretized position basis $|x_i, y_j, \sigma\rangle = |ia, ja, \sigma\rangle$ (with integers i, j labeling lattice sites and σ the spin index), the BdG Hamiltonian can be written in tight-binding form [57]

$$\begin{aligned}
H_{\mathrm{TB}} &= \hat{H}^{\mathrm{onsite}} + \left(\hat{V}^{\mathrm{up}} + \hat{V}^{\mathrm{right}} + \mathrm{H.c.}\right) , \\
\hat{H}^{\mathrm{onsite}} &= \sum_{i,j} h^{\mathrm{onsite}}(x_i) \, |x_i, y_j\rangle \langle x_i, y_j| \ , \\
\hat{V}^{\mathrm{up}} &= \sum_{i,j} v^{\mathrm{up}}(x_i) \, |x_i, y_j + a\rangle \langle x_i, y_j| \ , \\
\hat{V}^{\mathrm{right}} &= \sum_{i,j} v^{\mathrm{right}}(x_i) \, |x_i + a, y_j\rangle \langle x_i, y_j| \ .
\end{aligned} \tag{3.33}$$

In our geometry, the Hamiltonian parameters do not depend explicitly on y, so the onsite term h^{onsite} and the hopping matrices v^{up} and v^{right} depend only on x_i. They are given by

$$h^{\mathrm{onsite}}(x_i) = (4t-\mu)\,\tau_z\sigma_0 + \mathbf{B}\cdot\boldsymbol{\sigma} + \Delta(x_i)\tau_+ + \Delta^*(x_i)\tau_- \ , \tag{3.34}$$

$$v^{\mathrm{right}}(x_i) = -t\,\tau_z\sigma_0 + \frac{i\alpha}{2a}\tau_z\sigma_y - \frac{i\beta}{2a}\left(\cos 2\theta_c\,\tau_z\sigma_x - \sin 2\theta_c\,\tau_z\sigma_y\right) \ , \tag{3.35}$$

$$v^{\mathrm{up}}(x_i) = -t\,\tau_z\sigma_0 - \frac{i\alpha}{2a}\tau_z\sigma_x + \frac{i\beta}{2a}\left(\cos 2\theta_c\,\tau_z\sigma_y + \sin 2\theta_c\,\tau_z\sigma_x\right) \ , \tag{3.36}$$

where,

$$t = \frac{\hbar^2}{2m^* a^2} \tag{3.37}$$

is the nearest-neighbor hopping parameter.

Because each lattice site contains four BdG degrees of freedom, a lattice with N sites is represented by a $4N \times 4N$ tight-binding matrix Hamiltonian. We construct this matrix using the open-source Python package Kwant [59] and obtain its eigenenergies by numerical diagonalization using SciPy routines.

Table 2: Representative material parameters used in the numerical calculations. The Al/InAs and Al/InSb parameters follow Ref. [62], while the HgTe parameters follow Ref. [61].

Parameter	Al/InAs	Al/InSb	Al/HgTe
Effective mass (m^*/m_e)	0.033	0.013	0.038
Landé factor (g^*)	-10	-20	-10
Induced superconducting gap (Δ)	0.25 meV	0.21 meV	0.25 meV
SOC strength (λ)	16 meV nm	16 meV nm	16 meV nm
Chemical potential in S (μ_S)	2 meV	2 meV	1 meV
Chemical potential in N (μ_N)	2 meV	2 meV	1 meV
Junction width (W_N)	100 nm	100 nm	96 nm
Superconducting lead width (W_S)	500 nm	500 nm	240 nm
TB lattice constant (a)	10 nm	10 nm	8 nm
TB hopping parameter (t)	11.5 meV	29.3 meV	15.8 meV

The numerical calculations in this thesis use parameter sets corresponding to Al/InAs, Al/InSb, and Al/HgTe-based planar JJs, as summarized in Table 2. The Al/InAs and Al/InSb parameters are used in the calculations of the anisotropic SDE [see Chapter 6], while the HgTe parameters are used in the magneto-CPR calculations [see Chapter 5] [61, 62].

For the anisotropic SDE calculations in Al/InAs and Al/InSb junctions, the lattice spacing and junction dimensions are chosen as $a = 10$ nm, $W_N = 100$ nm, and $W_S = 500$ nm, with $\mu_S = 2$ meV, unless otherwise specified [62]. For the magneto-CPR calculations based on an HgTe 2DEG, we use $a = 8$ nm, $W_N = 96$ nm, and $\mu_S = 1$ meV [61].

In experiments, the chemical potential of the 2DEG can be on the order of $\mu \sim 50$–60 meV [20, 48, 55]. However, the tight-binding discretization accurately reproduces the continuum Hamiltonian only in the long-wavelength limit, where the relevant states remain close to the bottom of the tight-binding band. This condition requires $t \gg \mu$, or equivalently $k_F a \ll 1$. Using the experimentally inferred chemical potential would therefore require a substantially smaller lattice spacing, increasing the number of lattice sites needed to represent realistic junction dimensions and making the calculations computationally demanding. We consequently use reduced chemical potentials, together with lattice spacings of $a = 8$–10 nm, to keep the calculations within the long-wavelength regime while maintaining computationally tractable system sizes. This approximation is expected to yield accurate qualitative results as long as the chemical potential used is much larger than all other relevant energy scales in the system (superconducting gap, SOC splitting at E_F, and Zeeman energy), which is already satisfied for $\mu_S \sim 1$ meV. All numerical calculations are performed in the zero-temperature limit.

4 PHENOMENOLOGY OF SUPERCONDUCTING DIODE EFFECT IN PLANAR JOSEPHSON JUNCTIONS

This chapter is based in part on Ref. [62].

Nonreciprocal transport phenomena in superconductors have become a central topic in condensed matter physics due to their intimate connection with broken inversion and time-reversal symmetries and their potential applications in superconducting electronics. In particular, the SDE, characterized by different critical currents depending on the direction of flow, has been investigated in a wide variety of material platforms and heterostructures [16, 18, 20, 47, 63–92]. These diverse realizations highlight the richness of microscopic mechanisms leading to supercurrent rectification [15, 93, 94].

We focus on semiconductor-based planar JJs, where the SDE, also often referred to in the literature as the Josephson diode effect, has been robustly demonstrated experimentally [18, 20, 47, 69, 70, 83, 95]. A commonly invoked mechanism in these heterostructures is the formation of Cooper pairs with finite center-of-mass momentum under broken inversion and time-reversal symmetries, typically induced by Zeeman fields or exchange splitting [15, 18–20, 47, 69, 77, 83, 96–104]. However, the precise role of spin–orbit coupling (SOC) remains an open and actively debated question. While several theoretical works emphasize SOC-driven nonreciprocity [17–20, 47, 61, 69, 95, 97–100, 105], alternative approaches attribute the effect to mechanisms such as Doppler-shifted quasiparticle spectra, orbital contributions, edge reconstruction, diamagnetic effects induced by stray fields, among others [15, 77, 94, 102–104, 106]. Nevertheless, a wide range of experimental and theoretical studies across different JJ platforms [17–20, 47, 51, 61, 63, 64, 69–71, 80, 83, 87, 91, 95, 99, 102, 105, 107, 108] consistently highlight the crucial role of magnetic fields and spin–orbit textures in determining both the magnitude and directionality of the diode response.

This chapter examines the general phenomenology and microscopic mechanisms of the SDE in semiconductor-based planar JJs (such as Al/InAs and Al/InSb). We first define the nonreciprocal critical currents and the diode quality factor. We then discuss the general symmetry requirements for the SDE and explain how the combined action of Rashba SOC coupling and a Zeeman field can generate finite-momentum Cooper pairing. A phenomenological current–phase relation is introduced to identify the conditions required for unequal forward and reverse critical currents. Finally, the short-junction analytical model is used to determine how SOC and junction transparency influence the magnitude and magnetic-field dependence of the diode response. The detailed magnetic and crystalline anisotropies resulting from different field and junction orientations are examined separately in Chapter 6.

4.1 Diode Effect and Nonreciprocal Critical Currents

Having introduced the importance of nonreciprocal superconducting transport in planar JJs, we now define the diode response more explicitly. In a conventional semiconductor diode, such as a p-n junction, current flows preferentially in one direction while being suppressed in the opposite direction. This rectification effect is illustrated schematically in Fig. 10(a). In a JJ, an analogous nonreciprocal response can occur without dissipation: the junction can support different maximum supercurrents depending on the direction of current flow. The corresponding behavior is shown in Fig. 10(b). The forward and reverse critical currents, denoted by I_c^+ and I_c^-, mark the currents at which the junction switches

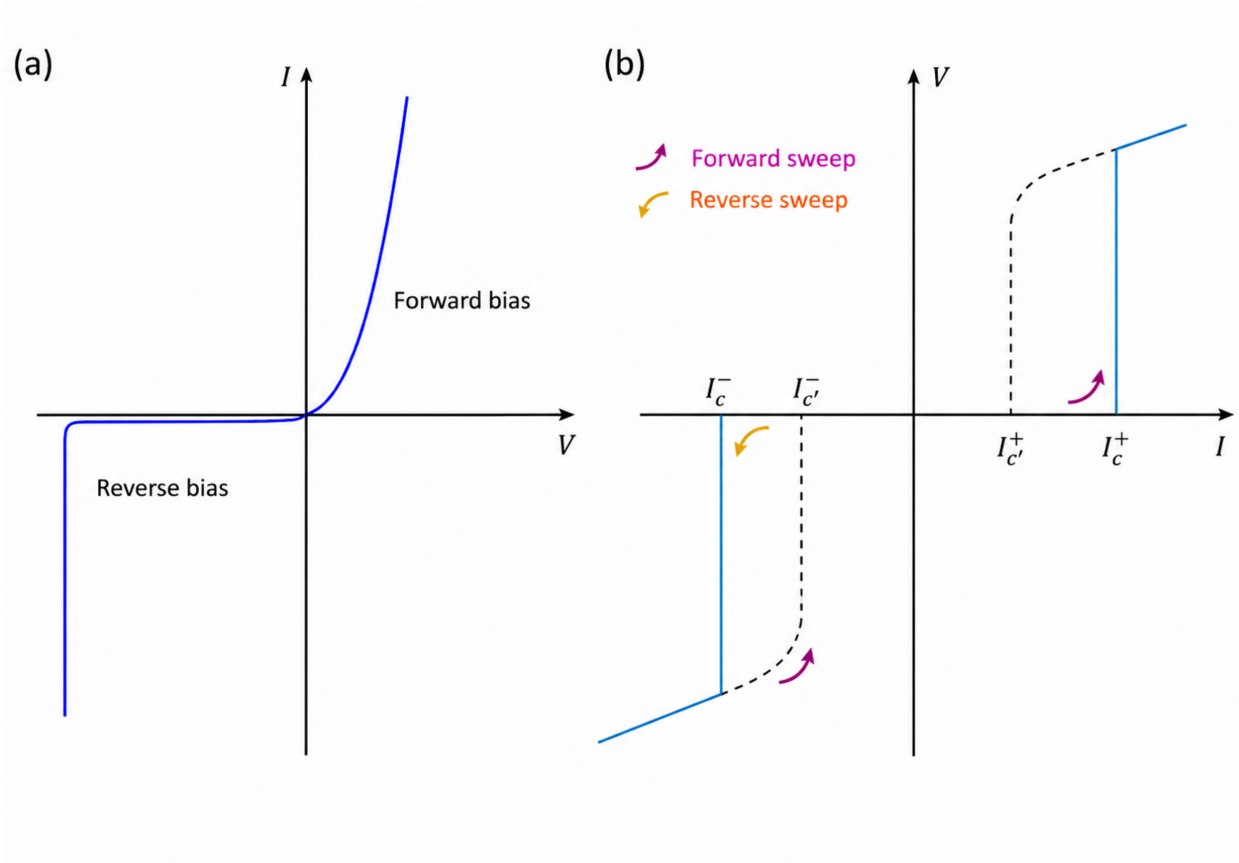

Figure 10: (a) I-V characteristics of a semiconductor p-n junction diode. The current flow is permitted in the forward direction, but in the reverse bias, the current is mostly suppressed [109]. (b) I-V characteristics of a JJ diode [110]. Two critical currents, denoted as I_c^+ (in the forward direction) and I_c^- (in the reverse direction), mark the transition of the superconductor into a normal metallic state. The SDE occurs when the forward and reverse critical currents are different, i.e., $I_c^+ \neq |I_c^-|$. If a current with an amplitude between $|I_c^-|$ and I_c^+ is applied, the current flows without resistance in one direction but becomes resistive in the other. Due to its finite capacitance, the JJ may exhibit a hysteresis behavior. In this case, the two return critical currents $I_{c'}^+$ and $I_{c'}^-$ can also have a non-reciprocal character with $I_{c'}^+ \neq |I_{c'}^-|$.

from the superconducting state to a resistive state for positive and negative current bias, respectively. The SDE occurs when these critical currents are unequal in magnitude,

$$I_c^+ \neq |I_c^-| \, . \tag{4.1}$$

In this regime, a bias current with magnitude between $|I_c^-|$ and I_c^+ can drive the junction normal in one direction while the junction remains superconducting in the opposite direction. Thus, the system behaves as a dissipationless superconducting diode over a finite current window.

To quantify the strength of the phase-unbiased SDE, we define the diode efficiency,

$$\eta(\mathbf{B}) = \frac{|I_c^+(\mathbf{B})| - |I_c^-(\mathbf{B})|}{I_0}, \tag{4.2}$$

where I_0 is the critical current at zero magnetic field. A finite value of η indicates unequal forward and reverse critical-current magnitudes. The sign of η determines the polarity of the diode response, while $\eta = 0$ corresponds to reciprocal critical currents. It is worth noting that an alternative definition of the diode efficiency, in which the denominator is given by the sum of the magnitudes of the forward and reverse critical currents, can also be adopted. However, the definition in Eq. (4.2) is more convenient for studying anisotropic effects, as it incorporates the critical-current anisotropies solely in the numerator.

Due to the finite capacitance of realistic JJs, the current-voltage characteristics may also exhibit hysteresis. In such cases, the return critical currents, denoted by $I_{c'}^+$ and $I_{c'}^-$, may also be nonreciprocal. However, throughout this chapter we focus primarily on the nonreciprocity of the critical currents and characterize the diode response using the difference between I_c^+ and $|I_c^-|$.

Diode effects are governed by the symmetry properties of the system. In ordinary semiconductor diodes, rectification originates from broken inversion symmetry at the p-n junction. In planar Josephson junctions, a finite SDE generally requires the simultaneous breaking of inversion and time-reversal symmetries. In semiconductor-superconductor planar junctions, inversion symmetry can be broken by structural inversion asymmetry, giving rise to Rashba spin-orbit coupling, or by bulk inversion asymmetry, giving rise to Dresselhaus spin-orbit coupling. Time-reversal symmetry is broken by an applied magnetic field or, more generally, by magnetic proximity effects. The combination of these ingredients produces an asymmetric Andreev spectrum and a nonreciprocal current-phase relation, ultimately leading to different forward and reverse critical currents.

The remainder of this chapter analyzes how these symmetry-breaking mechanisms appear in planar JJs. We first identify the symmetry conditions under which the SDE is suppressed, and then connect these conditions to the formation of finite-momentum Cooper pairs and to the anisotropy of the diode response.

4.2 Underlying Mechanisms

In this section, we discuss the considered mechanism responsible for the presence of SDE. We start by discussing the symmetry conditions of planar JJs, and general conditions at which the SDE is suppressed. Further, we also discuss the finite center-of-mass Cooper pair momentum.

4.2.1 Symmetry Analysis

To better understand the relation between forward and reverse currents in planar JJs, we perform a symmetry analysis based on Fig. 11. We consider junction with SOC and in-plane magnetic field $\mathbf{B} = (B_x, B_y, 0)$ and two different reference frames, $\mathbb{K}$ and $\mathbb{K}'$. Even in the absence of the magnetic field, the spin degeneracy is broken by the Rashba SOC and the spin orientation (blue and red arrows) at the single-particle Fermi contours (circles) depends on the wave-vector direction, as illustrated in Fig. 11. Note that Rashba SOC is invariant under rotations about the z-axis. Hence, the free energy measured by observers in the $\mathbb{K}$ and $\mathbb{K}'$ reference frames are given by $F(\mathbf{B}, \phi)$ and $F(-\mathbf{B}, -\phi)$, respectively. Since the free energy should not depend on the choice of reference frame, we conclude that,

$$F(\mathbf{B}, \phi) = F(-\mathbf{B}, -\phi) \ . \tag{4.3}$$

Furthermore, since the supercurrent is driven by the gradient of the phase, the current direction is determined by the phase difference across the junction and the forward and reverse supercurrents are determined by,

$$I^{\pm} = I(\mathbf{B}, \pm\phi) \ . \tag{4.4}$$

Non-reciprocity occurs when $I^+ \neq -I^-$, i.e.,

$$I(\mathbf{B}, \phi) \neq -I(\mathbf{B}, -\phi) \ . \tag{4.5}$$

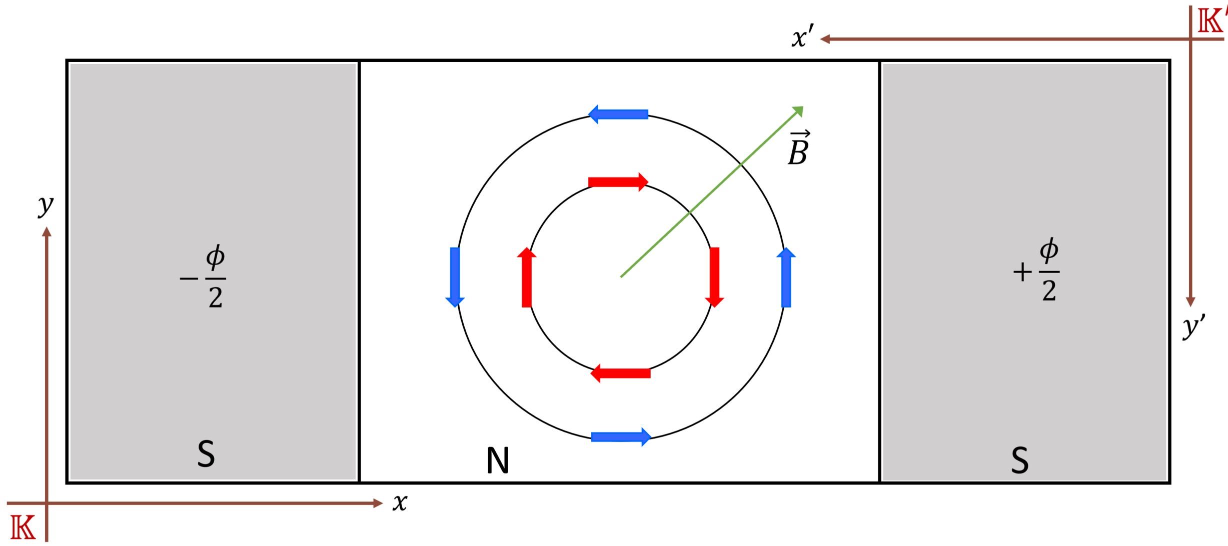

Figure 11: JJ with Rashba SOC in a constant in-plane magnetic field, $\mathbf{B}$. The spin orientation (blue and red arrows) induced by the Rashba SOC at the single-particle Fermi contours (circles) depends on the wave-vector direction. Since the Rashba SOC is invariant under rotations about the z-axis, the free energies measured by observers in the $\mathbb{K}$ and $\mathbb{K}'$ reference frames are $F = F(\mathbf{B}, \phi)$ and $F' = F(-\mathbf{B}, -\phi)$, respectively.

Let's now determine the implications of the non-reciprocal current-phase relation in Eq. (4.5) for the critical currents. Taking the phase derivative on both sides of Eq. (4.3) yields,

$$I(\mathbf{B}, \phi) = -I(-\mathbf{B}, -\phi) \ , \tag{4.6}$$

and, therefore,

$$I_c^+(\mathbf{B}) = \max_{\phi} I(\mathbf{B}, \phi) = \max_{\phi} \left[-I(-\mathbf{B}, -\phi)\right] = -\min_{-\phi} I(-\mathbf{B}, -\phi) = -I_c^-(-\mathbf{B}) \ . \tag{4.7}$$

Combining Eqs. (4.5) and (4.6) we obtain,

$$I_c^+(\mathbf{B}) = \max_{\phi} I(\mathbf{B}, \phi) \neq \max_{\phi} I(-\mathbf{B}, \phi) = I_c^+(-\mathbf{B}) \ . \tag{4.8}$$

Therefore, the non-reciprocity of the CPR leads to the inequality relation,

$$I_c^+(\mathbf{B}) \neq I_c^+(-\mathbf{B}) \ , \tag{4.9}$$

which, considering Eq. (4.7) can be rewritten as,

$$I_c^+(\mathbf{B}) \neq -I_c^-(\mathbf{B}) \ . \tag{4.10}$$

Since the current is proportional to the derivative of the free energy with respect to ϕ, the SDE condition [Eq (4.5)] in terms of the free energy reads,

$$F(\mathbf{B}, \phi) \neq F(\mathbf{B}, -\phi) \ , \tag{4.11}$$

which in the zero temperature limit reduced to,

$$E(\mathbf{B}, \phi) \neq E(\mathbf{B}, -\phi) \ . \tag{4.12}$$

We now analyze the requirements for realizing Eq. (4.12) [or equivalently, Eq. (4.5)] in the considered JJs. Let $H(\mathbf{B}, \phi)$ be the system Hamiltonian and $|\Psi\rangle$ and $|\tilde{\Psi}\rangle$ be eigenstates of the following eigenvalue problems,

$$\begin{aligned} H(\mathbf{B}, \phi)\,|\Psi\rangle &= E(\mathbf{B}, \phi)\,|\Psi\rangle \ , & (4.13) \\ H(\mathbf{B}, -\phi)|\tilde{\Psi}\rangle &= E(\mathbf{B}, -\phi)|\tilde{\Psi}\rangle \ , & (4.14) \end{aligned}$$

connected by the unitary transformation $|\psi\rangle = \mathcal{U}|\tilde{\psi}\rangle$. If the unitary transformation is such that,

$$\mathcal{U}^{\dagger} H(\mathbf{B}, \phi)\mathcal{U} = H(\mathbf{B}, -\phi) \ , \tag{4.15}$$

it follows from Eqs. (4.13)-(4.15) that,

$$E(\mathbf{B}, \phi) = E(\mathbf{B}, -\phi) \ . \tag{4.16}$$

Therefore, no SDE occurs as long as Eq. (4.15) holds. We then conclude that the SDE sufficient condition in Eq. (4.5) is realized when no unitary transformation obeying Eq. (4.15) exists. This observation can be particularly useful to determine whether the symmetry of the system Hamiltonian can support the existence of non-reciprocal effects.

Equation (4.15) provides a general symmetry criterion for the suppression of the SDE. The specific spatial and spin transformations that satisfy this condition depend on the Rashba and Dresselhaus SOC strengths and on the orientations of the magnetic field and junction. These orientation-dependent symmetry conditions that directly lead to the suppression of the SDE are discussed in more detail in Chapter 6.

4.2.2 Role of Cooper-Pair Momentum

A microscopic ingredient associated with the SDE in planar JJs is the formation of Cooper pairs with finite center-of-mass momentum (q) along the supercurrent direction [15, 20]. In the absence of spin splitting, electronic states at opposite momenta form Cooper pairs with zero total momentum. A magnetic field modifies the spin-dependent Fermi contours and can produce a mismatch between the momenta of the states participating in Cooper pairing. The resulting Cooper pairs may therefore acquire a finite center-of-mass momentum.

A finite Cooper-pair momentum alone, however, is not sufficient to produce unequal forward and reverse critical currents. For example, a current-phase relation containing only a phase shift,

$$I(\phi) = I_0 \sin(\phi - \phi_q),$$

has an anomalous phase ϕ_q, but its critical-current magnitudes remain equal,$I_c^+ = |I_c^-| = I_0$.

Therefore, the phase shift generated by finite Cooper-pair momentum does not, by itself, yield a phase-unbiased SDE. The additional asymmetry required for the SDE arises from the combined effects of SOC and the Zeeman interaction on the phase-dependent Andreev spectrum.

Figure 12 illustrates this distinction for four representative cases. The x direction is taken along the current flow, while the y direction is along the junction. Panel (a) demonstrates that a Zeeman-induced splitting of the Fermi contours can generate finite-momentum Cooper pairs while no SDE occurs. Panel (b) shows the limit in which Rashba SOC breaks spin degeneracy, but in the absence of a magnetic field, time-reversal symmetry requires pairing between opposite momenta, leading to $q = 0$ and the absence of SDE.

Figure 12(c) contains the combination relevant to the SDE considered in this thesis. Rashba SOC produces a momentum-dependent spin texture, while the magnetic field applied along the junction breaks time-reversal symmetry and modifies the two Fermi contours, and they are no longer concentric like the previous cases, giving a finite q along the

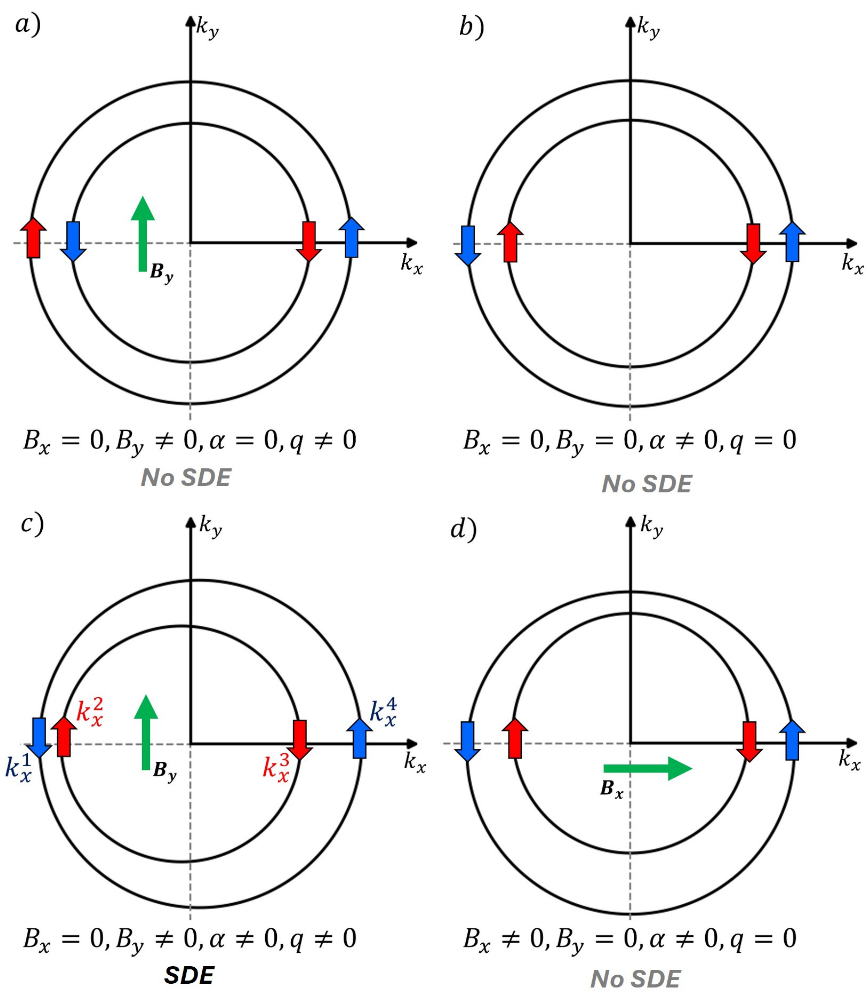

Figure 12: Qualitative comparison of the spin-split Fermi contours and Cooper-pair momentum for different combinations of Rashba SOC (α) and an in-plane magnetic field. The two black circles represent the spin-split Fermi contours. The arrows indicate the spin orientations of the electronic states at $k_y = 0$, while arrows of the same color identify the two states forming a Cooper pair. The green arrows indicate the direction of the applied magnetic field. **(a)** A Zeeman field along the junction produces finite Cooper-pair momentum, $q \neq 0$, even in the absence of Rashba SOC; however, no SDE occurs. **(b)** With Rashba SOC and zero magnetic field, time-reversed states form Cooper pairs with $q = 0$, and the SDE is absent. **(c)** When Rashba SOC and a magnetic field along the junction coexist, the blue arrows identify the pair formed by (k_x^1, k_x^4), while the red arrows identify the pair formed by (k_x^2, k_x^3). The mismatch between the momenta of the two states in each pair gives a finite center-of-mass momentum, $q \neq 0$, allowing a finite SDE. **(d)** When the magnetic field is applied along the current direction, the Cooper-pair momentum along the transport direction vanishes, and the SDE is suppressed.

current direction. In this case, the SDE is symmetry-allowed.

The orientation of the magnetic field is also important. As shown in Fig. 12(d), when the field is applied along the current direction, the Cooper-pair momentum along the current direction vanishes for a junction with purely Rashba SOC. Consequently, the SDE is suppressed.

Since the situation described in Fig. 12(c) is the only one that can generate a finite SDE, we focus on this particular case. The intersections of the two Fermi contours with the k_x axis are denoted by k_x^i, with $i = 1, 2, 3, 4$, as shown in Fig. 12(a). The states (k_x^1, k_x^4) and (k_x^2, k_x^3) define two intraband pairing channels. Cooper-pair center-of-mass momenta along the current direction can be approximated as,

$$q_x = \pm q \quad , \quad q = k_x^4 - |k_x^1| = \left|k_x^2\right| - k_x^3 \tag{4.17}$$

Rashba SOC-only case

Consider a planar JJ with Rashba SOC and an in-plane magnetic field. Within the narrow-junction approximation, and in the low-field regime,

$$E_Z \ll \alpha k_F, \tag{4.18}$$

the intersections of the spin-split Fermi contours with the k_x axis at $k_y = 0$ are approximately

$$
\begin{aligned}
k_x^1 &= -k_{\text{so}} - \sqrt{k_F^2 + k_{\text{so}}^2 + \kappa_B^2 \sin\theta_B},\\
k_x^2 &= k_{\text{so}} - \sqrt{k_F^2 + k_{\text{so}}^2 - \kappa_B^2 \sin\theta_B},\\
k_x^3 &= -k_{\text{so}} + \sqrt{k_F^2 + k_{\text{so}}^2 + \kappa_B^2 \sin\theta_B},\\
k_x^4 &= k_{\text{so}} + \sqrt{k_F^2 + k_{\text{so}}^2 - \kappa_B^2 \sin\theta_B}.
\end{aligned}
\tag{4.19}
$$

Substituting Eq. (4.19) into Eq. (4.17) and expanding to leading order in the Zeeman energy gives

$$
q \approx \frac{\kappa_B^2 \sin\theta_B}{\sqrt{k_F^2 + k_{\text{so}}^2}},
\tag{4.20}
$$

where

$$
\begin{aligned}
k_F &= \sqrt{\frac{2m^* E_F}{\hbar^2}},\\
\kappa_B &= \sqrt{\frac{2m^* E_Z}{\hbar^2}},\\
E_Z &= \frac{g^* \mu_B}{2}|B|,\\
k_{\text{so}} &= \frac{m^* \alpha}{\hbar^2}.
\end{aligned}
\tag{4.21}
$$

Equation (4.20) shows that the Cooper-pair momentum depends strongly on the magnetic-field orientation. It vanishes when the magnetic field is applied along the current direction, $\theta_B = 0$, and reaches its largest magnitude when the field is applied along the junction, $\theta_B = \pi/2$. Although Eq. (4.20) can remain finite in the limit $\alpha \to 0$, the SDE vanishes in that limit.

Coexisting Rashba and Dresselhaus SOC

For a junction containing both Rashba and Dresselhaus SOC, the two coupling strengths are parameterized as

$$\alpha = \lambda \cos\theta_{\text{so}}, \qquad \beta = \lambda \sin\theta_{\text{so}}, \tag{4.22}$$

where

$$\lambda = \sqrt{\alpha^2 + \beta^2}, \qquad \theta_{\text{so}} = \arctan\left(\frac{\beta}{\alpha}\right). \tag{4.23}$$

Here, λ represents the total SOC strength, while θ_{so} specifies the relative contributions of Rashba and Dresselhaus SOC.

At $k_y = 0$, the spin-dependent part of the Hamiltonian can be mapped onto an effective Rashba-only form through the spin rotation

$$\mathcal{U}_\theta = \exp\left[-\frac{i\theta}{2}\left(\sigma_0 \otimes \sigma_z\right)\right], \tag{4.24}$$

where the rotation angle satisfies

$$\cot\theta = \tan(2\theta_C) + \cot\theta_{\text{so}} \sec(2\theta_C). \tag{4.25}$$

Under this transformation, the effective Rashba SOC strength and magnetic-field orientation become

$$\widetilde{\alpha} = \lambda\Omega_+, \qquad \widetilde{\theta}_B = \theta_B - \theta, \tag{4.26}$$

where

$$\Omega_\pm = \sqrt{1 \pm \sin(2\theta_{\text{so}})\sin(2\theta_C)}. \tag{4.27}$$

The SOC-dependent Fermi-contour splitting is characterized by

$$\widetilde{k}_{\mathrm{so}} = k_\lambda \Omega_-, \qquad k_\lambda = \frac{m^* \lambda}{\hbar^2}. \tag{4.28}$$

Applying the Rashba result in Eq. (4.20) to the rotated Hamiltonian gives the Cooper-pair momentum for arbitrary Rashba and Dresselhaus SOC strengths,

$$\begin{aligned} q(\theta_B, \theta_C, \theta_{\mathrm{so}}) = & \frac{\kappa_B^2}{\Omega_+ \sqrt{k_F^2 + k_\lambda^2 \Omega_+^2}} \\ & \times \left[\sin\theta_B \cos\theta_{\mathrm{so}} - \sin\theta_{\mathrm{so}} \cos(\theta_B + 2\theta_C)\right]. \end{aligned} \tag{4.29}$$

Equation (4.29) contains both magnetic and crystalline anisotropies. The dependence on θ_B describes the response to rotations of the magnetic field, while the dependence on θ_C arises from the crystalline anisotropy introduced by Dresselhaus SOC. The expressions for two orthogonal magnetic-field orientations are summarized in Table 3.

For purely Rashba SOC, the Cooper-pair momentum is independent of the crystallographic orientation and vanishes for a magnetic field along the x direction. Dresselhaus SOC introduces an explicit dependence on θ_C, allowing q to vanish or reverse sign as the junction direction is rotated relative to the crystal axes. When both SOC contributions are present, the value and sign of q are controlled jointly by θ_B, θ_C, and θ_{so}.

The finite Cooper-pair momentum produces the phase shift

$$\phi_q = qW_N \tag{4.30}$$

in the Andreev spectrum. Therefore, the angular dependence of q directly determines the

Table 3: Cooper-pair momentum for representative SOC configurations and two orthogonal in-plane magnetic-field orientations. The expressions follow from Eq. (4.29). The orientation $\theta_B = 0$ corresponds to a magnetic field along the current direction, while $\theta_B = \pi/2$ corresponds to a field along the junction.

SOC configuration	$\theta_B = 0$	$\theta_B = \pi/2$
Pure Rashba SOC ($\theta_{\text{so}} = 0$)	$q = 0$	$q = \dfrac{\kappa_B^2}{\sqrt{k_F^2 + k_\lambda^2}}$
Pure Dresselhaus SOC ($\theta_{\text{so}} = \pi/2$)	$q = -\dfrac{\kappa_B^2 \cos(2\theta_C)}{\sqrt{k_F^2 + k_\lambda^2}}$	$q = \dfrac{\kappa_B^2 \sin(2\theta_C)}{\sqrt{k_F^2 + k_\lambda^2}}$
Coexisting Rashba and Dresselhaus SOC	$q = -\dfrac{\kappa_B^2 \sin\theta_{\text{so}} \cos(2\theta_C)}{\Omega_+ \sqrt{k_F^2 + k_\lambda^2 \Omega_+^2}}$	$q = \dfrac{\kappa_B^2 \left[\cos\theta_{\text{so}} + \sin\theta_{\text{so}} \sin(2\theta_C)\right]}{\Omega_+ \sqrt{k_F^2 + k_\lambda^2 \Omega_+^2}}$

magnetic and crystalline anisotropies of the SDE.

The analysis above shows that a junction with both Rashba and Dresselhaus spin-orbit coupling can be mapped onto an effective Rashba-only problem, as long as the spin-orbit strength and the magnetic-field direction are replaced by their effective values. This is useful because the analytical model developed earlier in section 3.1.1 for the Rashba case can now be applied to the general system. In this mapping, the Cooper-pair momentum produces a phase shift, which controls the asymmetry of the current-phase relation. Therefore, the dependence of q on θ_B, θ_C, and θ_{so} directly determines the magnetic and crystalline anisotropy of the SDE, which is discussed in Chapter 6.

To connect the symmetry and Cooper-pair momentum arguments to the current-phase relation, we now introduce a phenomenological model for the free energy and supercurrent.

4.3 Phenomenological Model

To better understand the magnetic and crystalline anisotropies of the supercurrents, we formulate a general phenomenological model capable of providing insight into the interrelation between the relevant orientations (θ_B and θ_C) and the superconducting phase difference, ϕ.

The free energy of the system is a periodic function of the superconducting phase difference and can be written as a trigonometric series,

$$
\begin{aligned}
F(\mathbf{B},\phi) &= F_0(\mathbf{B}) + \sum_{m=1}^{\infty} a_m(\mathbf{B}) \sin(m\phi) \\
&+ \sum_{m=1}^{\infty} b_m(\mathbf{B}) \cos(m\phi), \qquad (4.31)
\end{aligned}
$$

where the expansion coefficients, according to the symmetry relation $F(\mathbf{B},\phi) = F(-\mathbf{B},-\phi)$, obey

$$F_0(\mathbf{B}) = F_0(-\mathbf{B}),$$
$$a_m(\mathbf{B}) = -a_m(-\mathbf{B}),\ b_m(\mathbf{B}) = b_m(-\mathbf{B}). \tag{4.32}$$

It follows from Eq. (4.31) that, since the SDE requires $F(\mathbf{B},\phi) \neq F(\mathbf{B},-\phi)$, the expansion coefficients $a_n(\mathbf{B})$ are the ones responsible for the phase-biased SDE.

In what follows, we restrict our analysis to cases where the SOC cannot be gauged out, allowing the coefficients a_m and b_m to be expanded in powers of the magnetic field $\mathbf{B}$ and the SOC field components,

$$\mathbf{w}_x = \begin{pmatrix} \beta\cos 2\theta_C \\ -\beta\sin 2\theta_C - \alpha \\ 0 \end{pmatrix}, \tag{4.33}$$

and

$$\mathbf{w}_y = \begin{pmatrix} \alpha - \beta\sin 2\theta_C \\ -\beta\cos 2\theta_C \\ 0 \end{pmatrix}. \tag{4.34}$$

Note that these vectors define the SOC Hamiltonian $H_{\text{SO}} = H_R + H_D = \mathbf{W}_{so}\cdot\boldsymbol{\sigma}$, where the effective spin–orbit field is given by $\mathbf{W}_{so} = (p_x\mathbf{w}_x + \hbar k_y\mathbf{w}_y)/\hbar$. Taking into account the parity constraints in Eq. (4.32), a_m and b_m can generally be expanded as,

$$
\begin{aligned}
a_m &= \sum_{n,k_y} \left[c^{(1,1)}_{mnk_y} \left(\langle \mathbf{W}_{so} \rangle_n \cdot \mathbf{B} \right) + c^{(3,3)}_{mnk_y} \left(\langle \mathbf{W}_{so} \rangle_n \cdot \mathbf{B} \right)^3 + ... \right], \\
b_m &= \sum_{n,k_y} \left[d^{(0,0)}_{mnk_y} + d^{(2,0)}_{mnk_y} |\langle \mathbf{W}_{so} \rangle_n|^2 + d^{(0,2)}_{mnk_y} |\mathbf{B}|^2 + d^{(2,2)}_{mnk_y} \left(\langle \mathbf{W}_{so} \rangle_n \cdot \mathbf{B} \right)^2 + ... \right], \quad (4.35)
\end{aligned}
$$

where $\langle \cdots \rangle_n$ denotes the expectation value in the nth channel. Restricting to leading order in the magnetic field and spin–orbit coupling, one obtains the approximations

$$a_m \approx f_m(\mathbf{w}_x \cdot \mathbf{B}) \quad , \quad b_m \approx g_m, \tag{4.36}$$

where f_m and g_m are coefficients absorbing the sums over n and k_y. Note that the term involving $k_y \mathbf{w}_y$ is odd in k_y and therefore vanishes upon summation over k_y. In addition, this summation can induce a spin–orbit dependence in the coefficients a_m and b_m, arising from the mismatch between the Fermi wave vectors of the two spin channels (see Sec. 4.4 for explicit calculations). Using Eqs. (1.16), (4.31), and (4.36), the generic form of the supercurrent up to leading order in the magnetic and spin-orbit fields can be written as

$$I \approx \sum_{m=1}^{\infty} \left[g_m \sin(m\phi) - f_m(\mathbf{w}_x \cdot \mathbf{B}) \cos(m\phi) \right], \tag{4.37}$$

where, for brevity, the factor $-2em/\hbar$ has been absorbed into the coefficients f_m and g_m. It then follows from Eq. (4.37) that, except in the special cases where the SOC can be gauged out, the SDE is suppressed when

$$\mathbf{w}_x \cdot \mathbf{B} = B \left[\beta \cos(2\theta_C + \theta_B) - \alpha \sin \theta_B \right] = 0, \tag{4.38}$$

a condition that is consistent with the corresponding symmetry constraints derived in Chapter 6 (see Table 4). Using trigonometric identities, Eq. (4.37) can be recast as,

$$I \approx \sum_{m=1}^{\infty} I_m \sin(m\phi - \phi_m), \tag{4.39}$$

where,

$$I_m = \sqrt{g_m^2 + f_m^2(\mathbf{w}_x \cdot \mathbf{B})^2}, \tag{4.40}$$

and

$$\phi_m = \arctan\left[\frac{f_m(\mathbf{w}_x \cdot \mathbf{B})}{g_m}\right] \tag{4.41}$$

denote, respectively, the amplitude and anomalous phase associated with the mth harmonic.

Note that the presence of anomalous phases ensures that $I(\mathbf{B}, \phi) \neq -I(\mathbf{B}, -\phi)$, thereby yielding a finite phase-biased SDE even when only a single harmonic is present. Conversely, if the supercurrent is dominated by a single harmonic (say, the mth harmonic), the forward and reverse critical currents share the same amplitude, I_m, and the phase-unbiased SDE vanishes. Therefore, a finite phase-unbiased SDE requires contributions from at least two harmonics in the supercurrent. A minimal phenomenological model for the SDE, retaining only the two lowest harmonics, is given by [18, 47, 71],

$$I \approx I_1 \sin(\phi - \phi_1) + I_2 \sin(2\phi - \phi_2). \tag{4.42}$$

In the limit $I_1 \gg I_2$, the phases that maximize and minimize the supercurrent are approximately $\phi_{max} \approx \pi/2 + \phi_1$ and $\phi_{min} \approx -\pi/2 + \phi_1$, respectively. In the weak magnetic-field regime, the diode efficiency then reduces to

$$\eta \approx -2\sin(2\phi_2 - \phi_1) \approx \frac{2f_2 g_1 - 4f_1 g_2}{g_1 g_2}(\mathbf{w}_x \cdot \mathbf{B}). \tag{4.43}$$

This relation highlights the crystalline and magnetic anisotropies of the SDE, as reflected in its dependence on both the spin–orbit field (hence on the crystallographic orientation of the junction) and the direction of the applied magnetic field. As follows from Eq. (4.43), the SDE is expected to vanish when the condition in Eq. (4.38) is satisfied.

It is worth emphasizing that Eq. (4.43) represents a low-order approximation for the SDE and may fail to capture its behavior in certain systems, in particular perfectly transparent junctions, where the contribution to the diode efficiency linear in the magnetic field amplitude B has been shown to vanish [61]. Within the minimal model [Eq. (4.42)], quadratic-in-B contributions to η can arise by relaxing the condition $I_1 \gg I_2$ and/or by employing more accurate expressions for the critical phases ϕ_{max} and ϕ_{min}.

4.4 Analytical treatment of Superconducting Diode Effect

The narrow-junction δ-barrier model, the analytical Andreev spectrum, and the corresponding approximation for the free energy were developed in Sec. 3.1.1. In this section, those results are used to obtain quantities directly relevant to the SDE, including the forward and reverse critical currents, and the low-field diode efficiency.

The junction transparency can be characterized in terms of the parameter,

$$\tau = \frac{1}{1 + (V_0/2E_T)^2}, \tag{4.44}$$

where V_0 is the effective barrier strength and

$$E_T = \frac{\pi\hbar v_F}{2W_N} \tag{4.45}$$

is the Thouless energy. The analytical results derived below correspond to a junction with Rashba SOC only and a magnetic field applied along the y-direction (i.e., parallel to the junction) and are valid in the low-field, weak-SOC, and high-transparency regime.

4.4.1 Critical Currents and Diode Efficiency

The supercurrent is obtained by substituting the analytical Andreev spectrum in Eq. (3.24) into the approximate free energy in Eq. (3.30) and differentiating with respect to the superconducting phase difference. Expanding the resulting current in powers of the magnetic field and $\sqrt{1-\tau}$, followed by optimization with respect to the phase, gives the forward and reverse critical-current magnitudes [61],

$$\begin{aligned}\frac{|I_c^{\pm}|}{I_0} = 1 - \sqrt{1-\tau} \\ - \frac{1}{4}\left[1 \mp \frac{k_{so}}{k_F}\,\mathrm{sgn}(\phi_q \mp \phi_*)\right](\phi_q \mp \phi_*)^2.\end{aligned} \tag{4.46}$$

Here,

$$\phi_q = qW_N \tag{4.47}$$

is the phase shift generated by the finite Cooper-pair momentum derived in Sec. 4.2.2. The transparency-dependent phase shift

$$\phi_* \approx (1-\tau)^{1/4}\frac{k_{so}}{k_F} \tag{4.48}$$

determines the displacement of the two critical-current branches.

Equation (4.46) shows that the forward and reverse critical currents are controlled by two distinct effects. The finite Cooper-pair momentum produces the magnetic-field-dependent phase shift ϕ_q, while the combination of SOC and partial transparency introduces the additional scale ϕ_*.

Substituting Eq. (4.46) into the definition of the diode quality factor in Eq. (4.2) gives

$$\begin{aligned}\eta = & -\frac{1}{4}\left[(\phi_q-\phi_*)^2-(\phi_q+\phi_*)^2\right] \\ & +\frac{k_{so}}{4k_F}\left[\operatorname{sgn}(\phi_q-\phi_*)(\phi_q-\phi_*)^2+\operatorname{sgn}(\phi_q+\phi_*)(\phi_q+\phi_*)^2\right].\end{aligned} \tag{4.49}$$

The sign functions in Eq. (4.49) divide the result into three regimes determined by the relative magnitudes of ϕ_q and ϕ_*. Evaluating the expression in each regime yields

$$\eta = \begin{cases} \phi_q\phi_* - \dfrac{k_{so}}{2k_F}\left(\phi_q^2+\phi_*^2\right), & \phi_q < -\phi_*, \\ \phi_q\phi_*\left(1+\dfrac{k_{so}}{k_F}\right), & |\phi_q| \le \phi_*, \\ \phi_q\phi_* + \dfrac{k_{so}}{2k_F}\left(\phi_q^2+\phi_*^2\right), & \phi_q > \phi_*. \end{cases} \tag{4.50}$$

In the central region, $|\phi_q| \le \phi_*$, the leading contribution to the quality factor is linear in ϕ_q. Since the Cooper-pair momentum is linear in the Zeeman energy in the low-field regime, this term produces an approximately linear magnetic-field dependence. Its magnitude is proportional to $(1-\tau)^{1/4}$ and therefore decreases as the junction approaches perfect transparency.

For a fully transparent junction, $\tau = 1$, one has $\phi_* = 0$, and Eq. (4.46) reduces to

$$\frac{|I_c^\pm|}{I_0} = 1 - \frac{1}{4}\left[1 \mp \frac{k_{so}}{k_F}\,\mathrm{sgn}(\phi_q)\right]\phi_q^2. \tag{4.51}$$

Both critical-current magnitudes are therefore maximal at $\phi_q = 0$, corresponding to zero magnetic field. However, the SOC-dependent term gives the two branches different curvatures, producing a finite SDE whose leading magnetic-field dependence is quadratic.

When $\tau < 1$, the finite value of ϕ_* shifts the maxima of the two critical-current branches away from zero field. The approximate value of the finite magnetic field at which the critical current peaks is given by,

$$B_*^\pm = \pm\frac{2E_T}{|g^*|\mu_B}(1-\tau)^{1/4}\frac{k_{so}}{k_F}\sqrt{1+\left(\frac{k_{so}}{k_F}\right)^2}. \tag{4.52}$$

The shift in the critical current maximum increases with the SOC strength and with decreasing transparency. At the same time, reducing the transparency lowers the overall critical-current amplitude. The field positions of the critical-current maxima therefore provide experimentally accessible signatures of the combined effects of SOC and junction transparency. This is explored in detail in Chapter 5.

The analytical framework developed in this chapter will be extended in Chapter 6 to study the magnetic and crystalline anisotropies of the SDE in more detail. Before turning to that analysis, Chapter 5 examines the full magnetocurrent–phase relation and relates its phase-resolved asymmetry to the difference between the forward and reverse critical currents.

5 MAGNETO-CPR SIGNATURES OF THE SUPERCONDUCTING DIODE EFFECT

This chapter is based on Ref. [61].

The current-phase relation of a JJ contains information about the phase-dependent Andreev spectrum and the resulting supercurrent. When an external magnetic field is applied, both the quasiparticle spectrum and the current-phase relation acquire a magnetic-field dependence. The complete dependence of the supercurrent on the superconducting phase difference and the Zeeman energy is referred to here as the magneto-current-phase relation, or magneto-CPR. The system considered in this chapter is a proximitized planar JJ with Rashba spin-orbit coupling and an in-plane magnetic field applied along the junction direction. The magneto-CPR provides direct signatures of how Rashba SOC, junction transparency, and finite device geometry influence the SDE.

5.1 Magneto-CPR and Measures of Nonreciprocity

The phase-dependent Josephson current is given by

$$I(\phi, E_Z) = \frac{2e}{\hbar}\frac{\partial F(\phi, E_Z)}{\partial \phi}, \tag{5.1}$$

where ϕ is the superconducting phase difference and E_Z is the Zeeman energy. Since the quasiparticle spectrum depends on both ϕ and E_Z, the shape and amplitude of the current-phase relation evolve as the magnetic field is varied. For each value of E_Z, the forward (reverse) critical current is obtained by maximizing (minimizing) the magneto-CPR with respect to the phase,

$$I_c^+(E_Z) = \max_\phi I(\phi, E_Z), \qquad I_c^-(E_Z) = \min_\phi I(\phi, E_Z). \tag{5.2}$$

Here, $I_c^+ > 0$ and $I_c^- < 0$. In this chapter, we do not focus on anisotropic effects and therefore adopt a more conventional definition of the diode efficiency,

$$\eta(E_Z) = \frac{I_c^+(E_Z) - |I_c^-(E_Z)|}{I_c^+(E_Z) + |I_c^-(E_Z)|}. \tag{5.3}$$

A finite value of η indicates nonreciprocal supercurrent transport. Positive and negative values correspond to opposite diode polarities, while $\eta = 0$ occurs when the two critical currents have equal magnitudes and the SDE vanishes.

5.2 Fully Transparent Junctions

The fully transparent limit, $\tau = 1$, provides a useful reference for identifying the magnetic-field dependence of the SDE. The analytical expression for the critical currents in this limit was derived in Chapter 4.

Equation (4.51) shows that a finite diode response requires both Rashba SOC and a nonzero magnetic-field-induced phase shift. In the absence of either contribution, the forward and reverse critical currents become equal in magnitude, and the diode response vanishes. The two critical-current branches have an approximately quadratic dependence on the Zeeman energy in the low-field regime, but Rashba SOC produces different curvatures for the forward and reverse branches. Both critical-current magnitudes remain maximal at zero magnetic field.

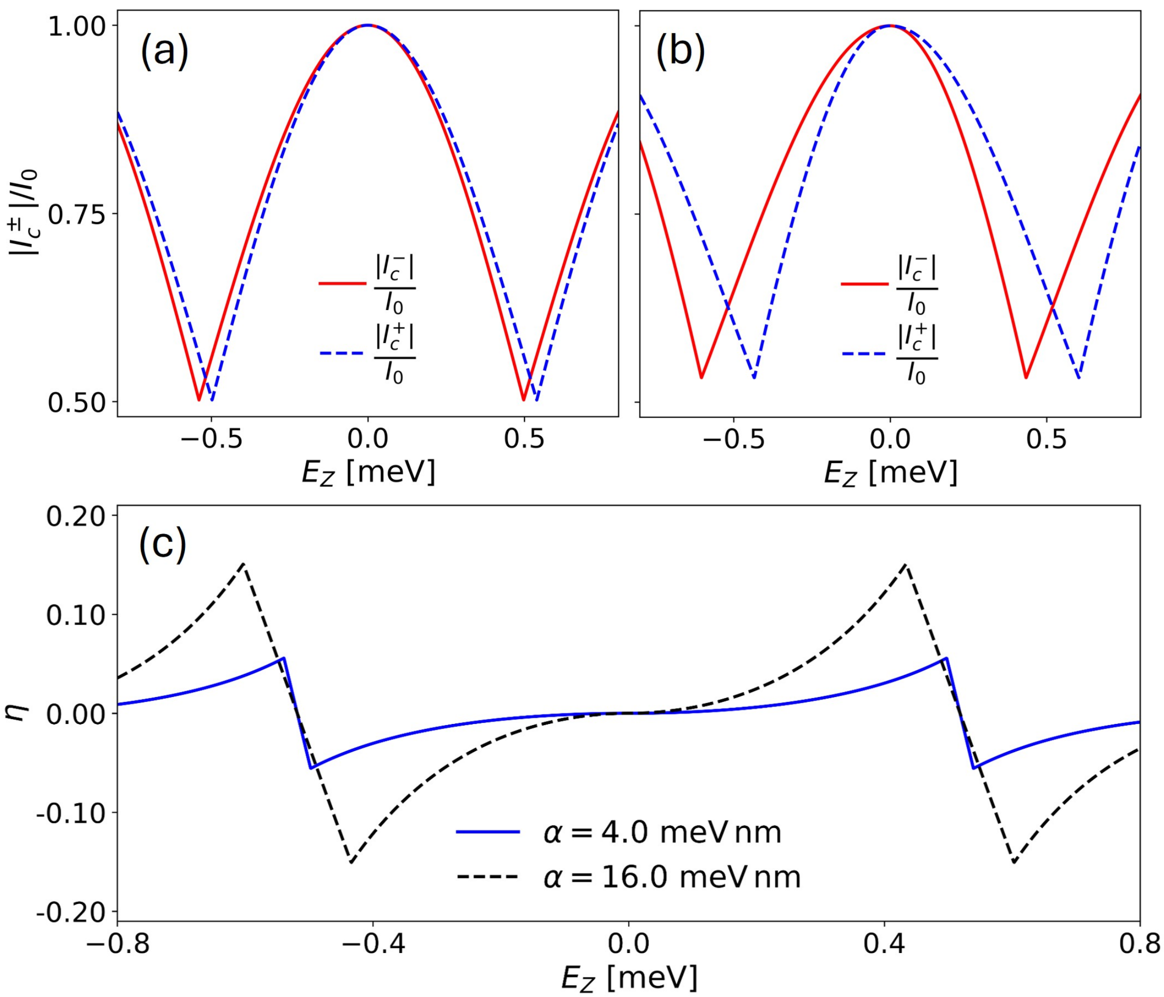

Figure 13: Superconducting diode effect in the fully transparent limit. **(a)** Zeeman-field dependence of the forward and reverse critical current magnitudes for a junction with Rashba SOC strength $\alpha = 4\ \mathrm{meV\,nm}$. **(b)** Same as panel (a), but for the larger Rashba SOC strength $\alpha = 16\ \mathrm{meV\,nm}$. Increasing the Rashba SOC strength enhances the difference between the two critical-current branches. **(c)** Diode efficiency η as a function of Zeeman energy for the two SOC strengths considered in panels (a) and (b). In the fully transparent limit, both critical-current magnitudes are maximal at zero field, while finite fields produce different curvatures and a nonzero diode response. Adapted from Ref. [61].

Figure 13 shows the magnetic-field dependence of the critical currents and diode efficiency in the fully transparent junction limit. Two different values of the Rashba spin-orbit coupling strength are considered. Figures 13(a) and 13(b) show that the forward and reverse critical-current magnitudes reach their maximum values at $E_Z = 0$. For $\alpha = 4$ meV nm [Fig. 13(a)], the two branches remain close to one another over most of the field range. Increasing the Rashba SOC strength to $\alpha = 16$ meV nm [Fig. 13(b)], produces a larger separation between the critical current branches and therefore a stronger nonreciprocal response.

The corresponding diode efficiencies are shown in Fig. 13(c). Increasing the Rashba SOC strength increases the magnitude of η in the low-magnetic-field regime. The diode efficiency changes sign whenever the magnitudes of the forward and reverse critical currents cross.

5.3 Partially Transparent Junctions

Perfect transmission is not generally realized in experimental planar JJs. Scattering from an electrostatic barrier, disorder, and reflections caused by the finite device dimensions can all lower the effective transparency of the weak link.

A reduction in transparency produces a clear change in the magnetic-field dependence of the critical currents. For $\tau = 1$, the magnitudes of both critical currents are largest at zero field. When $\tau < 1$, the maxima of the two branches move away from zero field in opposite directions. Thus, the magnetic fields at which the current maxima occur are no longer zero but determined by Eq. (4.52). According to that expression, the shift becomes larger when the Rashba SOC strength increases or when the junction transparency de-

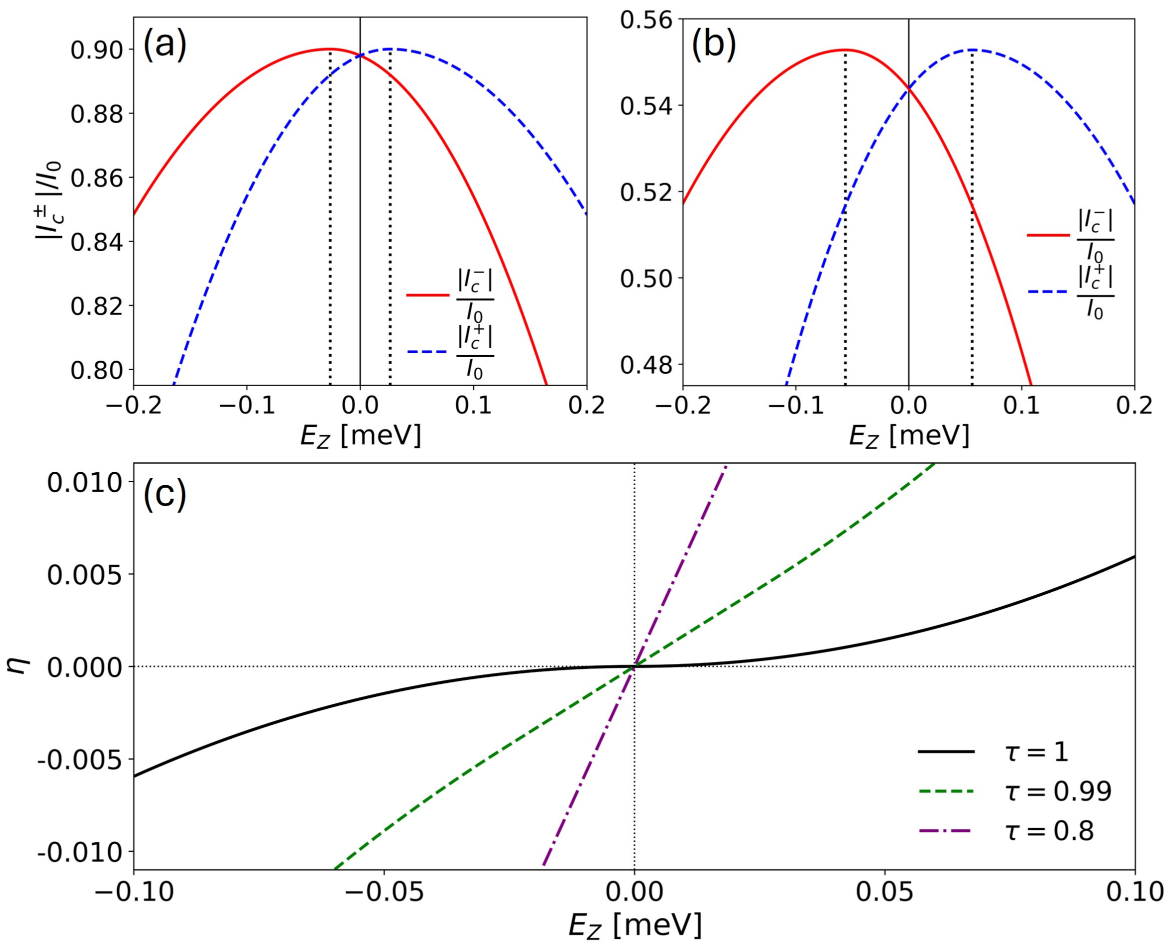

Figure 14: Superconducting diode effect in partially transparent junctions. **(a)** Zeeman-field dependence of the forward and reverse critical currents for a high-transparency junction with $\tau = 0.99$ and $\alpha = 16\,\mathrm{meV\,nm}$. The vertical dotted lines indicate the finite Zeeman energies at which the critical-current magnitudes reach their maximum values. **(b)** Same as panel (a), but for the lower transparency $\tau = 0.85$. Reducing the transparency shifts the critical-current maxima farther away from zero field and lowers the overall current amplitude. **(c)** Diode efficiency as a function of Zeeman energy for several junction transparencies. The low-field response changes from predominantly quadratic near $\tau = 1$ to increasingly linear as the transparency is reduced. Adapted from Ref. [61].

creases. At the same time, reducing the transparency decreases the overall critical-current amplitude because the barrier suppresses transmission through the junction.

These behaviors are illustrated in Fig. 14 for two values of the junction transparency. Figure 14(a) shows the forward and reverse critical currents for $\tau = 0.99$. Although the junction remains highly transparent, the two maxima are slightly displaced from zero field. The vertical dotted lines indicate their positions. For the lower transparency $\tau = 0.85$, shown in Fig. 14(b), the shift in the critical current maximum becomes considerably larger. The corresponding diode efficiencies are shown in Fig. 14(c). In the fully transparent limit, the low-field diode efficiency is predominantly quadratic because it originates from the different curvatures of the two critical-current branches. As the transparency is reduced, the shift of the critical current branches produces an increasingly linear field dependence near $E_Z = 0$. Apart from the sign reversal at zero magnetic field, the diode efficiency can also change sign at finite Zeeman energies. These reversals occur when the forward and reverse critical-current magnitudes cross, so that $\eta = 0$ and the polarity of the diode response changes. Finite-field crossings often occur near local minima of the critical currents and may be associated with 0–π-like changes in the equilibrium phase of a phase-unbiased junction [20, 47]. However, crossings may also occur away from the critical-current minima and therefore do not necessarily indicate a 0–π-like transition. Such sign reversals have been observed experimentally and reproduced theoretically in proximitized planar JJs [20, 47]. Junctions with wider normal regions can exhibit several critical-current minima as the magnetic field is increased, leading to multiple finite-field reversals of the SDE; this is explored in next section.

The field position of the critical-current maximum and the low-field slope of the diode efficiency therefore contain information about the effective junction transparency. Since these quantities also depend on Rashba SOC, an independent estimate of one parameter may allow the other to be inferred from magneto-CPR measurements. In addition, the complete phase dependence of the magneto-CPR can be used to reconstruct the ground-state phase of a phase-unbiased junction. As discussed in Sec. 5.4.1, the size of the resulting ground-state phase jumps can also be used to estimate the Rashba SOC strength [61].

5.3.1 Comparison with Experiment and Numerical Simulations

The transparency-dependent behavior discussed above has been observed experimentally in an epitaxial Al/InAs planar JJ [20]. This provides a direct comparison between the analytical predictions for partially transparent junctions and experimentally measured critical currents.

Figure 15 compares the experimental data with the analytical and tight-binding results. Figures 15(a)–15(c) show the absolute values of the forward and reverse critical currents in the low-field regime as functions of the in-plane magnetic field. Figure 15(b) shows the results from the analytical model. Here we use the dimensionless parameter introduced earlier, $Z_y = \frac{2m^* E_{Z,y} W_N}{\hbar^2 k_{F,S}}$ [Eq. 3.26] as a measure of the magnetic field strength. The gray dotted lines represent the critical currents for a fully transparent junction, $\tau = 1$, and show that both I_c^+ and $|I_c^-|$ reach their maxima at zero field. As the junction transparency decreases, however, the critical-current maxima shift to finite magnetic fields, denoted by B_*.

The approximately quadratic dependence of I_c^+ and $|I_c^-|$ on B_y, together with the shift of their maxima observed experimentally in the low-field regime, is well described

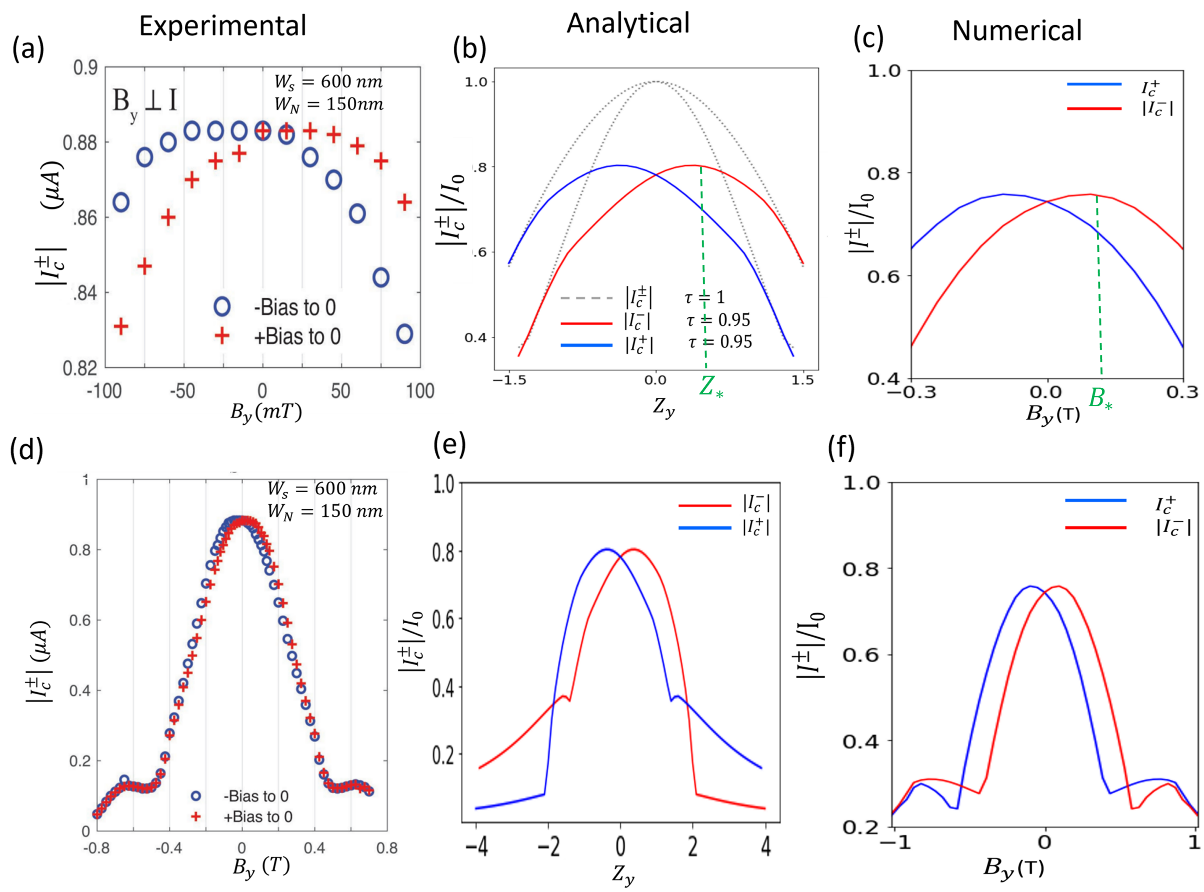

Figure 15: Magnetic-field dependence of the critical currents of an Al/InAs planar JJ in the low-field (top row) and high-field (bottom row) regimes. The field is applied along the junction direction, perpendicular to the current flow. **(a)** Experimentally measured forward (red crosses) and reverse (blue open circles) critical currents in a junction with $W_S = 600$ nm and $W_N = 150$ nm [20]. **(b)** Results from the analytical δ-barrier model for different values of the junction transparency. When the junction is perfectly transparent, the currents have their maximum amplitudes at zero field. When the transparency is smaller than one, the positions of the maxima shift to finite magnetic fields characterized by Z_* (vertical dashed lines). **(c)** Results from numerical tight-binding simulations performed using Kwant. The quantity B_* indicates the magnetic-field strength at which the critical-current maximum occurs. **(d)–(f)** Same as panels **(a)–(c)**, but over a wider field range that includes the high-field regime. The analytical and numerical critical currents are normalized to the zero-field critical current, $I_c(0)$. In the analytical model, the dimensionless quantity $Z_y = 2m^* E_{Z,y} W_N/(\hbar^2 k_{F,S})$, which is proportional to B_y, is used as a measure of the magnetic-field strength.

by both Eq. (4.46) and the more elaborate numerical simulations [see Figs. 15(a)–15(c)]. Furthermore, Eq. (4.52) predicts that the magnetic-field strength at which the critical-current maxima occur increases linearly with the SOC parameter and varies as $(1-\tau)^{1/4}$ with the junction transparency. These predictions could be tested experimentally using gated planar JJs, where the top gate can modify both the junction transparency and the Rashba SOC strength. Gate control could therefore provide an additional means of tuning the critical currents and the SDE, with possible applications in superconducting diodes and transistors.

In the high-field regime, the analytical model begins to deviate from the experimental observations [see Figs. 15(d) and 15(e)]. The numerical simulations, however, continue to reproduce the magnetic-field dependence of the critical currents more accurately [see Figs. 15(d) and 15(f)].

The comparison above concerns the extrema of the current–phase relation. Additional information about the origin of the nonreciprocity is contained in the complete phase dependence of the supercurrent. The following subsection examines this behavior using a phase-resolved current-asymmetry factor.

5.4 Phase-Resolved Current Asymmetry and Finite-Size Effects

The forward and reverse critical currents describe only the extrema of the magneto-CPR. To examine how nonreciprocity is distributed over the full phase range, we introduce the phase-resolved current asymmetry factor

$$Q_f(E_Z,\phi)=\frac{|I(E_Z,\phi)|-|I(E_Z,-\phi)|}{|I(E_Z,\phi)|+|I(E_Z,-\phi)|}. \tag{5.4}$$

A finite value of Q_f indicates that the magnitudes of the currents at opposite phase differences are unequal. The sign of Q_f identifies which phase direction supports the larger current.

From its definition, the phase-resolved asymmetry factor satisfies

$$Q_f(E_Z, \phi) = -Q_f(E_Z, -\phi) = -Q_f(E_Z, 2\pi - \phi), \tag{5.5}$$

which implies

$$Q_f(E_Z, \pi) = 0. \tag{5.6}$$

These symmetry relations provide a useful consistency check for numerical calculations and experimental magneto-CPR data. Figure 16 presents numerical tight-binding results for Q_f, the critical currents, and the diode efficiency. The three rows compare different Rashba SOC strengths and superconducting lead widths, allowing the effects of SOC and finite device geometry to be separated.

The left column of Fig. 16 shows the phase-resolved current asymmetry. Red and blue regions represent opposite signs of Q_f, while the boundaries between these regions mark changes in the direction of the phase-resolved asymmetry. The color maps show that the nonreciprocal response varies strongly with both phase and Zeeman energy.

The top row, Figs. 16(a)–16(c), corresponds to $\alpha = 16$ meV nm and $W_S = 40$ nm. The critical currents in panel (b) exhibit several crossings as the Zeeman energy is varied. These crossings produce the sign reversals of the diode efficiency shown in panel (c).

The middle row, Figs. 16(d)–16(f), corresponds to the stronger Rashba SOC $\alpha =$

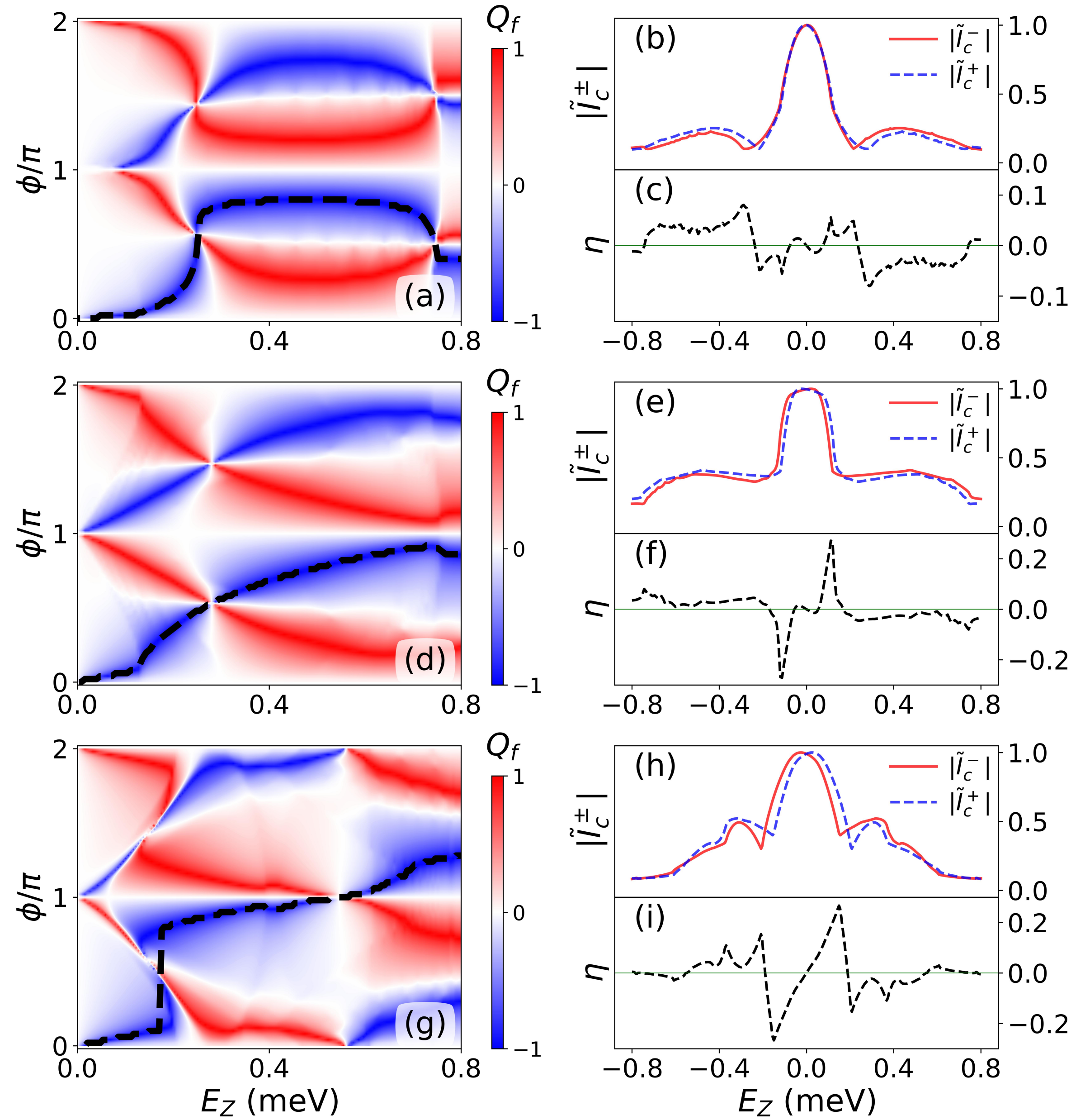

Figure 16: Numerical magneto-CPR signatures of the SDE. **(a)** Phase-resolved current asymmetry factor $Q_f(E_Z, \phi)$ for a junction with $\alpha = 16$ meV nm and superconducting lead width $W_S = 40$ nm. The black dashed curve marks the ground-state phase trajectory that the phase-unbiased junction would follow to minimize its free energy. **(b)** Forward and reverse critical currents extracted from the magneto-CPR represented in panel (a). **(c)** Diode efficiency calculated from the critical currents in panel (b). **(d)–(f)** Same quantities as in panels (a)–(c), but for the larger Rashba SOC strength $\alpha = 32$ meV nm and $W_S = 40$ nm. **(g)–(i)** Same quantities as in panels (a)–(c), but for $\alpha = 16$ meV nm and wider superconducting leads with $W_S = 240$ nm. The comparison shows that the diode response depends on both Rashba spin-orbit coupling and device geometry. Adapted from Ref. [61].

$32\ \mathrm{meV\,nm}$ with the same superconducting lead width. Increasing the Rashba SOC modifies the phase-resolved asymmetry pattern and generally increases the separation between the forward and reverse critical currents. The magnetic-field dependence of the diode efficiency is also modified, including the positions and amplitudes of its extrema.

The bottom row, Figs. 16(g)–16(i), uses the same SOC strength as the top row but increases the superconducting lead width to $W_S = 240\ \mathrm{nm}$. Comparing the top and bottom rows shows that the diode response is not determined by SOC strength alone. The finite width of the superconducting regions changes the critical-current oscillations and the positions of the diode polarity reversals.

This geometric dependence originates from reflections at the outer boundaries of the finite superconducting regions. These reflections can modify the effective junction transparency and introduce interference effects that are absent from the simplified analytical model, where the superconducting leads are assumed to be infinitely wide. As W_S is increased, the numerical results approach the analytical predictions more closely [compare Figs. 14(c) and 16(i)].

Besides revealing the phase-resolved current asymmetry, the magneto-CPR contains additional information about the equilibrium phase of a phase-unbiased junction. In particular, the black dashed curves in Figs. 16(a), 16(d), and 16(g) can be used to reconstruct the ground-state phase. The magnetic-field-dependent jumps of this phase also provide a method for estimating the Rashba SOC strength.

5.4.1 Ground-State Phase and Rashba SOC Strength

The magneto-CPR also provides access to the ground-state phase of a phase-unbiased junction. In the absence of an externally imposed phase bias, the superconducting phase difference self-adjusts to the value $\phi_{\rm GS}$ that minimizes the free energy. Therefore, the ground-state phase satisfies

$$\left.\frac{\partial F(\phi, E_Z)}{\partial \phi}\right|_{\phi=\phi_{\rm GS}} = 0, \qquad \left.\frac{\partial^2 F(\phi, E_Z)}{\partial \phi^2}\right|_{\phi=\phi_{\rm GS}} > 0. \tag{5.7}$$

Using the relation between the current and the free energy, these conditions can be expressed in terms of the magneto-CPR as

$$I(E_Z, \phi_{\rm GS}) = 0, \qquad \left.\frac{\partial I(E_Z, \phi)}{\partial \phi}\right|_{\phi=\phi_{\rm GS}} > 0. \tag{5.8}$$

For each value of E_Z, the first condition identifies phases at which the current vanishes, while the positive-slope condition selects phases corresponding to minima of the free energy. The resulting values of $\phi_{\rm GS}$ form a ground-state phase trajectory in the (E_Z, ϕ) plane, which can directly be determined from the magneto-CPR. Alternatively, it can also be extracted from the asymmetry factor Q_f. These ground-state phases are shown by the black dashed curves in Figs. 16(a), 16(d), and 16(g).

As the Zeeman energy is varied, $\phi_{\rm GS}$ can exhibit discontinuous 0–π-like jumps. The magnitude of these jumps depends strongly on the Rashba SOC strength. Within the analytical model and the weak-SOC approximation, the ground-state phase is approximately given by [61]

$$\phi_{\rm GS} \approx \begin{cases} 2\arctan\left[\dfrac{k_{\rm so}}{k_F}\tan\left(\dfrac{\phi_q}{2}\right)\right], & \phi_q \leq \dfrac{\pi}{2}, \\[2ex] 2\arctan\left[\dfrac{k_F}{k_{\rm so}}\tan\left(\dfrac{\phi_q}{2}\right)\right], & \phi_q > \dfrac{\pi}{2}. \end{cases} \tag{5.9}$$

Here,

$$k_{\mathrm{so}} = \frac{m^* \alpha}{\hbar^2} \tag{5.10}$$

is the Rashba SOC wave vector, while ϕ_q is the magnetic-field-induced phase shift. Equation (5.9) shows that the values of the ground-state phase on the two sides of the jump depend on the ratio k_{so}/k_F.

At $\phi_q = \pi/2$, the Rashba SOC wave vector can be estimated directly from the size of the ground-state phase jump, $\delta\phi_{\mathrm{GS}}$, according to

$$k_{\mathrm{so}} = \frac{m^* \alpha}{\hbar^2} \approx \left[\frac{1 - \sin\left(\delta\phi_{\mathrm{GS}}/2\right)}{\cos\left(\delta\phi_{\mathrm{GS}}/2\right)} \right] k_F. \tag{5.11}$$

A phase jump approaching $\delta\phi_{\mathrm{GS}} = \pi$ corresponds to a very small Rashba SOC strength, while a reduction in the jump size indicates stronger Rashba SOC. Measuring the ground-state phase trajectory from the magneto-CPR therefore provides an experimentally accessible method for estimating α.

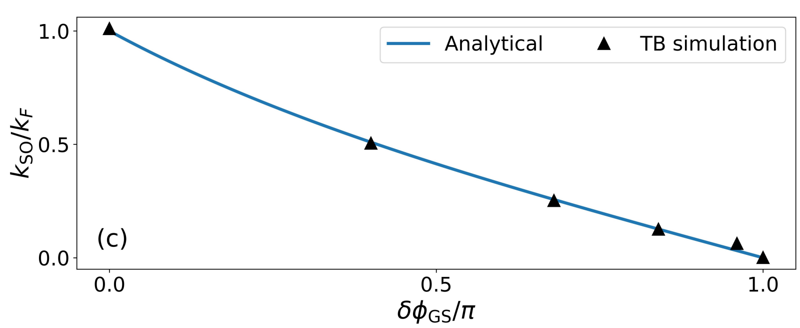

Figure 17: Normalized Rashba SOC wave vector k_{so}/k_F as a function of the normalized ground-state phase-jump size $\delta\phi_{\mathrm{GS}}/\pi$. The solid curve shows the analytical result obtained from Eq. (5.11), while the symbols represent numerical tight-binding results. The close agreement shows that the ground-state phase jump can be used to estimate the Rashba SOC strength. Adapted from Ref. [61].

Figure 17 compares the analytical prediction with the values obtained from the tight-binding simulations. The two approaches show close agreement over a broad range of SOC strengths. Although Eq. (5.11) was derived using the weak-SOC approximation, the agreement remains good even when k_{so}/k_F is not very small.

The magneto-CPR therefore provides complementary information about both nonreciprocal transport and the equilibrium properties of the junction. Its phase-resolved structure reveals the nonreciprocity quantified by Q_f, while its zero-current phase trajectories determine the ground-state phase and allow the Rashba SOC strength to be estimated from the size of the associated phase jumps. These results demonstrate that magneto-CPR measurements can be used not only to characterize the SDE, but also to extract important microscopic properties of planar JJs.

6 ANISOTROPIC SUPERCONDUCTING DIODE EFFECT

This chapter is based on Ref. [62].

This chapter examines the magnetic and crystalline anisotropies of the SDE using symmetry analysis, low-field analytical predictions, and numerical tight-binding calculations. Particular emphasis is placed on the angular dependence ($\theta_B, \theta_{so}, \theta_C$), symmetry-imposed zeros, and polarity reversals of the SDE response.

Semiconductor–superconductor heterostructures combine strong SOC with electrostatic control of the carrier density and junction transparency, allowing the diode response to be tuned experimentally [18, 20, 95, 102, 111–113]. Recent experiments have also revealed a dependence of the diode effect on the crystallographic orientation, indicating that crystalline anisotropy can play an important role in determining both its magnitude and sign [69, 95]. In junctions based on zinc-blende semiconductors, bulk inversion asymmetry gives rise to Dresselhaus SOC in addition to Rashba SOC induced by structural inversion asymmetry. The interplay between these two SOC contributions introduces a strong dependence on the crystal axes and can significantly modify the behavior of the SDE [69, 95].

6.1 Symmetry Conditions for Anisotropic Effects

A symmetry-based analysis can be used to identify geometric conditions involving the magnetic-field and junction orientations under which the diode effect vanishes independently of the field strength. These conditions provide robust and experimentally accessible signatures of the role of SOC in the emergence of the SDE.

As discussed in Chapter 4, the SDE is associated with the breaking of space-inversion and time-reversal symmetries. To determine the orientation-dependent conditions under which the SDE is suppressed, we consider transformations of the form

$$\mathcal{U}^{n}_{klm}(\mathbf{u}) = R_x^k R_y^l \left[\Gamma(\mathbf{u})\right]^m \mathcal{T}^n, \tag{6.1}$$

where R_x and R_y denote the spatial mirror operators with respect to the yz and xz planes, respectively. The operator

$$\Gamma(\mathbf{u}) = i\sigma_0 \otimes \left(\frac{\mathbf{u}}{|\mathbf{u}|} \cdot \boldsymbol{\sigma}\right) \tag{6.2}$$

represents a spin reflection through the plane perpendicular to the vector $\mathbf{u}$. The time-reversal operator is defined as

$$\mathcal{T} = -i\left(\sigma_0 \otimes \sigma_y\right)\mathcal{K}, \tag{6.3}$$

where $\mathcal{K}$ is the complex-conjugation operator. The exponents k, l, m, and n in Eq. (6.1) take the values 0 or 1.

Applying these transformations to the system Hamiltonian yields a set of conditions under which Eq. (4.15) is satisfied, leading to the suppression of the phase-biased SDE. These conditions are summarized as conditions (1)–(4) in Table 4.

Additional constraints arise in the special case $\beta = \pm\alpha$. Under suitable geometric conditions, the SOC can be completely gauged away through a position-dependent unitary transformation of the form

$$U = \exp\left\{i\mathbf{q}\cdot\mathbf{r}\left[\sigma_0 \otimes (\mathbf{v}\cdot\boldsymbol{\sigma})\right]\right\}. \tag{6.4}$$

As a representative example, for $\alpha = \beta$, $\theta_C = \pi/4$, and $\theta_B = \pi/2$, the SOC can be removed by choosing

$$\mathbf{q} = \left(\frac{2m^*\alpha}{\hbar^2}, 0, 0\right)^T, \qquad \mathbf{v} = (0, 1, 0)^T. \tag{6.5}$$

More generally, the SOC can be gauged away, and hence the SDE is suppressed, whenever $\beta = \pm\alpha \neq 0$ and the geometric condition

$$\sin(2\theta_C + \theta_B) = \mp\cos\theta_B \tag{6.6}$$

is satisfied [see condition (5) in Table 4].

The symmetry-imposed conditions summarized in Table 4 depend only on the symmetry of the SOC and the orientations of the magnetic field and junction. They can therefore be used to experimentally examine whether the interplay between SOC and the Zeeman interaction is responsible for the SDE in semiconductor-based planar JJs.

Table 4: Symmetry conditions for SDE suppression.

	α	β	$\{k, l, m, n\}$	$\mathbf{u}^T$	Condition
(1)	$\neq 0$	0	$\{1, 0, 1, 0\}$	$(1, 0, 0)$	$\sin\theta_B = 0$
(2)	0	$\neq 0$	$\{1, 0, 1, 0\}$	$(\sin 2\theta_C, \cos 2\theta_C, 0)$	$\cos(2\theta_C + \theta_B) = 0$
(3)	$\neq 0$	$\neq 0$ & $\neq \alpha$	$\{1, 0, 1, 0\}$	$(1, 0, 0)$	$\cos 2\theta_C = \sin\theta_B = 0$
(4)	$\neq 0$	$\pm\alpha$	$\{0, 0, 1, 1\}$	$(\pm\alpha\cos 2\theta_C, -\alpha[1 \pm \sin 2\theta_C], 0)$	$\cos(2\theta_C + \theta_B) = \pm\sin\theta_B$
(5)	$\neq 0$	$\pm\alpha$			$\sin(2\theta_C + \theta_B) = \mp\cos\theta_B$

6.2 Analytical Treatment of Anisotropic Effects

The critical-current expression obtained using the analytical model in Sec. 4.4.1 was obtained for a Rashba-only junction. The mapping derived in Sec. 4.2.2 allows the same analytical result to be extended to a junction containing both Rashba and Dresselhaus

SOCs. Under this mapping, k_{so} is replaced by the effective spin-orbit wave vector

$$\widetilde{k}_{so} = k_\lambda \Omega_-, \tag{6.7}$$

while the Cooper-pair momentum is given by Eq. (4.29).

Keeping only the leading contribution in the high-transparency and low-field regime, the diode efficiency becomes

$$\eta \approx (1-\tau)^{1/4} \frac{\widetilde{k}_{so}}{k_F} q W_N. \tag{6.8}$$

Note that, since we are now considering the anisotropic effects, we adopt the definition of diode efficiency given in Eq. (4.2). Equation (6.8) establishes the connection between the phenomenological and microscopic descriptions of the SDE. The phenomenological model [sec. 4.3] predicts that the leading nonreciprocal contribution is controlled by $\mathbf{w}_x \cdot \mathbf{B}$, while the analytical calculation shows that the same angular dependence is contained in the Cooper-pair momentum.

The angular dependence appearing in Eq. (4.29) can be written as

$$\mathcal{Q}(\theta_B, \theta_C, \theta_{so}) = \sin\theta_B \cos\theta_{so} - \sin\theta_{so} \cos(2\theta_C + \theta_B). \tag{6.9}$$

The numerator of the Cooper-pair momentum is therefore governed by the same combination of magnetic-field and crystallographic orientations that appears in $\mathbf{w}_x \cdot \mathbf{B}$. Figure 18(a) shows the result for purely Rashba SOC, $\theta_{so} = 0$. The diode efficiency is independent of θ_C, reflecting the in-plane rotational invariance of the Rashba interaction. The SDE vanishes for $\theta_B = 0$ and π, where the magnetic field is parallel to the current direction, and reaches

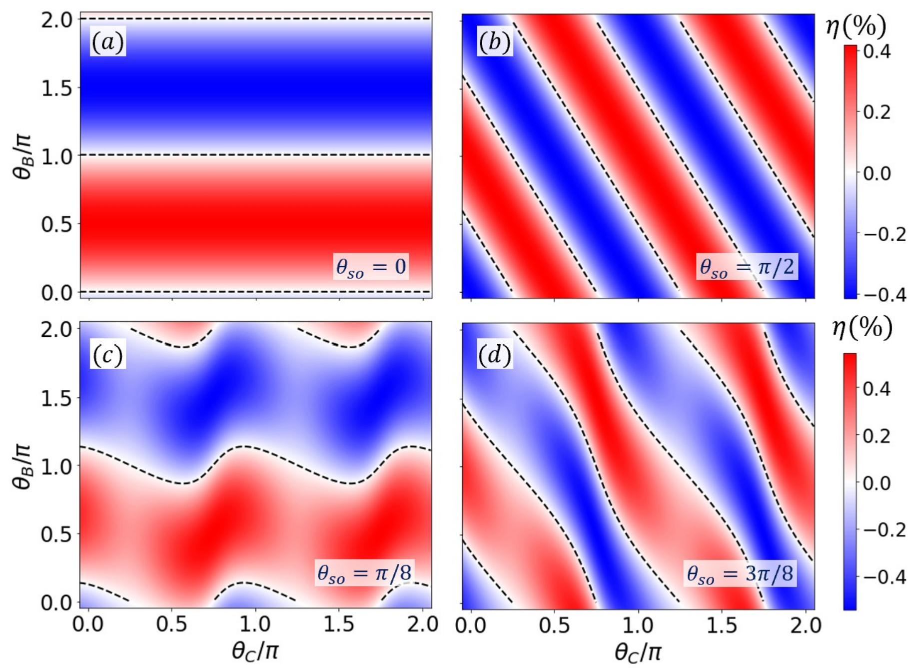

Figure 18: Diode efficiency as a function of the junction orientation θ_C and magnetic-field direction θ_B for SOC angles $\theta_{so} = 0$ **(a)**, $\pi/2$ **(b)**, $\pi/8$ **(c)**, and $3\pi/8$ **(d)**. The calculations are performed for an Al/InAs junction using the analytical model with $B = 0.1$ T and $\tau = 0.99$. The dashed curves indicate the field orientations for which the Cooper-pair momentum, and consequently the leading-order SDE, vanishes.

its largest magnitude when the field is directed along the junction. For purely Dresselhaus SOC, shown in Fig. 18(b), the efficiency depends on both θ_B and θ_C. This crystalline anisotropy arises because the Dresselhaus interaction depends on the crystallographic axes and is not invariant under arbitrary in-plane rotations.

Figures 18(c) and 18(d) show the Rashba-dominated and Dresselhaus-dominated regimes, respectively. The crystalline modulation is weaker when Rashba SOC dominates and becomes more pronounced as the Dresselhaus contribution increases.

Within the leading-order analytical approximation, the SDE is suppressed when the Cooper-pair momentum vanishes. Setting Eq. (6.9) equal to zero gives

$$\theta_B = \operatorname{arccot}\left[\tan(2\theta_C) + \cot\theta_{so}\sec(2\theta_C)\right], \tag{6.10}$$

a relation that is consistent with the conditions summarized in Table 4. The dashed curves in Fig. 18 describe zeros of the SDE and are determined by the solutions of Eq. (6.10). Unlike SDE zeros that depend on the magnetic-field strength, the zeros determined by Eq. (6.10) are only dictated by the structural symmetry and geometry of the junction.

Equation (6.10) also provides a method for estimating the ratio between the Rashba and Dresselhaus SOC strengths. If θ_B^* is a magnetic-field orientation at which the SDE is suppressed, then

$$\frac{\alpha}{\beta} = \cot\theta_{so} = \cot\theta_B^*\cos(2\theta_C) - \sin(2\theta_C). \tag{6.11}$$

Thus, measuring the angular positions of the SDE zeros for a junction with known crystallographic orientation can provide an estimate of the relative Rashba and Dresselhaus

SOC strengths. The validity of this procedure is restricted to the weak-field, weak-SOC, and high-transparency regime in which the leading-order analytical description applies.

6.3 Magneto-Anisotropic Superconducting Diode Effect

In this section, we focus on junctions where only Rashba SOC is present ($\theta_{so} = 0$), causing the SDE to become anisotropic with respect to the magnetic-field direction while remaining unaffected by the junction's crystallographic orientation. In this scenario, the SDE vanishes when the magnetic field is perpendicular to the junction ($\theta_B = n\pi$), as shown in Table 4, and reaches its maximum when $\mathbf{B}$ is aligned along the junction [$\theta_B = (2n+1)\pi/2$, with $n \in \mathbb{Z}$].

Figure 19 shows how the forward and reverse critical currents, along with the diode efficiency, depend on the magnetic field strength for an Al/InAs JJ at $\theta_B = \pi/2$. Panels (a) and (b) correspond to the numerical simulations and the analytical approach, respectively. The behavior of the critical currents and the diode efficiency is qualitatively consistent with recent experimental observations [20]. In particular, beyond the trivial SDE sign reversal near zero field, additional polarity changes appear at higher fields (around ± 0.75 T), close to the fields at which the junction in its equilibrium ground state would undergo 0–π transitions [20, 47]. We note that, while the numerical simulations include a magnetic field applied throughout the entire system, the analytical model assumes the field to be finite only in the N region for simplicity. As a consequence, a larger effective g-factor ($g^* = -40$) is required in the analytical model to reproduce the field values ($B \approx \pm 0.75$ T) at which the polarity changes associated with the 0–π transitions occur, as obtained in the numerical simulations with $g^* = -10$.

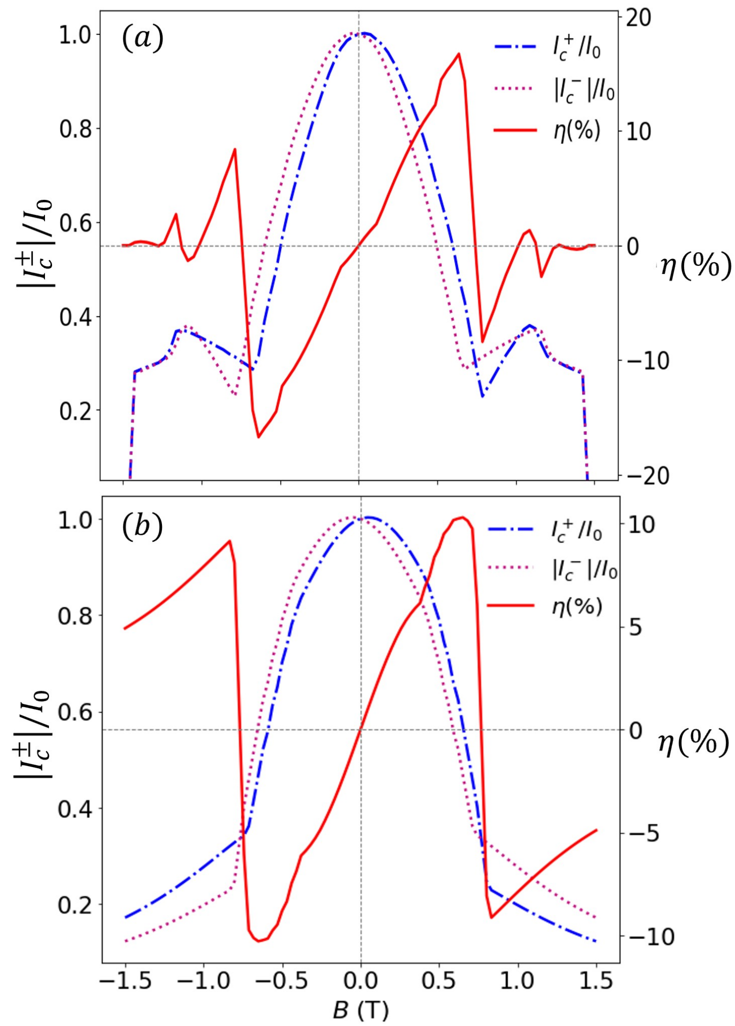

Figure 19: Magnetic-field dependence of the forward (I_c^+, dash-dotted line) and reverse (I_c^-, dotted line) critical currents, normalized to their zero-field value (I_0), for an Al/InAs junction with purely Rashba SOC ($\theta_{so} = 0$) and a magnetic field applied along the junction ($\theta_B = \pi/2$). The solid line denotes the corresponding diode efficiency. (a) Numerical simulations and (b) results from the analytical model (with $\tau = 0.99$ and $g^* = -40$) [62].

To illustrate the magnetoanisotropy of the SDE, we computed the diode efficiency as a function of the magnetic field angle θ_B (polar angle) and the field strength B (radial coordinate) for $\theta_{so} = 0$. The results, corresponding to the numerical simulations and the analytical approach, are shown as polar plots in Figs. 20(a) and (c), respectively. The analytical results agree well with the numerical simulations at weak magnetic fields. The

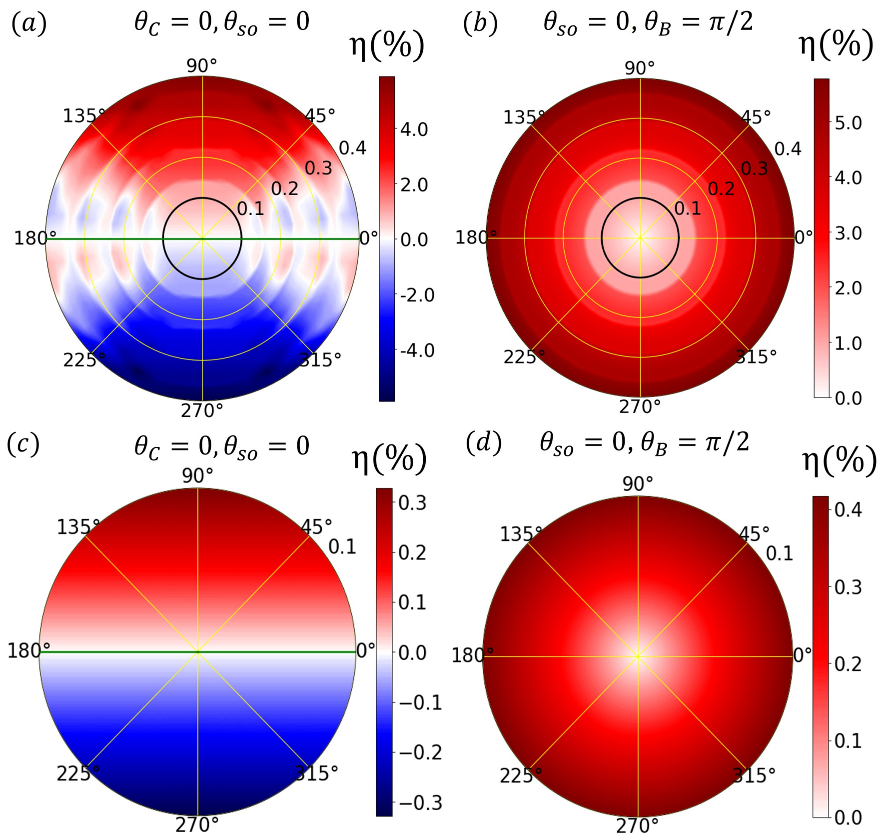

Figure 20: Magnetoanisotropy of the SDE in an Al/InAs junction with purely Rashba SOC ($\theta_{so} = 0$). **(a)** Diode efficiency from numerical simulations as a function of the magnetic-field angle θ_B (polar angle) and the field strength B (radial coordinate) for $\theta_C = 0$. **(b)** Diode efficiency from numerical simulations as a function of the junction orientation θ_C (polar angle) and the field strength B (radial coordinate) for $\theta_B = \pi/2$. Panels **(c)** and **(d)** show results from the analytical model (with $\tau = 0.99$) corresponding to (a) and (b), respectively, within the low-field region indicated by the black circles in (a) and (b). Green lines denote field directions for which symmetry-imposed suppression of the SDE occurs independently of the field strength [62].

green lines indicate field directions ($\theta_B = 0, \pi$) for which the SDE is suppressed regardless of field strength, confirming the predictions in Table 4.

This symmetry-imposed suppression has also been observed experimentally in epitaxial Al/InAs planar JJs [20]. Figure 21 provides an experimental demonstration of the predicted magnetoanisotropy. When the magnetic field is applied along the current direction, corresponding to $\theta_B = 0$, the critical-current magnitudes measured for the two

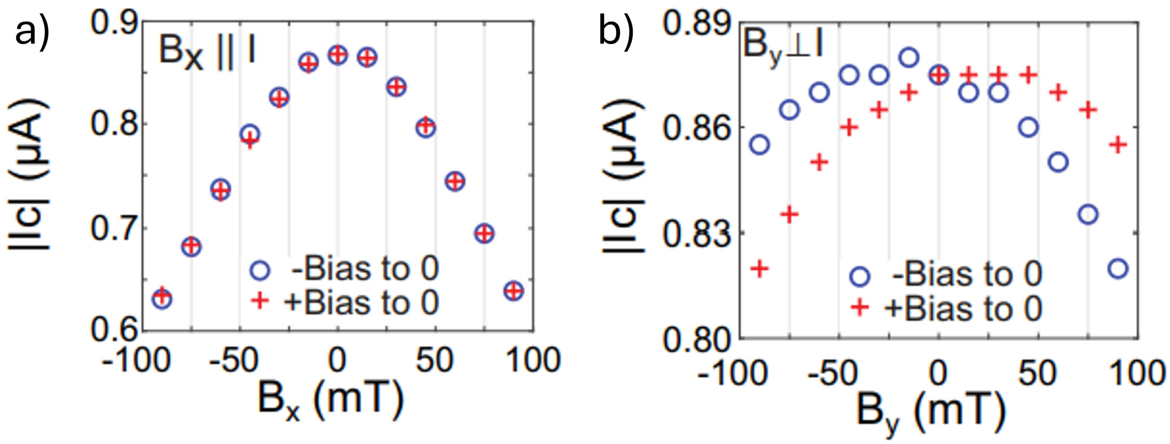

Figure 21: Experimental magnetic-field anisotropy of the SDE in an epitaxial Al/InAs planar JJ with superconducting lead width $W_S = 0.6\ \mu\mathrm{m}$. **(a)** Magnitudes of the critical currents as functions of the in-plane magnetic field applied parallel to the current, $B_x \parallel I$, corresponding to $\theta_B = 0$. The critical currents measured for the two bias directions nearly overlap, indicating suppression of the SDE. **(b)** Magnitudes of the critical currents for the magnetic field applied perpendicular to the current, $B_y \perp I$, corresponding to $\theta_B = \pi/2$. The separation between the two critical-current branches indicates a finite SDE. Adapted from Ref. [20].

current directions are nearly identical, and the SDE is suppressed [Fig. 21(a)]. In contrast, when the field is applied along the junction and perpendicular to the current, corresponding to $\theta_B = \pi/2$, the two critical-current branches become unequal and a finite SDE appears [Fig. 21(b)]. This angular dependence agrees with the symmetry analysis and with the numerical and analytical results in Fig. 20, which predict vanishing SDE for $\theta_B = 0$ and π in a junction dominated by Rashba SOC.

Additional zeros of the diode efficiency, which depend on B and are therefore not symmetry-related, appear in the numerical results shown in Fig. 20(a) and are attributed to finite-size effects and higher-order contributions. For completeness, we show in Figs. 20(a) and (b) the dependence of η on the junction crystallographic orientation θ_C and magnetic field strength for $\theta_B = \pi/2$. As expected, both the numerical [panel (b)] and

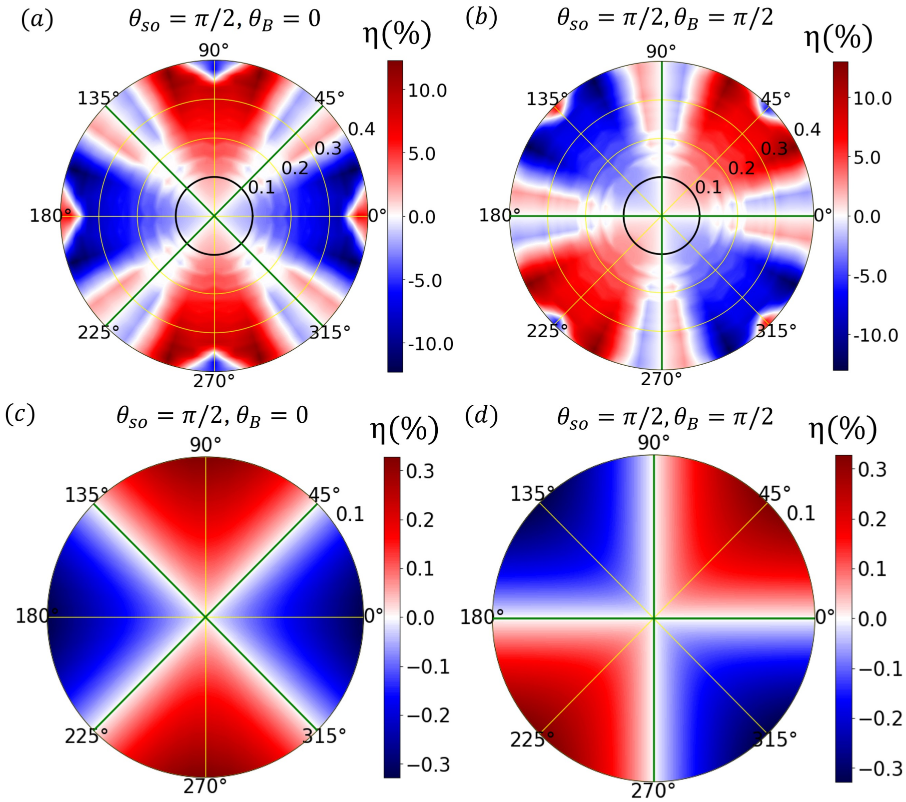

Figure 22: Crystalline anisotropy of the SDE in an Al/InSb junction with purely Dresselhaus SOC ($\theta_{so} = \pi/2$). **(a)** Diode efficiency from numerical simulations as a function of the junction orientation θ_C (polar angle) and the field strength B (radial coordinate) for a magnetic field applied along the current direction ($\theta_B = 0$). **(b)** Same as in (a), but for a magnetic field perpendicular to the current direction ($\theta_B = \pi/2$). Panels **(c)** and **(d)** show results from the analytical model (with $\tau = 0.99$) corresponding to (a) and (b), respectively, within the low-field region indicated by the black circles in (a) and (b). Green lines denote junction orientations for which symmetry-imposed suppression of the SDE occurs independently of the field strength [62].

analytical [panel (d)] results exhibit rotational symmetry, confirming the lack of crystalline anisotropy when Dresselhaus SOC is not present.

6.4 Magneto-Crystalline-Anisotropic Superconducting Diode Effect

Unlike Rashba SOC, Dresselhaus SOC is not invariant under in-plane rotations and therefore introduces, in addition to magnetoanisotropy, a crystalline anisotropy whereby the SDE depends on the current direction, which is set by the junction crystallographic orientation.

We first consider a junction with only Dresselhaus SOC ($\theta_{so} = \pi/2$). The diode efficiency, computed numerically as a function of the junction crystallographic angle θ_C (polar angle) and the magnetic field strength (radial coordinate), is shown in Figs. 22(a) and (b) for magnetic field orientations $\theta_B = 0$ and $\theta_B = \pi/2$, respectively. In the low-field regime, the analytical results, presented in panels (c) and (d), are in good agreement with the numerical simulations. A clear twofold crystalline anisotropy is observed for both selected magnetic field orientations. As the magnetic field increases, higher-order contributions become significant, leading to more complex and richer anisotropic patterns, as shown in panels (a) and (b). For $\theta_B = 0$ ($\theta_B = \pi/2$), the SDE is suppressed at junction orientations $\theta_C = (2n+1)\pi/4$ ($\theta_C = n\pi/2$), indicated by green lines, and in agreement with the predictions of Table 4. Figure 22 also shows that, in the presence of only Dresselhaus SOC, the SDE is maximized when the magnetic field is aligned with the current direction. This contrasts with the Rashba-only case, where the optimal SDE occurs when the magnetic field is oriented along the junction, i.e., perpendicular to the current flow.

The behavior of the diode efficiency for a junction with both Rashba and Dresselhaus SOC, with spin-orbit angle $\theta_{so} = \pi/8$ ($\beta \approx 0.4\alpha$), is shown in Fig. 23. The top and bottom rows correspond to the numerical simulations and the low-field analytical approach, respectively. The diode efficiency exhibits crystalline anisotropy for both $\theta_B = 0$ and $\theta_B = \pi/2$.

Despite the Rashba SOC strength being nearly twice that of the Dresselhaus SOC, the crystalline anisotropy at $\theta_B = 0$ closely resembles the Dresselhaus-only case (compare the left columns of Figs. 22 and 23). This occurs because, for $\theta_B = 0$, the magnetic

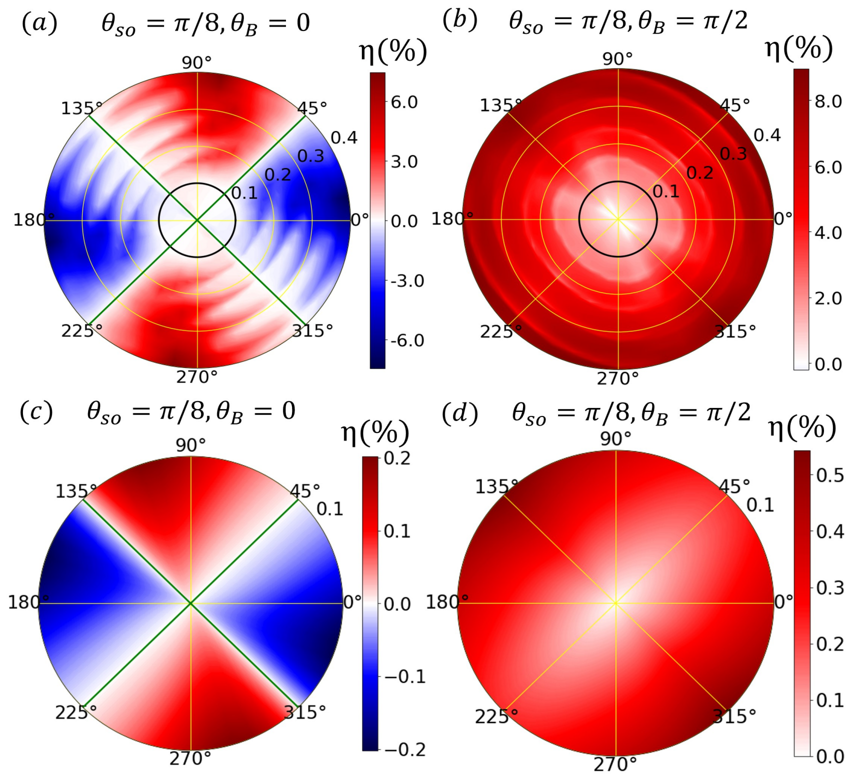

Figure 23: Crystalline anisotropy of the SDE in a Rashba-dominated Al/InAs junction with coexisting Rashba and Dresselhaus SOCs. (a)–(d) Same as in Figs. 22(a)–(d), but for $\theta_{so} = \pi/8$ ($\beta \approx 0.4\alpha$) [62].

field is aligned with the current direction, thereby reducing the influence of Rashba SOC. Consistent with the symmetry conditions in Table 4, for $\alpha \neq 0$, $\beta \neq \alpha$, and $\theta_B = 0$, the SDE is suppressed at junction orientations $\theta_C = (2n+1)\pi/4$, highlighted by green lines in Figs. 23(a) and (c).

In contrast, for $\theta_B = \pi/2$, the magnetic field is oriented along the junction (i.e., perpendicular to the current flow), enhancing the role of Rashba SOC. This results in a crystalline anisotropic SDE pattern that more closely resembles the Rashba-only case, while still retaining anisotropy. In this configuration, the symmetry conditions in Table 4 cannot be satisfied for any θ_C, and therefore no junction orientation leads to symmetry-

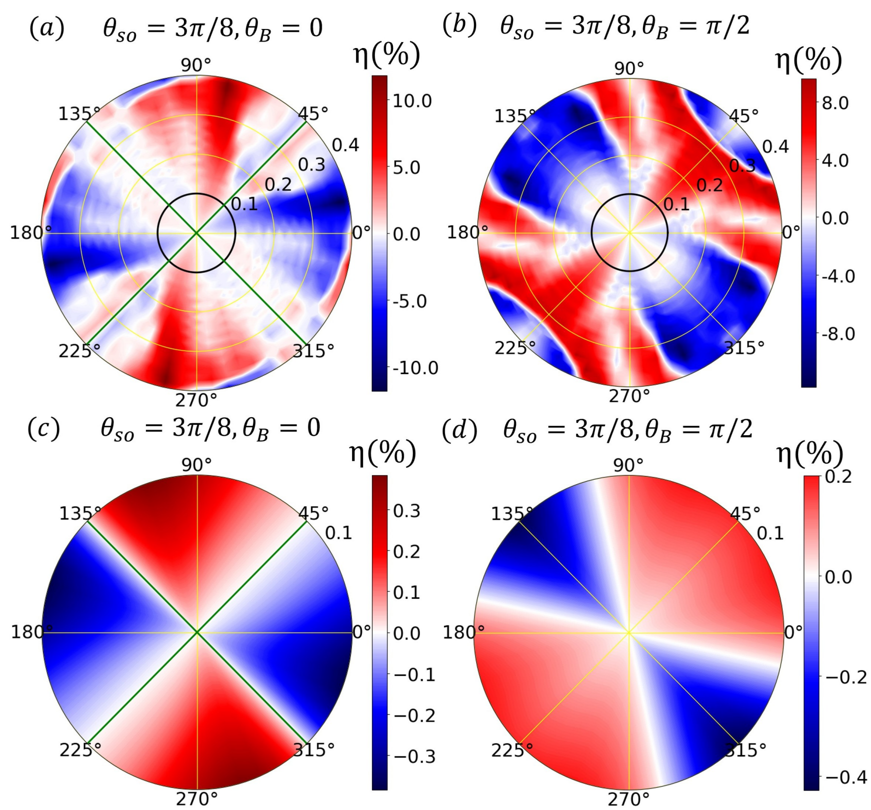

Figure 24: Crystalline anisotropy of the SDE in a Dresselhaus-dominated Al/InSb junction with coexisting Rashba and Dresselhaus SOCs. (a)–(d) Same as in Figs. 22(a)–(d), but for $\theta_{so} = 3\pi/8$ ($\beta \approx 2.4\alpha$) [62].

imposed suppression of the SDE.

Complementary to Fig. 23, Fig. 24 illustrates the crystalline anisotropy of the SDE in a Dresselhaus-dominated Al/InSb junction with $\theta_{so} = 3\pi/8$ ($\beta \approx 2.4\ \alpha$). The top and bottom rows present the numerical and low-field analytical results, respectively. The symmetry-imposed zeros of the SDE—indicated by the green lines in Figs. 24(a) and (c)—are consistent with condition (3) in Table 4. Note, however, that the SDE zeros in Figs. 24(b) and (d) depend explicitly on the magnetic field strength and are therefore not symmetry-protected.

Notably, in the Dresselhaus-dominated regime, polarity reversals of the SDE can

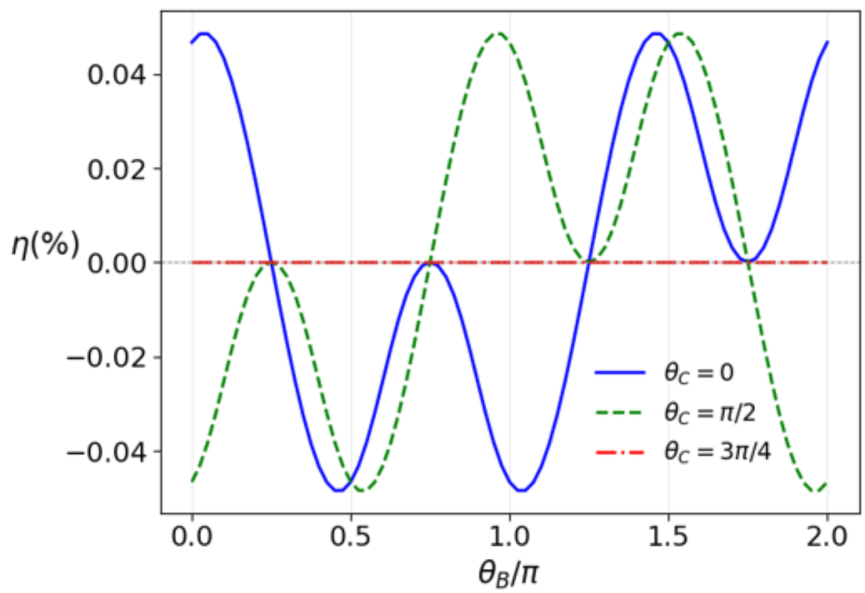

Figure 25: Diode efficiency from numerical simulations as a function of the magnetic-field angle θ_B for Al/InSb junctions at $B = 0.1$ T, with $\alpha = \beta$ ($\theta_{so} = \pi/4$), and junction orientations $\theta_C = 0$ (blue solid line), $\theta_C = \pi/2$ (green dashed line), and $\theta_C = 3\pi/4$ (red dash-dotted line). Zeros of η accompanied by sign reversals satisfy condition (4) in Table 4, while those occurring without sign changes satisfy condition (5).

occur at low magnetic fields for specific crystallographic orientations, even in the absence of a junction potential. This effect is particularly pronounced when the magnetic field is aligned with the junction ($\theta_B = \pi/2$), as clearly illustrated in Fig. 24(b), where SDE polarity reversals appear at $B \approx 0.13$ T and $B \approx 0.31$ T for junction orientations $\theta_C = (4n+3)\pi/4$ and $\theta_C = (4n+1)\pi/4$, respectively. These results demonstrate that SDE sign reversals at low magnetic fields can arise not only from electrostatic gating but also from the geometric configuration of the magnetic field and junction orientation.

As an illustration, Fig. 25 shows the dependence of η on the magnetic field orientation for a junction with $\alpha = \beta$ ($\theta_{so} = \pi/4$), for different junction orientations and $B = 0.1$ T. The figure reveals a distinctive signature: zeros associated with the fulfillment of Eq. (4.15) [condition (4) in Table 4] are accompanied by polarity reversals of the SDE, whereas those related to gauging out the SOC [condition (5) in Table 4] do not involve a change of sign. A particularly notable case occurs for $\alpha = \beta$ and $\theta_C = 3\pi/4$, where the SDE vanishes for any in-plane orientation of the magnetic field.

The numerical results presented in this chapter are in agreement with the magnetic and crystalline anisotropies predicted by the symmetry analysis and low-field analytical model. The predicted magnetic and crystalline anisotropies can serve as signatures of the interplay between SOC and the Zeeman interaction and their impact on the SDE, thereby helping to identify the physical mechanism responsible for the SDE in proximitized planar JJs.

7 GATE AND DISORDER DEPENDENCE OF DIODE EFFECT IN PLANAR JOSEPHSON JUNCTIONS

7.1 Gate Dependence of the Superconducting Diode Effect

This section is based in part on Ref. [62].

In planar semiconductor-superconductor JJs, electrostatic gates provide an important experimental control knob for tuning the transport properties of the junction. A gate voltage applied to the normal region modifies the chemical potential, and therefore changes the transparency of the weak link. Since the SDE is sensitive to the Andreev bound states, even modest changes in the junction potential can strongly affect the forward and reverse critical currents. In addition, the Rashba SOC strength in semiconductor heterostructures can also depend on electrostatic gating, providing another mechanism through which the diode response can be controlled.

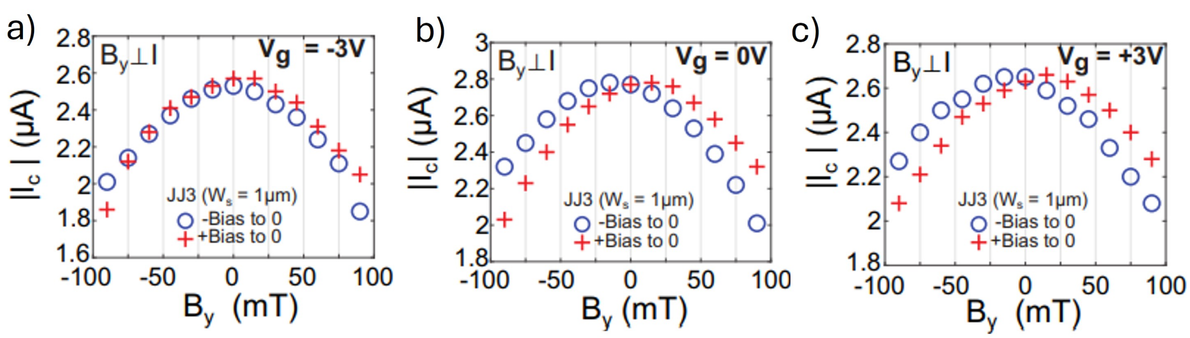

Figure 26: Gate dependence of the SDE in a gated Al-InAs planar JJ. The absolute values of the forward and reverse critical currents are shown as functions of the in-plane magnetic field applied perpendicular to the current direction for three gate voltages: **(a)** $V_g = -3$ V, **(b)** $V_g = 0$ V, and **(c)** $V_g = +3$ V. A clear splitting between the two critical-current branches is observed at $V_g = 0$ and $V_g = +3$ V, indicating a finite SDE. At $V_g = -3$ V, the two branches nearly overlap, showing a rather weak diode response [20].

Recent experiments on gated epitaxial Al-InAs JJs have demonstrated that the SDE depends strongly on the applied gate voltage. Figure 26 shows the measured critical currents as a function of in-plane magnetic field for a gated junction at three different gate voltages. At $V_g = 0$ and $V_g = +3$ V, the magnitudes of the forward and reverse critical currents are clearly different when the in-plane magnetic field is applied perpendicular to the current direction, indicating a finite SDE. In contrast, at $V_g = -3$ V, the two critical-current branches are nearly identical, showing that the diode effect is strongly suppressed. This behavior suggests that gate voltage modifies the microscopic ingredients responsible for nonreciprocity.

To investigate this effect theoretically, we model the gate voltage as a local electrostatic potential applied to the normal region of the planar JJ, as described earlier in Sec. 4.4. In the short-junction δ-barrier model, this gate dependence enters through the junction potential

$$V_0(V_g) = \mu_S - \mu_N(V_g), \tag{7.1}$$

where μ_S and $\mu_N(V_g)$ are the chemical potentials in the superconducting and normal regions, respectively. Equivalently, the gate voltage controls the dimensionless barrier strength

$$Z_0(V_g) = \frac{V_0(V_g)}{2E_T} = \frac{V_0(V_g)W_N}{\pi\hbar v_F}. \tag{7.2}$$

The corresponding junction transparency is then

$$\tau(V_g) = \frac{1}{1 + Z_0^2(V_g)} = \frac{1}{1 + \left[\frac{V_0(V_g)}{2E_T}\right]^2}. \tag{7.3}$$

Thus, varying the gate voltage changes the effective barrier strength and therefore the junction transparency.

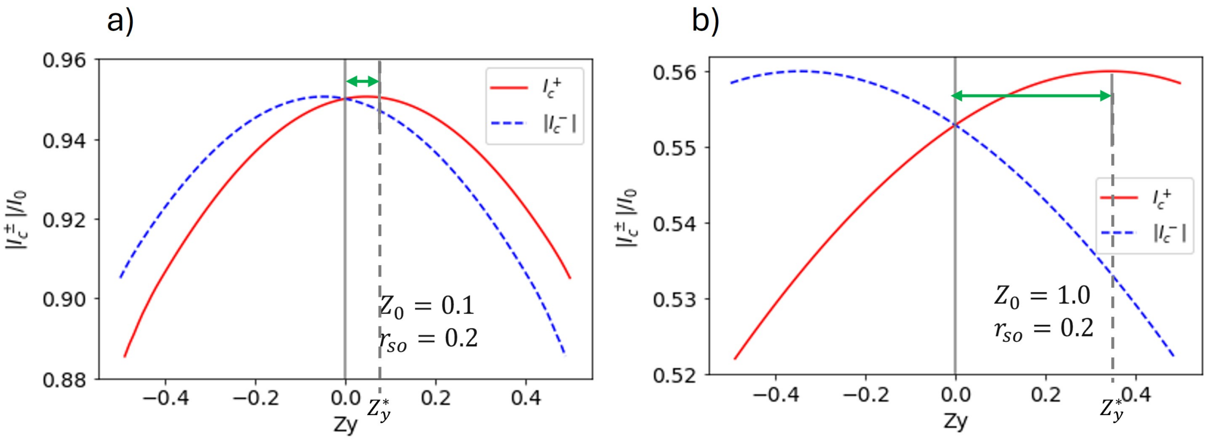

Figure 27: Gate-dependent shift of the critical-current maximum in the short-junction δ-barrier analytical model. The normalized forward and reverse critical currents, $|I_c^+|/I_0$ and $|I_c^-|/I_0$, are plotted as functions of the dimensionless Zeeman parameter Z_y for two junction-barrier strengths: (a) $Z_0 = 0.1$ and (b) $Z_0 = 1.0$, with fixed spin-orbit ratio $r_{\rm so} = k_{\rm so}/k_F = 0.2$. For a nearly transparent junction, the maxima of the two critical-current branches occur close to zero magnetic field. As the barrier strength increases, corresponding to reduced junction transparency, the critical-current maxima shift to finite and opposite values of Z_y. This illustrates that gate tuning of the junction potential can control the magnetic-field position of the critical-current maximum.

Figure 27 shows how the gate-controlled junction potential affects the magnetic-field position of the critical-current maximum. Increasing Z_0 corresponds to reducing the junction transparency τ, as shown in Eq. (7.3).

For the nearly transparent case, $Z_0 = 0.1$, shown in Fig. 27(a), the forward and reverse critical-current branches reach their maximum close to $Z_y = 0$. This is expected because, in the perfect-transparency limit, the critical current is maximized when the magnetic-field-induced phase shift is nearly zero. However, when the barrier strength is increased to $Z_0 = 1.0$, as shown in Fig. 27(b), the maxima of the forward and reverse critical currents shift to finite and opposite values of Z_y. This critical current maximum shift can be understood from the analytical expression for the field $B_*^\pm$. From Eq. (4.52), the position of the maximum scales as

$$Z_y^* \propto (1-\tau)^{1/4}\frac{k_{\rm so}}{k_F}. \tag{7.4}$$

Thus, when the gate voltage changes the junction potential, it modifies the transparency $\tau(V_g)$, which in turn shifts the magnetic field at which the critical current reaches its maximum. For $\tau \approx 1$, the factor $(1-\tau)^{1/4}$ is small and the maximum remains close to zero magnetic field. As τ decreases, this factor increases, pushing the critical-current maxima to finite magnetic fields.

Assuming that the barrier height, $V_0(V_g)$, varies monotonically with the gate voltage V_g, increasing V_g reduces the junction transparency. As a result, the maximum of the critical-current amplitudes shifts to higher magnetic-field strengths. This qualitatively explains the behavior observed experimentally and shown in Fig. 26.

Determining the precise dependence of V_0 on the gate voltage requires self-consistent band-structure and charge-density calculations together with the Poisson equation for the electrostatic potential. Such an approach would substantially increase the complexity of the numerical simulations and is therefore beyond the scope of the present study. Instead, we perform numerical simulations for different values of V_0, disregarding its explicit dependence on V_g. Although this approach precludes a direct quantitative comparison with the experimental results, it provides valuable insight into the role of gate-induced changes in the barrier height and offers a qualitative description of the gate-voltage dependence of the SDE.

Figure 28 shows the Kwant simulation results for different barrier potentials for a

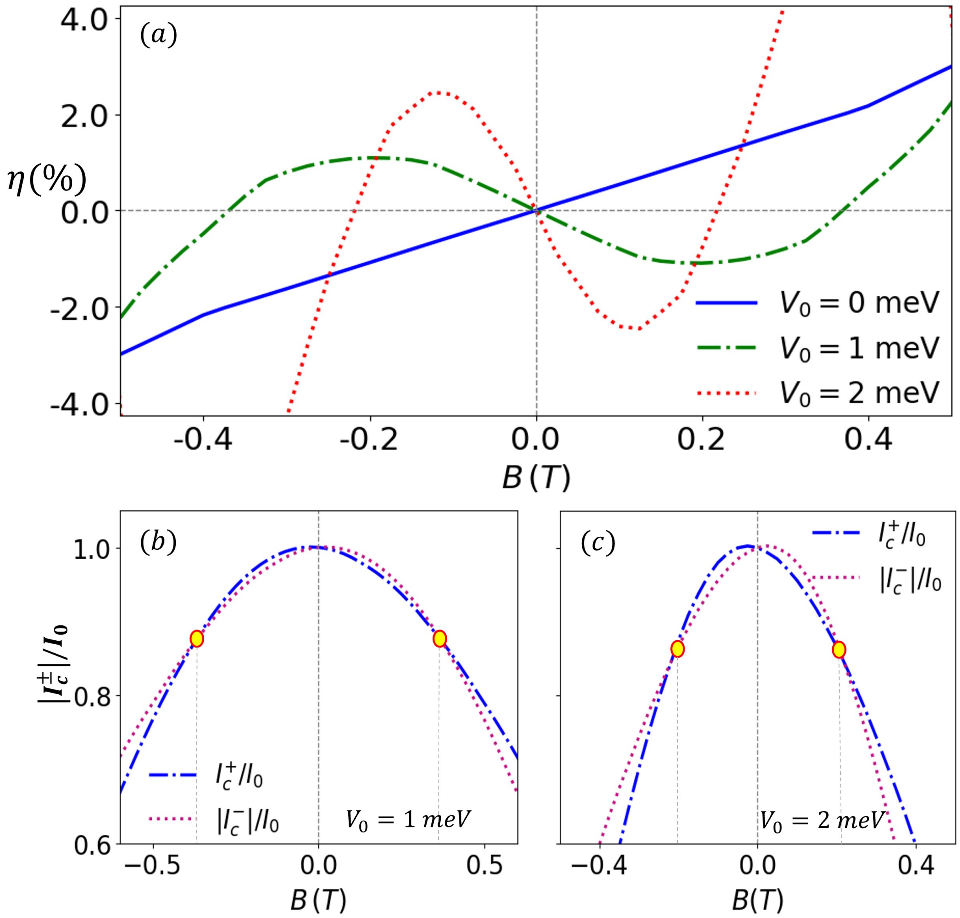

Figure 28: (a) Magnetic-field dependence of the diode efficiency in an Al/InAs junction with purely Rashba SOC ($\theta_{so} = 0$), shown for different values of the junction potential V_0. Panels (b) and (c) display the corresponding magnetic-field dependence of the forward (dash-dotted line) and reverse (dotted line) critical currents. Yellow markers indicate the crossing points of the critical currents at $B \approx \pm 0.37$ T and $B \approx \pm 0.22$ T, which coincide with the SDE polarity reversals shown in panel (a) for $V_0 = 1$ meV and $V_0 = 2$ meV, respectively. The numerical simulations correspond to a junction with $W_N = 100$ nm and $W_S = 40$ nm, with the magnetic field oriented perpendicular to the current flow (i.e., $\theta_B = \pi/2$)[62].

system with only Rashba SOC. Interestingly, SDE sign reversals can also occur at lower fields when the potential V_0 is properly tuned, which can be achieved using a top gate on the N region of the junction. This behavior has recently been observed experimentally [95] and attributed to the interplay between linear and cubic Dresselhaus SOC and gate-induced modulations of the Rashba SOC strength. However, our numerical calculations indicate that SDE polarity switches at low fields can also arise in junctions with narrow superconducting contacts solely from gate-induced changes in the junction potential, even in the absence of Dresselhaus SOC. This is illustrated in Fig. 28(a), which shows the diode efficiency as a function of magnetic field strength for an Al/InAs junction with only Rashba SOC and $\theta_B = \pi/2$, at three different values of the junction potential. The figure illustrates how tuning V_0 from 0 to 1 meV and 2 meV modifies the magnetic-field dependence of the diode efficiency, leading to the emergence of SDE polarity reversals at lower fields. As shown in Figs. 28(b) and (c), which display the magnetic-field dependence of the critical currents, these low-field polarity reversals occur when the forward and reverse critical currents intersect due to differences in the curvature of their B dependence. Importantly, this mechanism is not related to 0–π transitions, which typically arise at higher magnetic fields (see Fig. 19).

Our numerical simulations highlight the role of electrostatic control in superconducting non-reciprocity, suggesting that gate tuning can serve as a practical way to control not only the magnitude of the Josephson critical currents but also the sign and strength of the diode response.

7.2 Disorder Effects on the Superconducting Diode Effect

Disorder is unavoidable in realistic semiconductor-superconductor planar JJs. Charge impurities and electrostatic fluctuations produce spatial variations of the local chemical potential. Since the SDE depends on the phase-dependent Andreev bound-state spectrum, these variations can modify the magnitude, magnetic-field dependence, and polarity of the diode response [114].

To model electrostatic disorder, a fixed number of impurities, N_{imp}, is placed randomly on the tight-binding lattice. The impurity concentration is defined as

$$n_{\text{imp}} = \frac{N_{\text{imp}}}{N} \times 100\%, \tag{7.5}$$

where N is the total number of lattice sites. Following Ref. [115], the unnormalized impurity potential at the lattice site $\mathbf{r}_i$ is written as

$$\nu_i = \sum_{j=1}^{N_{\text{imp}}} (-1)^j \exp\left[-\frac{|\mathbf{r}_i - \mathbf{R}_j|}{\lambda_{\text{imp}}}\right], \tag{7.6}$$

where $\mathbf{R}_j$ is the position of the jth impurity and λ_{imp} is the characteristic decay length of its potential. The factor $(-1)^j$ produces alternating positive and negative contributions.

The spatial average and standard deviation of ν_i are

$$\overline{\nu} = \frac{1}{N}\sum_{i=1}^{N} \nu_i, \qquad N_\nu = \left[\frac{1}{N}\sum_{i=1}^{N} \left(\nu_i - \overline{\nu}\right)^2\right]^{1/2}. \tag{7.7}$$

The position-dependent impurity potential is then defined as

$$V_{\text{imp}}(\mathbf{r}_i) = \frac{V_{\text{dis}}}{N_\nu}\left(\nu_i - \overline{\nu}\right), \tag{7.8}$$

where V_{dis} determines the overall disorder strength. This normalization gives

$$\langle V_{\text{imp}} \rangle = 0, \qquad \left\langle V_{\text{imp}}^2 \right\rangle = V_{\text{dis}}^2. \tag{7.9}$$

In the numerical implementation, the local chemical potential is modified according to

$$\mu(\mathbf{r}_i) = \mu + V_{\rm imp}(\mathbf{r}_i) \ . \tag{7.10}$$

Because disorder removes translational invariance, the numerical calculation becomes computationally demanding. We therefore consider a reduced-size planar JJ in which the left and right superconducting regions each have width $W_S = 400$ nm, the normal region has width $W_N = 100$ nm, and the finite junction length is $L = 50$ nm. The lattice constant is $a = 10$ nm, giving a total of $N = 455$ lattice sites. The calculations use $N_{\rm imp} = 91$, corresponding to an impurity concentration of 20%.

The same random seed and, therefore, the same spatial arrangement of impurities are used for all nonzero values of $V_{\rm dis}$. Changing $V_{\rm dis}$ modifies only the amplitude of the disorder potential.

Figure 29 shows the spatial structure of the impurity potential. The superposition of the exponentially decaying impurity contributions produces a nonuniform electrostatic landscape containing both positive and negative regions.

The magnetic-field dependence of the SDE is first calculated for a junction with purely Rashba SOC and a magnetic field applied along the junction, $\mathbf{B} \parallel \hat{\mathbf{y}},\ \theta_B = \pi/2$. For this field orientation, the clean junction supports a finite SDE, as predicted by the symmetry analysis in Chapter 6.

Figure 30(a) shows the clean-junction result. For weak disorder, $V_{\rm dis} = 0.5$ meV, the critical currents and diode efficiency are only slightly modified, as shown in Fig. 30(b). The overall magnetic-field dependence remains similar to that of the clean junction.

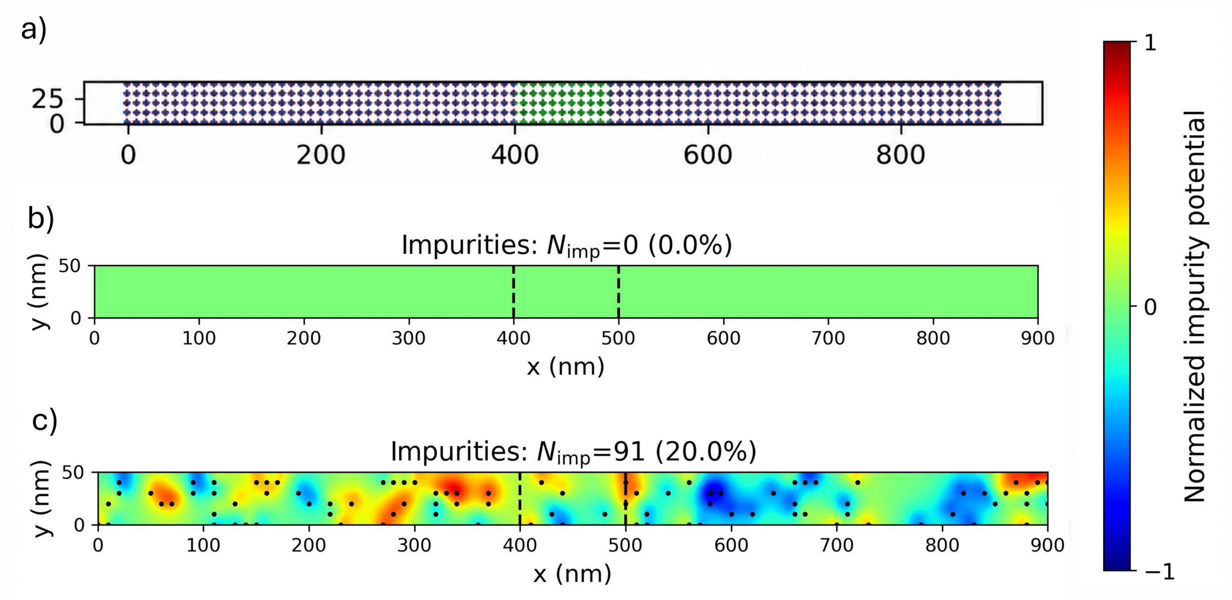

Figure 29: **(a)** Tight-binding lattice used in the disorder calculations. The superconducting regions each have width $W_S = 400$ nm, the normal region has width $W_N = 100$ nm, and the junction length is $L = 50$ nm. **(b)** Clean junction with $N_{\text{imp}} = 0$. **(c)** Representative normalized impurity-potential profile for $N_{\text{imp}} = 91$, corresponding to an impurity concentration of 20%. The black dots on (c) indicate the positions of the impurities. The color bar represents the normalized impurity potential, $V_{\text{imp}}(\mathbf{r})/\max|V_{\text{imp}}(\mathbf{r})|$.

For the larger disorder strength, $V_{\text{dis}} = 4$ meV, the changes become more pronounced. The critical-current profiles, the magnitude of η, and its low-field slope are modified. A small additional polarity reversal also appears close to zero field due to a crossing between the forward and reverse critical-current magnitudes.

Calculations were also performed for a magnetic field applied along the current direction, in which case the SDE was found to remain strongly suppressed, even in the presence of disorder. This behavior is consistent with the experimental results reported in Ref. [20] [see Fig. 21(a)].

The present preliminary results are based on a single impurity arrangement. For the same impurity concentration, different random seeds generate different spatial distributions and may produce different diode responses. Therefore, a more systematic study

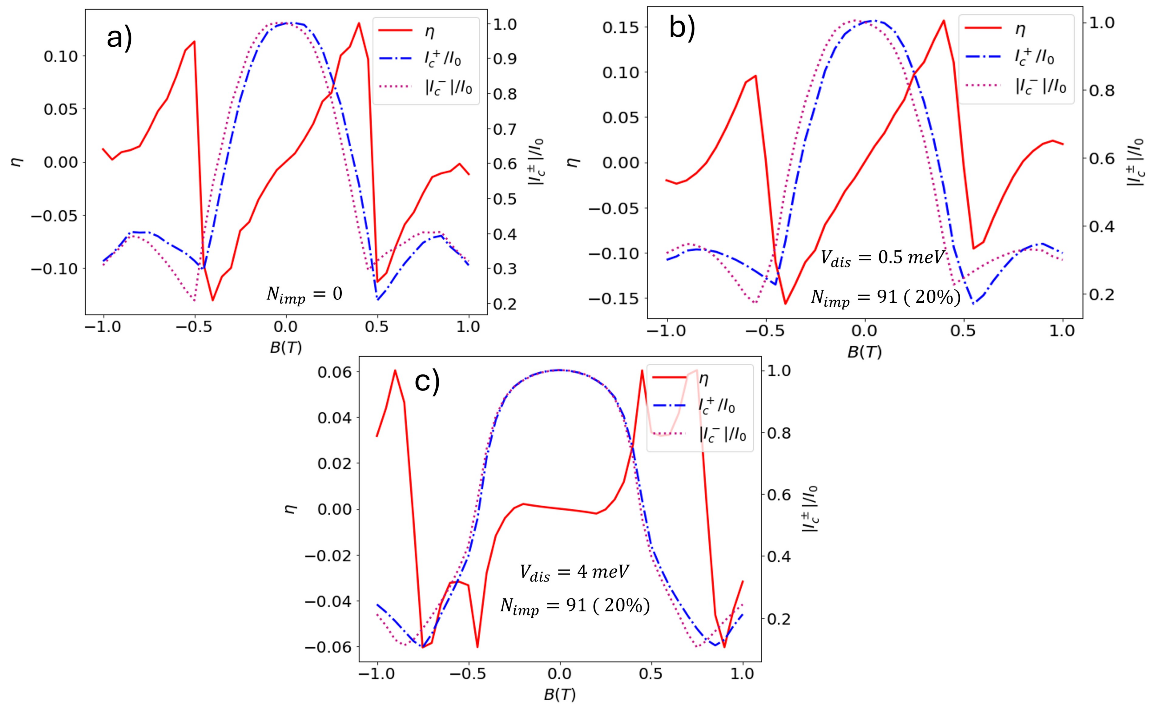

Figure 30: Magnetic-field dependence of the diode efficiency η and the normalized forward and reverse critical-current magnitudes for a junction with purely Rashba SOC and $\mathbf{B} \parallel \hat{\mathbf{y}}$. **(a)** Clean junction. **(b)** Weak disorder with $V_{\mathrm{dis}} = 0.5\ \mathrm{meV}$. **(c)** Stronger disorder with $V_{\mathrm{dis}} = 4\ \mathrm{meV}$. The same impurity arrangement is used in panels **(b)** and **(c)**.

incorporating multiple random impurity configurations is still needed to provide a comprehensive understanding of the effects of disorder on the SDE. Such an investigation is left for future work.

8 CONCLUSION

We theoretically investigated the magnetic and crystalline anisotropies of the SDE in proximitized planar JJs with Rashba and Dresselhaus SOC under an in-plane magnetic field. A symmetry analysis revealed geometric constraints on field and crystallographic orientations for which the SDE is suppressed, independent of field strength. These predicted conditions provide experimentally testable signatures to assess whether the interplay between SOC and Zeeman interaction underlies the SDE in semiconductor-based planar JJs.

We developed a phenomenological model showing that the diode efficiency depends on the relative alignment between the spin–orbit and magnetic fields. This behavior was confirmed in the narrow-junction, low-field regime by an analytical model, which linked the anisotropy of the diode efficiency to the SOC-induced deformation of the Fermi contours and the anisotropy of the Cooper-pair momentum. These results were supported by tight-binding simulations of the Bogoliubov–de Gennes equation and were comparable with experimental measurements in epitaxial Al/InAs planar JJs, where the SDE is suppressed when the in-plane magnetic field is parallel to the current and becomes finite when the field is perpendicular to the current [20]. Notably, the simulations showed that electrostatic gating can induce SDE polarity reversals in the low-field regime even with only Rashba SOC, in agreement with recent experiments [20, 95]. Additional polarity reversals were predicted for specific magnetic-field orientations, junction geometries, and SOC ratios α/β.

Preliminary disorder calculations showed that weak electrostatic disorder produced relatively small changes in the critical currents and diode quality factor, whereas stronger disorder could significantly modify the magnitude, slope, and polarity of the SDE. Since these calculations were based on a single impurity arrangement, a systematic study involving multiple disorder realizations is required to determine the disorder-averaged behavior.

Overall, our results clarify how SOC and Zeeman coupling generate anisotropic nonreciprocal transport and provide guidance for experimentally probing the mechanisms behind the SDE through its predicted anisotropic diode response.

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ABSTRACT

NON-RECIPROCAL EFFECTS IN GATED SUPERCONDUCTOR/SEMICONDUCTOR PLANAR JOSEPHSON JUNCTIONS

by

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We theoretically investigate nonreciprocal superconducting transport in proximitized planar Josephson junctions (JJs). In these systems, the superconducting diode effect (SDE) can emerge when inversion and time-reversal symmetries are simultaneously broken, leading to different forward and reverse critical-current magnitudes. A central objective of this study is to determine how the SDE depends on the magnetic-field orientation, crystallographic direction, junction transparency, and electrostatic gating. In particular, we show how magnetic and crystalline anisotropies can provide experimentally testable information about the mechanism underlying the SDE. We consider junctions with Rashba and Dresselhaus spin–orbit couplings (SOCs) under an in-plane magnetic field and identify symmetry-imposed field and junction orientations for which the SDE is suppressed independently of the magnetic-field strength. Phenomenological and analytical models show how the diode response depends on the relative orientation of the spin–orbit and Zeeman fields. Numerical tight-binding calculations based on the Bogoliubov–de Gennes formalism support the analytical results and yield critical-current behavior in agreement with re-

cent experiments on Al/InAs planar JJs. We also investigate the magnetic-field-dependent current–phase relation and show that its phase-resolved asymmetry provides signatures of the SDE, while the associated ground-state phase jumps can be used to estimate the Rashba SOC strength. Junction transparency strongly modifies the magnetic-field dependence of the forward and reverse critical currents, including the positions of their maxima and the low-field behavior of the diode efficiency. Electrostatic gating provides additional control by tuning the chemical potential and transparency of the junction, thereby changing the magnitude of the SDE and inducing reversals of its polarity. These results establish magnetic and crystalline anisotropies, magneto-CPR measurements, and gate tunability as complementary tools for probing and controlling the SDE in semiconductor-based planar JJs.

AUTOBIOGRAPHICAL STATEMENT

Abhishek Chilampankunnel Prasannan (Abhishek C P) was born and raised in Kerala, India. He earned his Bachelor of Science degree in Physics from Mahatma Gandhi University, Kerala, in 2017. After briefly working as a Technical Supervisor in the Research and Development Department of MRF Limited, he returned to academia to pursue his interest in physics and scientific research. He completed his Master of Science degree in Physics at Mahatma Gandhi University in 2020. During the following year, he worked as a laboratory assistant in the Materials Science Research Laboratory at St. Thomas College, Palai, where he gained experience in the structural characterization of nanomaterials using powder X-ray diffraction. This experience further strengthened his interest in condensed matter and materials physics.

In 2021, Abhishek moved to the United States to pursue graduate studies at Wayne State University in Detroit, Michigan. He received a Master of Science degree in Physics from Wayne State University in 2024 and continued his doctoral research under the supervision of Dr. Alex Matos-Abiague. His research focuses on non-reciprocal superconducting transport in superconductor–semiconductor planar Josephson junctions. Using analytical methods and tight-binding quantum-transport simulations.

Abhishek also served as a teaching assistant, laboratory instructor, and instructor of record at Wayne State University. His long-term goal is to pursue a teaching-focused academic career and make physics accessible and meaningful to students from diverse educational backgrounds.